

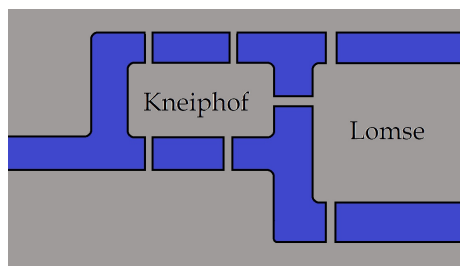
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Lecture 1 Introduction: Bridges of Königsberg

In the 18th century, the Prussian city of Königsberg (today: Kaliningrad) was divided into four different landmasses by the Pregel river: The islands Kneiphof and Lomse and the mainland to either side of the river. Those landmasses were connected by seven bridges:

- Two bridges each going from either part of the mainland to Kneiphof
- One bridge each going from either part of the mainland to Lomse
- and one more bridge connecting Kneiphof and Lomse

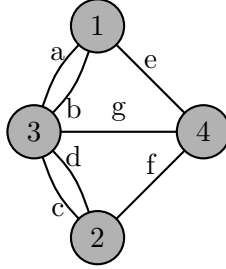


Given this situation, the mathematician Leonard Euler became invested in determining whether one could give a tour through the city traversing all bridges exactly once. In 1736, he gave a rigorous proof that one cannot, in a paper that is now regarded as the founding moment of graph theory. His argument was based on the following abstractions: Since the exact position of a walker on a landmass is irrelevant, we can represent the landmasses by abstract points (which we call *vertices*) and the bridges by line segments (which we call *edges*) connecting those points. These are the basic ingredients for what we call a *multigraph* in modern language:

Definition 1. A *(multi-)graph* G is a triple (V, E, ι) consisting of a set V of *vertices*, a set E of *edges*, and a map $\iota : E \rightarrow \mathcal{P}_1(V) \cup \mathcal{P}_2(V)$ assigning to each edge e its endpoints¹.

In this way, after Euler's abstraction we obtain something like this:

¹Here, we used the following notation: For a set S , the set of all subsets of S with exactly i elements is denoted by $\mathcal{P}_i(S)$. Thus, an edge can have either one or two distinct endpoints.



In other words, we can model the landmasses of Königsberg as a set $V = \{1, 2, 3, 4\}$ of four vertices, and the bridges as a set of seven edges $E = \{a, b, \dots, g\}$, where

$$\begin{aligned} \iota(a) = \iota(b) &= \{1, 3\}, & \iota(c) = \iota(d) &= \{2, 3\} \\ \iota(e) &= \{1, 4\}, & \iota(f) &= \{2, 4\}, & \iota(g) &= \{3, 4\}. \end{aligned}$$

Some terminology. The definition of a multigraph explicitly allows $\iota(e) = \{v\}$ for some $e \in E, v \in V$ — thus, an edge might connect a vertex to itself. In this case, we say that the edge is a *loop*. Perhaps more relevant to the bridges of Königsberg, we say that two edges $e, e' \in E$ are *parallel* if $\iota(e) = \iota(e')$. The two endpoints of a common edge are called *adjacent* or *neighbours*. We also sometimes say that a vertex v is *incident* to an edge e , by which we mean that v is an endpoint of e . Finally, a graph is *finite* if $|V| + |E| < \infty$.

Some notation. Given a graph G , we often write $V(G)$ and $E(G)$ for the vertex set and the edge set of G , respectively. We let $v(G) = |V(G)|$ and $e(G) = |E(G)|$ be the number of vertices and edges in G , respectively.

Having successfully encoded the topography of Königsberg into a mathematical structure, how do we proceed with the notion of a city tour?

Definition 2. Let $G = (V, E, \iota)$ be a multigraph. A *walk* of length k is a sequence $v_0 e_1 v_1 e_2 v_2 \dots e_k v_k$, where $e_1, \dots, e_k \in E$ and $v_0, v_1, \dots, v_k \in V$, such that $\iota(e_i) = \{v_{i-1}, v_i\}$ for all $i = 1, \dots, k$. A *trail* is a walk that uses no edge twice. Moreover, a *circuit* is a trail where the first and last vertex coincide.

Note that a walk of length k involves k edges and $k + 1$ vertices.

Definition 3. A trail that uses every edge in the graph exactly once is called an *Eulerian trail*. Analogously, a circuit using every edge exactly once is called an *Eulerian circuit* (or *Euler tour*). If a graph admits an Eulerian circuit, it is simply called *Eulerian*.

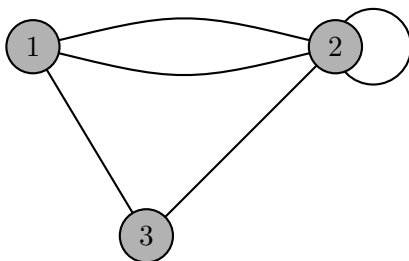
Hence, we are specifically interested in whether or not the graph we obtained earlier for Königsberg admits an Eulerian trail (or even an Eulerian circuit). Our answer will be in form of a general characterisation of such graphs, for which we need yet another two concepts:

Definition 4. Let $G = (V, E, \iota)$ be a graph. A vertex x is *connected to* a vertex y if there is a walk (equivalently, a trail or – defined next time – a path) starting in x and ending in y . If this is satisfied for all $x, y \in V$, then we say G is connected.

Remark 5. Every vertex is connected to itself via a ‘lazy’ walk of length 0 using no edges at all. It thus follows that connectedness is an equivalence relation on V , and the equivalence classes are called the (*connected*) *components* of G .

Definition 6. Let $G = (V, E, \iota)$ be a graph and $v \in V$ a vertex. The degree $\deg(v)$ is the number of edges v is incident to, with loops counted twice.

For example, in this graph, $\deg(1) = 3$, $\deg(2) = 5$, and $\deg(3) = 2$:



Let us now (as Euler did 286 years ago) imagine a connected graph with an Eulerian circuit. As we traverse along that circuit, we have to leave every visited vertex by a different edge than we entered (even if we have loops!). If a vertex v is visited t times, it has degree $\deg(v) = 2t$. In particular, the degree of every vertex must be even. This is the first implication in the following theorem:

Theorem 7 (Euler, 1736). *A finite connected (multi-)graph is Eulerian if and only if all of its vertex degrees are even.*

Proof of Theorem 7. We already showed the easy direction, so assume now that $G = (V, E, \iota)$ is a finite connected graph with only even vertex degrees.

To simplify the argument, we assume for now that the graph contains no loops. Consider a trail $\mathcal{T} = v_0 e_1 v_1 e_2 v_2 \dots e_k v_k$ of maximal length (i.e. such that no edge is used twice, and such that there is no strictly longer trail in G). Since this trail cannot be extended by an additional edge at v_k , all edges incident to v_k must already be in $\mathcal{T}$. We now show that this implies $v_0 = v_k$.

Assume that there are $2s$ ($s \in \mathbb{N}$) edges incident to v_k . One of those edges is required to be e_k . Where do the other $2s - 1$ go? If v_k occurs as an internal vertex of $\mathcal{T}$, that is, say $v_j = v_k$ for $1 \leq j < k$, then e_j and e_{j+1} must both be edges incident to v_k . Thus, any internal occurrence of v_k takes up 2 edges

(We are really using here that there are no self-loops). This explains where $2s - 2$ of the edges at v_k go, leaving one edge – and that one has to go out of v_0 to $v_1 \neq v_k$. Hence $v_0 = v_k$, and the trail $\mathcal{T}$ is really a circuit.

Next, we prove that this trail $\mathcal{T}$ is Eulerian: Assume for a contradiction that it wasn't. Then there is an edge e , say with endpoints $\iota(e) = \{x, y\}$, that is not on $\mathcal{T}$, but one of its endpoints, say $x \in V$, is – this follows from connectedness. Hence, x occurs in some position on $\mathcal{T}$, say $x = v_i$. Now consider the following trail:

$$\mathcal{T}' := y e x e_{i+1} v_{i+1} \dots e_k v_k e_1 v_1 \dots e_i v_i.$$

This is indeed a trail, since e was not on $\mathcal{T}$, and it contains $k + 1$ edges, contradicting the assumption that $\mathcal{T}$ is of maximal length. Therefore $\mathcal{T}$ is the desired Eulerian trail.

Finally, what happens if G contains loops? Simply cut off the loops (this doesn't change the connectedness or the parity of the degrees by Definition 6) and consider the Eulerian trail $\mathcal{T}$ obtained from the proof in the loop-less case. Now walk alongside $\mathcal{T}$. As $\mathcal{T}$ necessarily passes through all vertices in the graph, stop whenever you reach a vertex for the first time, and traverse all loops at that vertex, before continuing with $\mathcal{T}$. $\square$

We also have the following corollary:

Corollary 8. *A finite connected graph admits an Eulerian trail if and only if either 0 or 2 of its vertices have odd degree.*

Proof. If all vertices have even degree, there is nothing left to show. Otherwise, introduce a new edge connecting the two vertices with odd degree. In the modified graph, all degrees are even, hence there is an Eulerian circuit. Removing the newly inserted edge from the circuit turns it into an Eulerian trail.

Conversely, if a graph admits an Eulerian trail that is closed, i.e. a circuit, we simply apply the theorem. Otherwise, such an Eulerian trail must have a starting and an ending vertex both having odd degree. $\square$

In particular, the bridges of Königsberg did not allow for walking through the city along an Eulerian trail, as all four vertices in the corresponding graph have odd degree. (If you're curious, check out Kaliningrad on Google Maps to see whether it's possible nowadays).

Corollary 8 might raise the question “What happens if a graph has exactly one vertex of odd degree?” – as it turns out, this can't happen:

Lemma 9 (Handshake lemma). *Let $G = (V, E, \iota)$ be a finite graph. Then*

$$\sum_{v \in V} \deg(v) = 2|E| \tag{1}$$

In particular, G must contain an even number of vertices of odd degree.

Proof. We use a trick called *double counting*: On the one hand, every edge contributes two half-edges, thus there are $2|E|$ half-edges in total. On the other hand, every vertex contributes $\deg(v)$ half-edges by definition, and thus both sides of equation (1) count the same quantity. $\square$

Food for thought: If $G = (V, E, \iota)$ is a connected graph with 4 vertices of odd degree, can we always partition the edge set as $E = E_1 \uplus E_2$ such that $(V, E_1, \iota|_{E_1})$ and $(V, E_2, \iota|_{E_2})$ admit Eulerian trails? In other words, is it possible to colour every edge either red or blue such that the graph with red edges and the graph with blue edges each contain an Eulerian trail? What if we have $2k$ vertices of odd degree (for $k \geq 2$)?

Lecture 2 Simple graphs and subgraphs

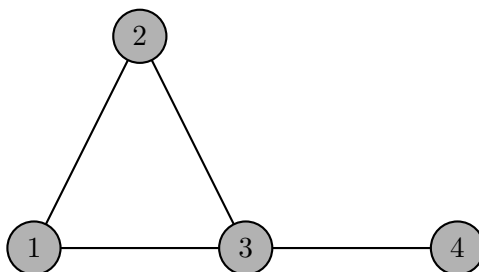
For the bridges of Königsberg problem last lecture, we considered multigraphs, as the relevant information was *how* the landmasses are connected with each other. However, often in applications it is only relevant *if* vertices are neighbours to each other.

Definition 10. A *simple graph* is a multigraph without double edges or loops. Equivalently, it is a pair $G = (V, E)$ where $E \subseteq \mathcal{P}_2(V)$.

Let's discuss this definition: A multigraph $G = (V, E, \iota)$ has no double edges or loops iff the incidence map ι sends no edge to a single-vertex set, and doesn't send any two edges to the same set of vertices. Hence, a graph is simple iff ι is an injective map $E \rightarrow \mathcal{P}_2(V)$. But in that case, we can directly identify an edge e with its image $\iota(e)$, so we can just as well assume $E \subseteq \mathcal{P}_2(V)$.

Most of the remaining lectures will be concerned with simple graphs, but we will occasionally require multigraphs as well.

Given an edge $e = \{v, w\} \in \mathcal{P}_2(V)$, we often abbreviate $e = vw$. So we might specify a graph G as having vertex set $V = \{1, 2, 3, 4\}$ and edge set $E = \{12, 23, 31, 34\}$.



Since every simple graph is a multigraph, everything we did in the last lecture also applies to simple graphs, and we will use the same terminology as introduced then — for example vertex degrees and connectivity. Given $v \in V$, let

$$N(v) = \{w \in V : vw \in E\}$$

be the *neighbourhood* of v , then for simple graphs we have

$$\deg(v) = |N(v)|.$$

Last time we defined trails, which are walks where no edge is repeated, and circuits, which are trails which start and end in the same vertex. Quite often we also want to avoid repeating vertices, which is captured in the notion of a *path* or a *cycle*.

Definition 11. Let G be a (simple or multi-) graph, and $k \geq 0$. A *path* of length k is a walk

$$v_0 e_1 v_1 e_2 v_2 \dots v_k$$

so that no two vertices are the same. A *cycle* of length k is a walk of length k so that $v_0 = v_k$, but no other pair of vertices are the same.

Warning: In this course, a path of length k contains k edges and $k + 1$ vertices. Some authors define it has having k vertices and $k - 1$ edges instead.

Every path is a trail, and every cycle is a circuit, but not the other way around. Avoiding to visit the same vertex twice means that we can now define the *distance* between two vertices in a graph: In a connected graph G , let $d_G(v, w)$ denote the length of the shortest path between two vertices v and w . We will return to this concept in Lecture 5. Note that the neighbourhood of v are exactly those vertices at distance 1 from v .

One of the great mathematical benefits of simple graphs is that they are a much more restricted class of graphs, allowing us to make statements such as:

Lemma 12. Any simple graph on n vertices has at most $\binom{n}{2}$ edges.

Proof. There are $\binom{n}{2}$ different 2-element subsets of V , hence $|E| \leq \binom{n}{2}$. $\square$

From this lemma, one might suspect that there exist only a finite number of different simple graphs on n vertices. This is true, but how many there are depends on what we mean by “different” – graphs (V, E) and (V', E') might “look the same”, but be different according to definition simply because the sets V and V' are not the same. There are two ways to circumvent this - one by fixing the underlying vertex set, leading to labelled graphs, the other by formalising what is meant by “looking the same”, leading to the notion of isomorphic graphs.

Definition 13. A *labelled* graph is a graph with a fixed vertex set, commonly (if V is finite) $V = \{1, 2, \dots, n\}$.

Corollary 14. There are $2^{\binom{n}{2}}$ different labelled graphs on n vertices.

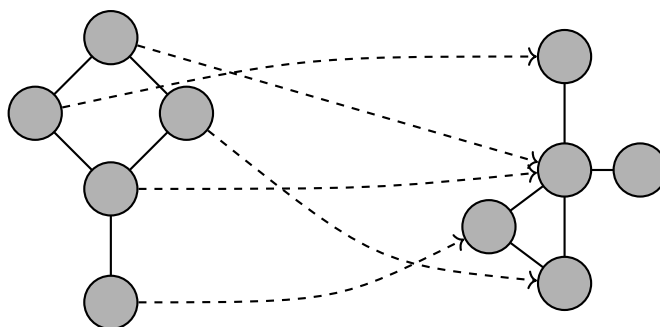
Proof. Since $V = \{1, \dots, n\}$ is fixed, graphs on this vertex set are the same iff their edge sets $E \subseteq \mathcal{P}_2(V)$ coincide. Conversely, any subset $E \subseteq \mathcal{P}_2(V)$ defines a labelled graph on V . Thus, the labelled graphs on n vertices are in bijection with the power set of $\mathcal{P}_2(V)$, hence there are $|\mathcal{P}_2(V)| = 2^{\binom{n}{2}}$ of them. $\square$

Definition 15. Let $G = (V, E)$ and $G' = (V', E')$ be simple graphs. A *morphism* $\varphi : G \rightarrow G'$ of simple graphs is a map $\varphi : V \rightarrow V'$ such that $\{\varphi(v), \varphi(w)\} \in E'$ whenever $\{v, w\} \in E$.

Warning: This notion of graph morphisms is only viable for simple graphs. While it is possible to define graph morphisms for multigraphs as well, the definition is considerably different.

Observe that every simple graph $G = (V, E)$ comes with an *identity morphism* id_G : This is the map $\text{id} : V \rightarrow V$ which sends every vertex to itself, which satisfies the condition in Definition 15. Moreover, if G, G', G'' are simple graphs with vertex sets V, V', V'' respectively, and $\varphi : G \rightarrow G'$ and $\varphi' : G' \rightarrow G''$ are morphisms, then there is a morphism $\varphi' \circ \varphi : G \rightarrow G''$ given by the map $\varphi' \circ \varphi : V \rightarrow V''$. (The students knowing what a category is might now verify that simple graphs with morphisms do indeed form a category, all other students don't need to care as it will play no further role in this course).

Example 16. Here is an example of a graph morphism, the dashed arrow indicating which vertices in the left graph are sent to which vertices on the right:



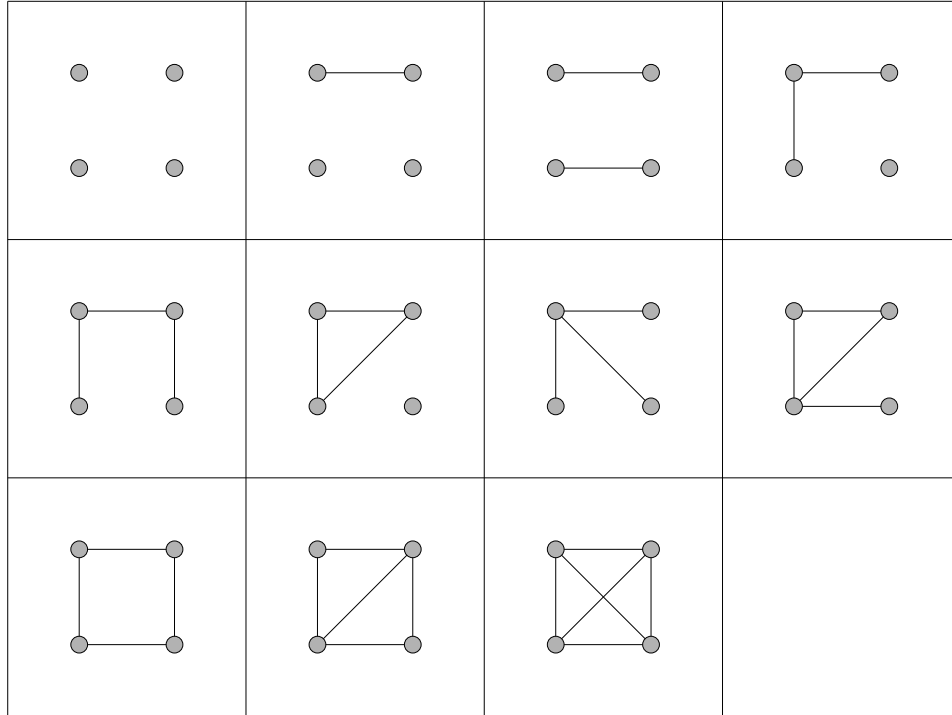
Definition 17. Two simple graphs G, G' as above are *isomorphic* if there is a bijection $\varphi : V \rightarrow V'$ such that φ is a morphism $G \rightarrow G'$ and φ^{-1} is a morphism $G' \rightarrow G$. In this case, we call φ an *isomorphism*.

It follows from this definition that a morphism φ is an isomorphism if and only if it is a bijection on vertices, such that additionally $\{\varphi(v), \varphi(w)\} \in e'$ if and only if $\{v, w\} \in E$. We may simply think of an isomorphism as a *relabelling* of the vertices: We give every vertex a new name while keeping the edges the same.

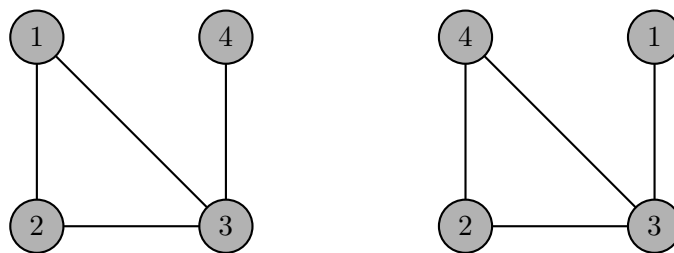
Being isomorphic provides an equivalence relation on the class of all simple graphs, and we will not always distinguish rigorously between an isomorphism class (i.e. an equivalence class under this relation) and a graph. So, a statement like ' G contains a copy of G' ' will mean: ' G contains a graph isomorphic to G' '.

Observe that graph-theoretic notions are typically invariant under isomorphisms, this includes e.g. vertex degrees, connectedness, Eulerianity, and many more we have yet to encounter.

Example 18. To further illustrate the point of isomorphic graphs, here is the complete list of 11 non-isomorphic graphs on 4 vertices:



As we can see, the number of non-isomorphic graphs on 4 vertices (11) is different from the number of labelled graphs on 4 vertices, which is $2^{\binom{4}{2}} = 64$. For example, the following two graphs are isomorphic, but different as labelled graphs:



There is no exact formula for the number g_n of non-isomorphic graphs on n vertices, but we can give a lower bound.

Proposition 19. *The number g_n satisfies*

$$g_n \geq \frac{2^{\binom{n}{2}}}{n!},$$

where $n! = 1 \cdot 2 \cdot \dots \cdot n$.

Proof. There are $2^{\binom{n}{2}}$ labelled simple graphs on the vertex set $V = \{1, \dots, n\}$. Pick one such graph $G = (V, E)$, then every graph $G' = (V, E')$ that is isomorphic to G with vertex set V is defined by a bijection $\sigma : V \mapsto V$. (The edges in E' are then exactly the ones given by the isomorphism σ , that is, $vw \in E'$ iff $\sigma^{-1}(v)\sigma^{-1}(w) \in E$.) There are $n!$ different bijections $\sigma : V \mapsto V$, so there are at most $n!$ labelled graphs G' which are isomorphic to G (note that some bijections σ, σ' might map G to the same labelled graph, so there might be fewer than $n!$ such labelled graphs G').

Imagine sorting the $2^{\binom{n}{2}}$ labelled graphs on V into ‘buckets’, where G and G' go in the same bucket if they are isomorphic. Then the argument above shows that each bucket contains at most $n!$ labelled graphs. Thus the number of buckets — which is the number of non-isomorphic graphs on n vertices — is at least $2^{\binom{n}{2}}/n!$. $\square$

There are asymptotic expansions of $g(n)$, like the following, where we write $f(n) \sim g(n)$ to express $\lim_{n \rightarrow \infty} f(n)/g(n) = 1$. This shows that the lower bound from above is close to the truth for large n .

Theorem 20. *The number g_n satisfies*

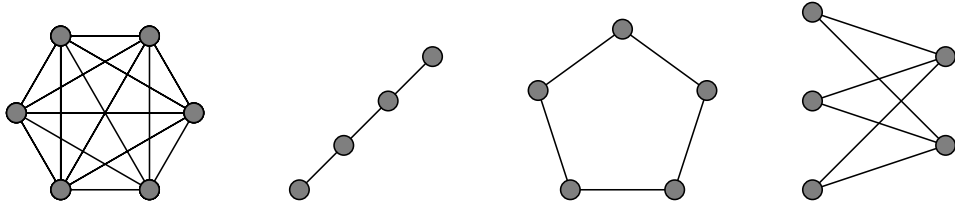
$$g_n = \frac{2^{\binom{n}{2}}}{n!} \left(1 + \frac{n^2 - n}{2^{n-1}} + O\left(\frac{n^3}{2^{3n/2}}\right) \right).$$

In particular, $g_n \sim 2^{\binom{n}{2}}/n!$.

Let us turn our attention towards special graphs, that is, isomorphism classes of graphs that turn up very often.

- The *complete graphs* on n vertices, denoted K_n : Those are the graphs containing all $\binom{n}{2}$ potential edges.
- The *path graphs* of length ℓ , denoted P_ℓ : If we label the vertices by $0, 1, \dots, \ell$ from left to right, our edges are precisely of the form $\{i-1, i\}$ for all $i = 1, \dots, \ell$.
- The *cycle graphs* on n vertices, denoted C_n : This is a path of length $n-1$ with an additional edge joining the first and last vertex.
- The *complete bipartite graphs* on $a+b$ vertices, denoted $K_{a,b}$: This graph’s vertex set is partitioned into two disjoint sets, $V = V_1 \uplus V_2$, where $|V_1| = a$ and $|V_2| = b$. There are no edges between two vertices both belonging to the same set of the partition, but all ab possible edges between V_1 and V_2 are present.

- The *complete multipartite graphs* $K_{a_1, \dots, a_r}$: Fix $r \in \mathbb{N}$, and for a given number of vertices n choose r positive integers $a_1, \dots, a_r$ such that $a_1 + \dots + a_r = n$. The graph's vertex set is a partition $V = \bigsqcup_{i=1}^r V_i$ with $|V_i| = a_i$, where two vertices are neighbours iff they belong to different sets of this partition. For $r = 2$, we get the complete bipartite graphs, for $r = n$ we get the complete graph on n vertices.



The figure above shows, from left to right, K_6 , P_3 , C_5 and $K_{2,3}$.

Proposition 21. *The complete multipartite graph $K_{a_1, \dots, a_r}$ on n vertices has $\frac{1}{2}(n^2 - a_1^2 - \dots - a_r^2)$ edges.*

Proof. We use the Handshake Lemma (Lemma 9) from Lecture 1. Since a vertex in V_i has an edge to every vertex not in V_i , it has degree exactly $n - a_i$, and there are a_i such vertices. Hence,

$$2|E| = \sum_{v \in V} \deg(v) = \sum_{i=1}^r a_i(n - a_i) = n \sum_{i=1}^r a_i - \sum_{i=1}^r a_i^2 = n^2 - \sum_{i=1}^r a_i^2,$$

proving the claim.

Alternatively, we could also count the edges in the following way: There are $\binom{n}{2}$ possible edges in total, and $K_{a_1, \dots, a_r}$ is missing exactly the edges where both endpoints are in the same vertex set V_i for $i = 1, \dots, r$. There are $\binom{a_i}{2}$ such edges for the set V_i . So

$$\begin{aligned} |E| &= \binom{n}{2} - \sum_{i=1}^r \binom{a_i}{2} = \frac{n(n-1)}{2} - \sum_{i=1}^r \frac{a_i(a_i-1)}{2} \\ &= \frac{1}{2}(n^2 - n - \sum_{i=1}^r a_i^2 + \sum_{i=1}^r a_i) = \frac{1}{2}(n^2 - \sum_{i=1}^r a_i^2), \end{aligned}$$

using that $\sum_{i=1}^r a_i = n$. □

In particular, $K_{a,b}$ has $\frac{1}{2}(n^2 - a^2 - b^2) = ab$ edges, and $K_n = K_{1, \dots, 1}$ has $\frac{1}{2}(n^2 - n) = \binom{n}{2}$ edges.

Let us now introduce the important concept of subgraphs:

Definition 22. Let $G = (V, E)$ be a simple graph. A simple graph $H = (V', E')$ is a *subgraph* of G if $V' \subseteq V$ and $E' \subseteq E$. An *induced subgraph* is a subgraph $H = (V', E')$ of G such that $E' = \{\{x, y\} \in E : x, y \in V'\}$.

Several remarks are necessary for the proper appreciation of this definition:

Remark 23. The definition requires H to be a graph in its own right. It is thus not good enough to choose any subsets $V' \subseteq V$ and $E' \subseteq E$ to obtain a subgraph; one has to make sure these choices are compatible: including an edge in E' requires that its endpoints are included in V' .

Remark 24. By definition of an induced subgraph H , it is enough to choose V' , since then the edge set of H is then given by all edges in G with both their endpoints in V' . For this reason, we will also write $G[V']$ for the subgraph induced by V' .

Remark 25. The notion of (induced) subgraphs extends to multigraphs in a natural way, but for a formal definition, one has to make sure that the edges are incident to the same vertices in both graphs. That is, for $H = (V', E', \iota')$ to be a sub-multigraph of $G = (V, E, \iota)$, we have to require that H is a multigraph, that $V' \subseteq V$ and $E' \subseteq E$, but also that $\iota' = \iota|_{E'}$.

What we introduced above as an ‘induced subgraph’ is sometimes called the vertex-induced subgraph. Indeed, there is the dual notion of the edge-induced subgraph:

Definition 26. Let $G = (V, E)$ be a simple graph, $E' \subseteq E$ a subset. The edge-induced subgraph $G \langle E' \rangle$ is the subgraph (V', E') where $V' := \{v \in V : v \text{ is incident to some } e \in E'\}$.

Finally, the subgraphs that include the entire vertex set of the host graph are often of special interest and therefore deserve a name of their own:

Definition 27. A subgraph $H = (V', E')$ of $G = (V, E)$ is called *spanning subgraph* if $V = V'$.

The same definition applies when G or G and H are multigraphs. If G is a simple graph, then the only subgraph of G that is both spanning and induced is G itself. It follows that a spanning subgraph H of G is uniquely determined by the choice of the set $E' \subseteq E$, hence G has $2^{|E|}$ spanning subgraphs.

Lecture 3 Trees

Definition 28. A *tree* is a graph $T = (V, E)$ that is connected and contains no cycles.

Observe how in a multigraph, any loops or parallel edges introduce cycles immediately, so restricting the definition to simple graphs makes no difference.

Trees occur everywhere! As an example, consider QuickSort, an algorithm sorting a list of numbers into the correct (say, ascending) order. The algorithm works as follows:

1. Fix an arbitrary pivot element from the list.
2. Compare all non-pivot elements a with the pivot, and if the pivot is larger, move a to the left of the pivot; but if the pivot is smaller, move a to the right of the pivot. (After this step, at least the pivot element will be in its correct position, but the lists to its left and right might not be).
3. Repeat steps 1 and 2 for both sublists to either side of the pivot.

Example 29. Consider the list $(3, 7, 1, 4, 9, 8, 6, 2, 5)$, and let's choose 7 as our pivot element. After rearranging, the list might look like

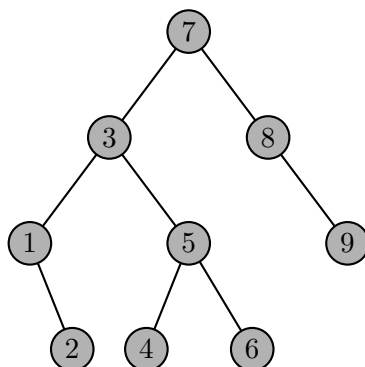
$$(3, 1, 4, 6, 2, 5, 7, 9, 8)$$

– this depends on how exactly the partitioning step is implemented. Now, the 7 is in correct position, and we are left to sort $(3, 1, 4, 6, 2, 5)$ and $(9, 8)$ independently of each other, and might choose 3 and 8 as pivots for the left and right part, respectively. After rearranging with respect to the pivots, we could then end up with

$$(1, 2, 3, 4, 6, 5, 7, 8, 9)$$

and it only remains to sort the part to the right of 3 and to the left of 7, so $(4, 6, 5)$. Let's pick 5 as the pivot element, and obtain $(4, 5, 6)$ after rearranging, which sorts the list as required.

We might also choose to represent the algorithm in a more picturesque way:



As you can see, the final result is a labelled tree (with some additional properties, like the choice of a root node, and an orientation between left and right children, but we will choose to ignore this).

Lemma 30. *Every finite tree with at least 2 vertices contains at least two vertices of degree 1. Such vertices are called leaves.*

Proof. Let T be a finite tree on at least 2 vertices. Consider a path $P = xe_1x_1e_2\dots y$ of maximum length in T . This path begins with a vertex x , and ends with a different vertex y . Assume that one of x, y (w.l.o.g. x) has degree at least 2. Thus x has a neighbour w different from x_1 . If w is not contained in P , then the path going from w to x and then via P to y is strictly longer than P , contradicting the maximality of P . Hence, w is in P , but then T contains a cycle: Starting at x , follow along P until w , then go back to x via the edge $\{x, w\}$.

From this contradiction, we conclude $\deg(x) = \deg(y) = 1$. $\square$

Proposition 31. *Any tree on n vertices has $n - 1$ edges.*

Proof. We use induction over the number of vertices. There is exactly one tree (up to isomorphism) consisting of exactly one vertex, $\bullet$, and it contains no edges. Now, assume that the statement is true for all trees on $n \geq 1$ vertices, and consider an arbitrary tree $T = (V, E)$ on $n + 1$ vertices. By Lemma 30, there is a vertex x in T with $\deg(x) = 1$ (so, it has a unique neighbour which we will call y). Remove x from T by considering the induced subgraph $T' := T[V \setminus \{x\}]$. This removes exactly the vertex x and the edge $\{x, y\}$ from T . Then T' is still connected and has no cycles, so it is a tree. As T' has n vertices, by the induction hypothesis it has $n - 1$ edges, so T had $n + 1$ vertices and n edges, as required. $\square$

Theorem 32 (Cayley). *There are n^{n-2} labelled trees on n vertices.*

Proof. We will provide a bijection between labelled trees on n vertices and sequences of labels of length $n - 2$, so called Prüfer sequences. This bijection

will take the form of two algorithms, one to convert a tree into a sequence, the other to get the tree back from the sequence. Since there are exactly n^{n-2} Prüfer sequences (sequences of length $n - 2$ containing elements from $\{1, 2, \dots, n\}$), this will prove the theorem. To avoid the annoying small cases, assume $n \geq 2$.

Algorithm 1. Let T be a tree on n vertices, with labels $\{1, \dots, n\}$. As long as T has at least 3 vertices, remove the leaf with the smallest label and write down the label of its (unique!) neighbour as the next term in the sequence. Stop when there are only 2 vertices in T left.

Algorithm 2. For every subset $S \subset \{1, \dots, n\}$ with $|S| \geq 2$, we define the notion of a ‘Prüfer sequence on S ’: let this be any sequence of length $|S| - 2$ containing only elements of S . (So in this terminology, what we called just a ‘Prüfer sequence’ earlier is a Prüfer sequence on $S = \{1, \dots, n\}$.) We will give an algorithm that, given a Prüfer sequence on S , outputs a tree with vertex set S .

So let $s = |S|$. If $s = 2$, the only possible Prüfer sequence is the empty sequence, and in this case simply connect the two elements of S , giving a tree (the only possible tree).

For $s \geq 3$, let $A = (a_1, \dots, a_{s-2})$ be a Prüfer sequence on S . Let j be the smallest element of S that does not appear in the sequence (such an element must exist because at most $s - 2$ different elements of S appear in A). Then connect vertex j to vertex a_1 (since j does not appear in A , we must have $j \neq a_1$) and remove j from S , setting $S' = S \setminus \{j\}$. Remove the first entry a_1 from A , setting $A' = (a_2, \dots, a_{s-2})$. Then A' is a Prüfer sequence on S' . Recursively apply the algorithm to construct a tree on S' . Together with the vertex j , which is connected to $a_1 \in S'$, this defines a tree on S .

To conclude the proof we need to show that applying algorithm 1 and then algorithm 2 to a labelled tree T returns this labelled tree, and that applying algorithm 2 and then algorithm 1 to a Prüfer sequence will return the sequence. This will be done in the exercises. $\square$

Example 33. The tree we obtained from the introductory example has Prüfer sequence $(1, 3, 5, 5, 3, 7, 8)$.

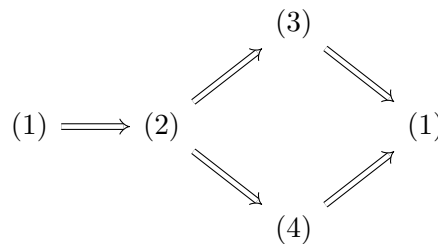
Recall that we defined connectiv

Theorem 34 (Characterisation of trees). *The following statements are equivalent:*

- (1) T is a tree.
- (2) For any two vertices x, y of T there exists a unique path connecting x and y .

- (3) *T is edge-minimal among connected graphs. In other words, if we remove any edge from T , the resulting graph is disconnected.*
- (4) *T is edge-maximal among cycle-free graphs. In other words, adding any edge between two non-neighbouring vertices creates a cycle.*

Proof. Let $T = (V, E)$ be a simple graph. We will show the following implications:



(1) $\implies$ (2): Assume T is a tree, $x, y \in V$. Since T is connected, there is at least one path from x to y . If there was more than one path, the union of two of those paths would create a cycle in T . Hence, the path is unique.

(More formally: Let C be the minimal length walk consisting of a sub-path of P_1 from some vertex v to some vertex $w \neq v$, and then a subpath of the reverse of P_2 from w to v , so that the two subpaths are not identical. At least one such closed walk exists, namely the whole path P_1 then (reverse) P_2 . By minimality you can conclude C is a cycle.)

(2) $\implies$ (3): Consider any edge $\{x, y\} \in E$. By (2), this edge is the unique path from x to y , hence removing it creates one connected component containing x and one containing y .

(2) $\implies$ (4): Consider two non-neighbouring vertices $x, y \in V$. By (2), there is a unique path in T from x to y . Hence, introducing the new edge $\{x, y\}$ creates a cycle (by concatenating the path from x to y with the new edge).

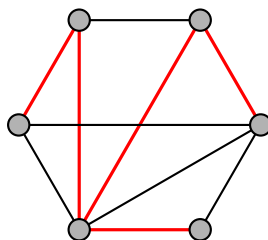
(3) $\implies$ (1): If T is edge-minimal among connected graphs, then T is in particular connected. Assume now that T were to contain a cycle. Deleting any edge on this cycle would not disconnect T , contradicting edge-minimality. Hence T is a tree.

(4) $\implies$ (1): By assumption (4), T is cycle-free. Assume that T is disconnected, then adding an edge between two different connected components does not introduce a cycle, contradicting edge-maximality. Hence T is a tree. $\square$

Recall from last lecture that given a graph G , a spanning subgraph of G is a subgraph containing all vertices of G .

Definition 35. Let G be a graph (simple or multi-). A *spanning tree* of G is a spanning subgraph that is also a tree.

Example 36. The following graph contains a spanning tree whose edges are marked in red:



If the host graph G is disconnected, then any spanning subgraph must be disconnected as well, so there can be no spanning trees of G . The following theorem shows that the obviously necessary condition of connectivity is also sufficient:

Theorem 37. *Let G be a connected (multi-)graph. Then G contains a spanning tree.*

The existence of spanning trees for finite connected G is a straightforward consequence of Theorem 34: If G is finite, and if there is an edge e so that after taking away e , the resulting graph is still connected, remove e from G . Continue like this until you can't anymore, i.e., when removing any edge would disconnect the graph. Then the subgraph of G you are left with is an edge-minimal connected graph, so it is a (spanning) tree.

Similarly, you could 'build up' to a spanning tree by starting from the empty set and adding edges from G until adding any further edge from G would create a cycle. The resulting graph H must be connected (as G is connected and H is edge-maximal) and cycle-free and hence a tree.

For infinite graphs, the situation is less clear, and to prove Theorem 37 the axiom of choice is needed. Below is a proof using Zorn's lemma (which is equivalent to the axiom of choice) — this is bonus material, we did not cover it during the lectures.

Theorem 38 (Zorn's Lemma). *Let $(A, \leq)$ be a non-empty partially ordered set. A subset $C \subseteq A$ is a chain if for any two elements $c, c' \in C$, we either have $c \leq c'$ or $c' \leq c$. Assume that for every chain C in A there exists an upper bound $b \in A$ (i.e. $c \leq b$ for all $c \in C$). Then there exists a maximal element $m \in A$, in the sense there is no element $a \in A$ with $a \neq m$ and $m \leq a$.*

To quote Fields medalist Tim Gowers from his blog: *If you are building a mathematical object in stages and find that (i) you have not finished even*

after infinitely many stages, and (ii) there seems to be nothing to stop you continuing to build, then Zorn's lemma may well be able to help you.

This is exactly the situation we find ourselves in with infinite connected graphs if we try to repeat our construction from the finite case! We can keep removing edges that do not disconnect the graph, or alternatively we can keep adding edges from G to the empty graph until we would have to create a cycle, but this process does not necessarily finish (even after infinitely many steps).

Let's apply Zorn's lemma to the setting where we try to find an edge-maximal cycle-free subgraph of some infinite multi-graph G . So let A be the set of all cycle-free spanning subgraphs H of $G = (V, E, \iota)$, and for $H, H' \in A$, define $H \leq H'$ if H is a subgraph of H' . This gives $(A, \leq)$, the structure of a partially ordered set. Furthermore, A is non-empty since $H = (V, \emptyset)$ is spanning and cycle-free. Let C be a chain in A , consisting of elements $H_i = (V, E_i)$ for $i \in I$. Define $H_b := (V, \bigcup_{i \in I} E_i)$. We want to show that H_b is an upper bound for C .

By construction, H_b is a spanning subgraph of G . Assume it contains a cycle consisting of edges $e_1, \dots, e_r$. Then, for every $\ell = 1, \dots, r$ there is an $H_{i(\ell)}$ that contains e_ℓ in C . Let $i_{\max} = \max(i(\ell) \mid 1 \leq \ell \leq r)$. Since C is a chain, one of the graphs $H_{i_{\max}}$ contains all the $H_{i(\ell)}$, and therefore all the edges $e_1, \dots, e_r$. But then $H_{i_{\max}}$ contains a cycle, which is a contradiction. Hence H_b is cycle-free, so $H_b \in A$. Finally, since $E_i \subseteq \bigcup_{i \in I} E_i$, we have $H_i \leq H_b$ for all $H_i \in C$.

But now all the assumptions of Zorn's lemma are satisfied! Hence there must be a maximal (with respect to $\leq$, i.e. edge-maximal) cycle-free spanning subgraph H of G . H is also connected because G is: if H had two or more connected components, then there must be an edge e in G , not already contained in H , which joins two of these components. But adding e to H would not create a cycle (otherwise the two components of H would have been connected before adding e), contradicting the edge-maximality of H . So H is connected and cycle-free, hence a (spanning) tree.

Lecture 4 Counting spanning trees

Given a labelled, connected, simple graph G , how many spanning trees does G have? Let $t(G)$ be the number of spanning trees of G .

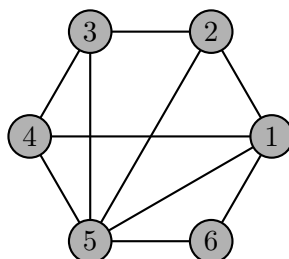
If G is the complete graph K_n , then Cayley's theorem immediately tells us $t(K_n) = n^{n-2}$.

To provide a general answer to the question, we will develop some of the basic language in *algebraic graph theory*.

Let us begin by introducing two ways of encoding a graph algebraically. For the entire lecture, we will assume that $G = (V, E)$ is a simple graph with the finite labelled vertex set $\{1, \dots, n\}$ and edge set $\{e_1, \dots, e_m\}$.

Definition 39. The *adjacency matrix* A of a graph G is the $n \times n$ -matrix having entries $A_{ij} = 1$ if i and j are neighbours, and $A_{ij} = 0$ otherwise.

Example 40. For example, the graph



has adjacency matrix

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Observe that, since ‘being neighbours’ is a symmetric relation on vertices, the matrix A is symmetric. Moreover, no vertex is a neighbour to itself, so all diagonal entries of A are 0.

There is another important way to capture the relevant information about a simple graph in a matrix; however, it turns out to be more useful to *orient* the edges for this: So, for every edge $e_1, \dots, e_m$, designate one of its endpoint as the positive end, and the other as the negative end, in some arbitrary way.

Definition 41. The *incidence matrix* D of a graph G with a fixed orientation

is the $n \times m$ -matrix having entries D_{ij} ($1 \leq i \leq n, 1 \leq j \leq m$), where

$$D_{ij} = \begin{cases} 1 & \text{if the vertex } i \text{ is the positive end of the edge } e_j \\ -1 & \text{if the vertex } i \text{ is the negative end of the edge } e_j \\ 0 & \text{otherwise} \end{cases}$$

Example 42. The matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is one possible incidence matrix for the graph G from Example 40. The edges were ordered in the following way:

$$(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 1), (1, 4), (1, 5), (2, 5), (3, 5)$$

Let's see why the incidence matrix is interesting:

Lemma 43. *Let D be the incidence matrix of a finite simple graph G . Then:*

- (i) *The sum of any column of D is zero, and so $\text{rank } D \leq n - 1$.*
- (ii) *If G is connected, then $\text{rank } D = n - 1$.*
- (iii) *If G has c components, then $\text{rank } D = n - c$.*

Proof. The column sum of D is zero since in any column, there is exactly one entry 1 and one entry -1 , representing the positive and negative endpoint of the edge to which the column is associated. So the sum of all row-vectors in D is 0, which is a non-trivial linear combination, and so $\text{rank } D \leq n - 1$.

To show part (ii), we show that if G is connected, this is (up to scaling) the only non-trivial linear combination of 0 with row-vectors. So, let r_i for $i = 1, \dots, n$ denote the row-vectors of D , and suppose we have a non-trivial linear combination $\sum_{i=1}^n \alpha_i r_i = 0$. Consider a row k for which $\alpha_k \neq 0$. In this row, there is a non-zero entry ($+1$ or -1) in every column corresponding to an edge incident with the vertex k . For each of these columns, there is exactly one other row (say, r_ℓ) with a non-zero entry in that column, and the two entries will be of opposite signs. But for our linear combination to come out as 0, we need to have $\alpha_\ell = \alpha_k$. So we know that if $\alpha_k \neq 0$, then $\alpha_\ell = \alpha_k$ for all ℓ which are neighbours to k . But as G is connected, this argument extends to all vertices of G , so the linear combination is a scalar multiple of $\sum_{i=1}^n r_i = 0$. So $\text{rank } D = n - 1$.

For part (iii), observe that it is possible to re-label the vertex and edge sets in such a way that D is in block-diagonal form, with every block being the incidence matrix of a connected component. The claim then follows from (ii). $\square$

Lemma 44. *Any square submatrix of an incidence matrix D has determinant 0, 1, or -1 .*

Proof. This is obviously true for square submatrices of size 1×1 . Assume that it is true for submatrices of size $k \times k$, and consider (if possible within D) a square submatrix M of size $(k+1) \times (k+1)$. If every column of M has either two or no non-zero entries, then $\det M = 0$ (because in this case every column sums to 0, so $\text{rank } M < k+1$, which implies $\det M = 0$). Otherwise, there's a column of M with exactly one non-zero entry ($+1$ or -1). Expanding $\det(M)$ along this column yields $\det M = \pm \det M'$ where M' is a $k \times k$ -submatrix of D . Hence $\det M \in \{-1, 0, 1\}$. $\square$

The following lemma follows easily from the last lecture's results on trees.

Lemma 45. *A simple graph T with n vertices and $n-1$ edges is a tree if and only if it is connected.*

Proof. In exercise session. $\square$

Now observe that, if $S \subseteq E$ is a set of edges from G , and D_S denotes the matrix consisting of exactly those columns that correspond to edges in S , then D_S is the incidence matrix of the spanning subgraph (V, S) of G . In particular, if S contains exactly $n-1$ edges, then by part (iii) of the previous lemma, (V, S) is a spanning tree of G iff D_S has rank $n-1$. The following lemma shows that in that case indeed every $(n-1) \times (n-1)$ -submatrix of D_S is non-singular.

Proposition 46. *Let $S \subseteq E$ with $|S| = n-1$, and let D_S be the $n \times (n-1)$ -submatrix of D consisting of the columns corresponding to edges in S . Let M be any $(n-1) \times (n-1)$ -submatrix of D_S . Then (V, S) is a spanning tree of G if and only if $\det M \neq 0$.*

Proof. First suppose (V, S) is a spanning tree. We have already observed that D_S has rank $n-1$. Exactly as in the proof of Lemma 43, (ii) we know that the only non-trivial linear combinations of the n row vectors $r'_1, \dots, r'_n$ of D_S that give the 0-vector are scalar multiples of $\sum_{i=1}^n r'_i = 0$. We obtain M from D_S by removing one of the row vectors, say r'_i . Suppose for a contradiction that $\det M = 0$. Then there is a non-trivial linear combination of the vectors $\{r'_1, \dots, r'_n\} \setminus \{r'_i\}$ which gives the 0-vector. We can extend this to a linear combination of all vectors $r'_1, \dots, r'_n$ by giving r'_i the coefficient 0, still giving

the 0-vector. But that's a non-trivial linear combination of the vectors $r'_1, \dots, r'_n$ which is not a scalar multiple of $\sum_{i=1}^n r'_i = 0$, a contradiction. So we must have $\det M \neq 0$.

For the other direction, if $\det M \neq 0$, then D_S contains at least $n - 1$ linearly independent rows and the same number of linearly independent columns. Hence $\text{rank } D_S = n - 1$ and (V, S) must be connected. But a connected graph on n vertices with $n - 1$ edges is a tree. $\square$

Let us introduce the third important matrix of a graph, which ties D and A together.

Lemma 47. *Let G be a finite simple graph. Denote by A its adjacency matrix and by D its incidence matrix (with respect to some fixed orientation). Let Δ be the diagonal $n \times n$ -matrix having diagonal entries $\Delta_{ii} = \deg(i)$ for $1 \leq i \leq n$. Then*

$$DD^T = \Delta - A =: L,$$

and this matrix L is called the Laplacian matrix of G . In particular, L is independent from the chosen orientation on G .

Proof. The entry $(DD^T)_{ij}$ is the Euclidean inner product of rows r_i and r_j of D . If $i = j$, this amounts to summing up the squares of the entries in r_i . The non-zero entries in r_i are ± 1 , and there are exactly $\deg(i)$ -many of them. So $(DD^T)_{ii} = \deg(i) = (\Delta - A)_{ii}$. For $i \neq j$, the rows r_i and r_j have a non-zero entry in the same column iff this column corresponds to an edge $\{i, j\}$ (and there can be at most one such edge). In that case, the scalar product is $r_i \cdot r_j = (+1)(-1) = -1$, since the entries in this column must be of opposite signs. So

$$(DD^T)_{ij} = -1 = (\Delta - A)_{ij}$$

if $\{i, j\}$ is an edge, and

$$(DD^T)_{ij} = 0 = (\Delta - A)_{ij}$$

if $\{i, j\}$ is not an edge. $\square$

Recall that a *cofactor* of a square-matrix M is

$$\tilde{m}_{ij} := (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant of the submatrix of M where the row i and the column j have been removed.

We are finally ready to state the following theorem:

Theorem 48 (Kirchhoff's matrix-tree-theorem). *Let G be a finite simple graph having Laplacian matrix L . Then the number $t(G)$ of spanning trees of G is equal to any cofactor of L .*

Furthermore, if L has eigenvalues $\lambda_1, \dots, \lambda_n$ where $\lambda_n = 0$ and the remaining eigenvalues are non-zero, then

$$t(G) = \frac{1}{n} \lambda_1 \cdots \lambda_{n-1}.$$

This theorem gives a fast way to count the number of spanning trees of any graph G : Given G , it is easy to work out the Laplacian $L = \Delta - A$, and then any cofactor (obtained by deleting any row and column from L and calculating the determinant of the resulting matrix) is equal to the number of spanning trees.

For the proof we will need the following theorem from linear algebra. To state it, assume that A is an $n \times m$ -matrix with $n \leq m$. Then for $S = \{s_1, \dots, s_n\} \subseteq \{1, \dots, m\}$, we write A_S for the $n \times n$ -submatrix of A consisting of exactly the columns number $s_1, \dots, s_n$.

Theorem 49 (Cauchy-Binet). *Let A, B be two $n \times m$ -matrices, with $n \leq m$. Then*

$$\det(AB^T) = \sum_{\substack{S \subseteq \{1, \dots, m\} \\ |S|=n}} \det A_S \det B_S.$$

(Partial) proof of the matrix-tree-theorem. We are only going to show the statement for symmetrical co-factors $\tilde{m}_{ii}$ of L .

Let D be the incidence matrix of G for a fixed orientation, let $1 \leq i \leq n$, and delete the i^{th} row from D , calling the resulting $(n-1) \times m$ matrix $\tilde{D}$. Then $\tilde{D}\tilde{D}^T$ is $L = DD^T$ with the i^{th} row and column deleted, so $(-1)^{n+n} \det(\tilde{D}\tilde{D}^T) = \det(\tilde{D}\tilde{D}^T)$ is exactly the cofactor $\tilde{m}_{ii}$ of L . By the Cauchy-Binet theorem we have

$$\det \tilde{D}\tilde{D}^T = \sum_{\substack{S \subseteq E \\ |S|=n-1}} (\det \tilde{D}_S)^2.$$

By Lemma 44 the summands on the right-hand side are either 0 or 1, and by Proposition 46 a summand is 1 iff (V, S) is a spanning tree of G . So the sum above simply counts the (edge sets S of) spanning trees of G .

The second part follows from general statements from linear algebra about the characteristic polynomial of a matrix where all rows and columns sum to 0. $\square$

Lecture 5 Weights and distances

In the previous two lectures, we proved that in a connected graph G , spanning trees exist, and found a way to count how many there are. The important question left open is: How do we actually find one? As it turns out, we can even answer this question in a more general setting, where we assume that every edge in G comes at a certain cost (the *weight* of an edge), and we want to find a spanning tree with minimal weight. Depending on the application at hand, these weights might be distance, resistance, capacity, For convenience, we will work with simple graphs (but everything is easily generalised to multigraphs).

Definition 50. A *weighted graph* is a finite simple graph $G = (V, E)$ together with a *weight function* $w : E \rightarrow (0, \infty)$. If $H = (V', E')$ is a subgraph of G , then its *weight* is defined as $w(H) := \sum_{e \in E'} w(e)$, the sum of the weights of the edges in H . A *minimum spanning tree* (MST) is a spanning tree T of G such that $w(T)$ is minimal among all spanning trees of G .

For a connected weighted graph G , we know that there exist finitely many spanning trees, and so in particular, a minimum spanning tree. In general the MST does not have to be unique — there could be two minimum spanning trees with the same weight. But it turns out that this cannot happen if all edge weights are different:

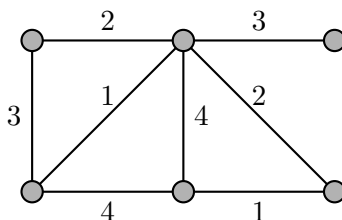
Lemma 51. *Let G be a connected weighted graph. If w assigns a different weight to every edge, the MST is unique.*

We will prove this lemma later, together with Theorem 53.

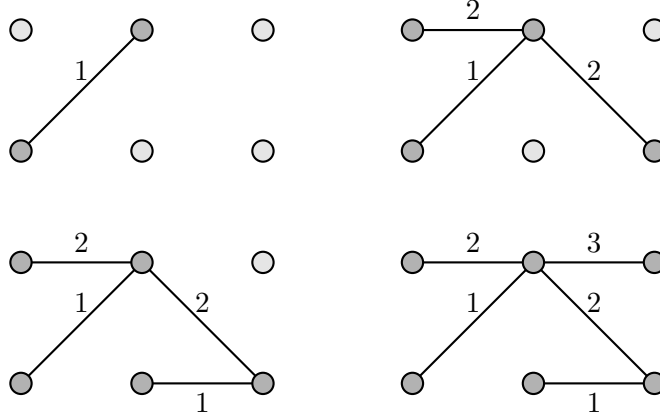
We will now proceed to present two famous and surprisingly simple algorithms for finding minimum spanning trees:

Prim's algorithm: Let $G = (V, E)$ be a connected weighted graph. Set $T = (\{v\}, \emptyset)$ for any vertex $v \in V$. As long as T is not a spanning subgraph of G , find an edge e of G between $V(T)$ and $V \setminus V(T)$ of minimal weight. Add this edge together with its endpoint in $V \setminus V(T)$ to T .

Example 52. Prim's algorithm, applied to the graph



takes the following steps, after being initialized in the top middle vertex (some steps have been executed simultaneously):



Theorem 53. *Let $G = (V, E)$ be a connected weighted graph with weight function w . Prim's algorithm generates a minimum spanning tree.*

Proof of Theorem 53 and Lemma 51. Prim's algorithm generates a connected spanning subgraph T , and it is inductively clear that T contains $n - 1$ edges for $n = |V|$. Hence T is a spanning tree. To show that T is an MST, consider any other spanning tree T' . We will show that $w(T) \leq w(T')$, which proves Theorem 53. Furthermore, we will show that if all edge weights are distinct we will show $w(T) < w(T')$, which proves Lemma 51.

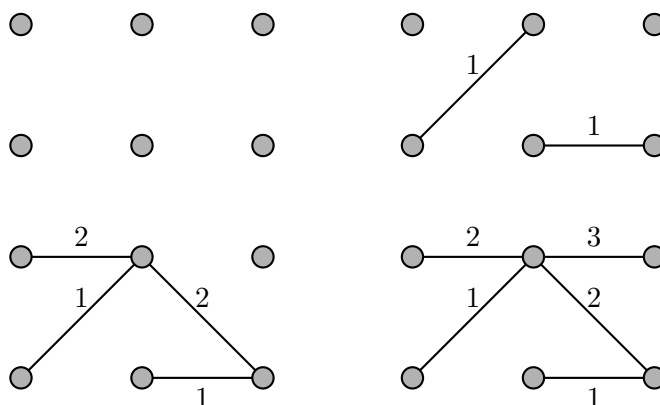
Consider the earliest edge e that belongs to T but not to T' , according to the order in which Prim's algorithm added edges to T . Partition the vertex set into two parts V_1 and V_2 (and $V_1 \uplus V_2 = V$) such that $T[V_1]$ is the tree that Prim's algorithm had constructed before adding e , and let $e = uw$ with $u \in V_1$, $w \in V_2$. Note that the tree $T[V_1]$ is also a subgraph of T' . Since T' is a tree, there must be a unique path in T' connecting u and w and at some point this path must cross from V_1 into V_2 . Call the edge it crosses with $f = u'v'$. By construction of Prim's algorithm, e has minimal weight amongst all edges between V_1 and V_2 , so $w(e) \leq w(f)$. Now update T' by adding the edge e and removing f , then the updated T' is still a spanning tree. Furthermore the weight $w(T')$ has strictly decreased (if $w(f) > w(e)$) or stayed the same (if $w(f) = w(e)$) — but the latter case cannot happen if all edge weights are distinct.

Now repeat this procedure with the updated T' , and continue until at the end $T = T'$. It follows that for the original tree T' that we started this procedure with, $w(T') \geq w(T)$. Furthermore, if all edge weights are distinct, we have $w(T') > w(T)$. $\square$

Remark 54. The runtime of Prim's algorithm depends on the implementation, specifically on how the graph is represented, and how one finds the next edge between $V(T)$ and $V \setminus V(T)$. A good implementation has a runtime of $O(|E| + |V| \log |V|)$.

Kruskal's algorithm: Let $G = (V, E)$ be a connected weighted graph. Let S be a list of edges of G , sorted in order of increasing weight. Let $T = (V, \emptyset)$. While T is disconnected, delete the first entry of S and add this edge to T unless doing so creates a cycle.

Example 55. Kruskal's algorithm, applied to the graph in Example 52, takes the following steps (with edges of the same weight being added simultaneously, if possible):



Theorem 56. Let $G = (V, E)$ be a connected weighted graph with weight function w . Kruskal's algorithm generates a minimum spanning tree.

Proof. Kruskal's algorithm generates a cycle-free subgraph T that is also edge-maximal amongst all cycle-free subgraphs of G (as any edge that we did not add would have created a cycle). Therefore T must be a spanning tree: T is connected, because if it were not and had several connected components, then in G , there would be an edge between two of these connected components, which we could then add to T without creating a cycle, and that's a contradiction to T being edge-maximal. So T is cycle-free and connected and therefore a tree.

Let T' be another spanning tree, then we will show that $w(T) \leq w(T')$.

Let e be the earliest edge (according to Kruskal's algorithm) in T that is not contained in T' . Adding e to T' creates a cycle, and somewhere on this cycle there must be an edge f belonging to T' but not to T . Update T' by adding e and removing f — note that the updated T' is still a spanning tree because it is connected and has $n - 1$ edges. Since Kruskal's algorithm chose to include e but not f (at the time that e was added, all edges present were also in T' , so f would not have created a cycle), e must have come first in the ordered list S , hence $w(e) \leq w(f)$. Therefore the weight $w(T')$ of the updated tree T' can only have decreased (if $w(e) < w(f)$) or stayed the same (if $w(e) = w(f)$).

Continue with the updated tree T' , until at the end $T = T'$. It follows that $w(T) \leq w(T')$ for the original tree T' that we started with. $\square$

Kruskal's algorithm has a runtime of $O(|E| \log |V|)$.

As mentioned above, we can interpret edge-weights as distances. This gives rise to the following notion of distance in a graph:

Definition 57. Let $G = (V, E)$ be a weighted graph with weight function $w : E \rightarrow (0, \infty)$. For vertices $v, v' \in V$, we define the *graph distance* between v and v' via

$$d_G(v, v') = \min \left\{ \sum_{e \in E(P)} w(e) : P \text{ is a path from } v \text{ to } v' \right\}$$

where we declare $d_G(v, v') = \infty$ if no such path exists (i.e. if v, v' are from different connected components). This definition extends also to non-weighted graphs by introducing the weight function $w(e) = 1$ for any edge $e \in E$.

It is not hard to show that graph distances satisfy the axioms of a *metric space*:

- (i) For all v, v' , we have $d_G(v, v') \geq 0$, and $d_G(v, v') = 0$ if and only if $v = v'$,
- (ii) $d_G(v, v') = d_G(v', v)$ for all $v, v' \in V$, and (iii) $d_G(v, v') + d_G(v', v'') \geq d_G(v, v'')$ for all $v, v', v'' \in V$ (the triangle inequality).

Definition 58. The *diameter* of a weighted graph $G = (V, E)$ is the maximum distance between any pair of vertices, i.e.

$$\text{diam}(G) = \max \{ d_G(v, v') : v, v' \in V \}.$$

Again, this definition extends to graphs without weight functions by setting $w(e) = 1$ for any edge $e \in E$.

Given a weighted graph, how can we actually find the distances between vertices, and how can we find the shortest path? Dijkstra's algorithm below gives us the answers; in fact it gives us the distances between a selected starting vertex and all other vertices in the graph.

Dijkstra's algorithm: Given a weighted graph $G = (V, E)$, select an initial vertex v_0 , and initialise a distance function $d(v_0, \cdot)$ by

$$d(v_0, v) = \begin{cases} 0 & \text{if } v = v_0 \\ \infty & \text{if } v \neq v_0 \end{cases}$$

Also define a set Q of unvisited vertices, which is initially defined to be $Q = V$. To begin with, consider v_0 to be the currently visited vertex. Now proceed as follows:

1. Remove the currently visited vertex v from the set Q .
2. For all neighbours v' of v in Q , check whether $d(v_0, v) + w(\{v, v'\}) < d(v_0, v')$. If this is the case, then going through v to v' provides a shorter path from v_0 to v' than previously known; hence we update $d(v_0, v')$ to be the smaller value $d(v_0, v) + w(\{v, v'\})$. Otherwise, no change to $d(v_0, \cdot)$ is needed.
3. Pick as the new currently visited vertex the vertex v'' in Q with the smallest value $d(v_0, v'')$.
4. Repeat steps 1 - 3 until $Q = \emptyset$, then return the distance function $d(v_0, \cdot)$.

Then the returned distance function coincides with $d_G(v_0, v)$. (Exercises in L7.)

Remark 59. Dijkstra's algorithm can easily be modified:

- If we are only interested in the distance $d_G(v, v')$ for one particular v' , then run the algorithm with $v_0 = v$, but stop as soon as the currently visited vertex is v' .
- If not just the distance, but the actual path realising d_G is of interest, it is possible to modify step 2 as follows: Suppose you have a currently visited vertex v with a neighbour v' in Q , and $d(v_0, v) + w(\{v, v'\}) < d(v_0, v')$. Additionally to updating $d(v_0, v')$, assign to v' the precursor vertex v . Updating this precursor vertex together with $d(v_0, \cdot)$ enables us to backtrack the shortest path after the algorithm has terminated.

Example 60. Here's how Dijkstra's algorithm runs on the graph from Example 52, with vertices being labelled $A, B, \dots, F$ from top left to bottom right:

Step	Current vertex	$d(A, \cdot)$					
		A	B	C	D	E	F
Init	–	0	∞	∞	∞	∞	∞
1	A	0	2	∞	3	∞	∞
2	B	0	2	5	3	6	4
3	D	0	2	5	3	6	4
4	F	0	2	5	3	5	4
5	C	0	2	5	3	5	4
6	E	0	2	5	3	5	4

With the right implementation, Dijkstra's algorithm has a runtime of $O(|E| + |V| \log |V|)$.

Lecture 6 Hamilton cycles

Recall from all the way back in Lecture 2 that a *cycle* in a graph G is a walk $v_0e_1v_1e_2v_2\ldots v_{k-1}e_kv_0$ such that all the vertices $v_0, \ldots, v_{k-1}$ and all the edges $e_1, \ldots, e_k$ are distinct.

Definition 61. Let $G = (V, E)$ be a finite simple graph. A *Hamilton cycle* in G is a cycle that visits every vertex of G . If G has a Hamilton cycle, we also simply call G *Hamiltonian*.

There was a simple criterion (even vertex degrees) by which we could almost immediately decide which graphs have Euler trails and which ones don't. Unfortunately the situation is much more complicated for Hamilton cycles, and there is no such criterion. In fact, the problem of determining whether a graph has a Hamilton cycle is NP-complete.

Finding a Hamilton cycle is a special case of the *travelling salesman problem*. Here we are given a weighted graph, and the task is to find a minimum weight Hamilton cycle (the weights are usually interpreted as distances in this context, so we are asking for the 'shortest' Hamilton cycle). This is usually asked on the complete graph to ensure the existence of a Hamilton cycle, but this is unnecessary: if the graph is not complete, we could simply add all missing edges and set their weights to infinity. Of course this problem is also NP-complete (since finding *any* Hamilton cycle already is).

However, there are some sufficient conditions to guarantee the existence of a Hamilton cycle, and we will take a look into some of them:

Theorem 62 (Dirac 1952). *Let $G = (V, E)$ be a simple graph on $n \geq 3$ vertices, such that every vertex has degree at least $n/2$. Then G is Hamiltonian.*

Proof. First note that G is connected, because otherwise the vertices in the smallest component would have degree less than $n/2$.

Now consider a path P of maximum length, say $P = v_0e_1v_1e_2v_2\ldots v_{k-1}e_kv_k$, where $k \leq n - 1$. As P is maximal, it must contain all $\geq n/2$ neighbours of v_0 and all $\geq n/2$ neighbours of v_k , since otherwise P could be extended.

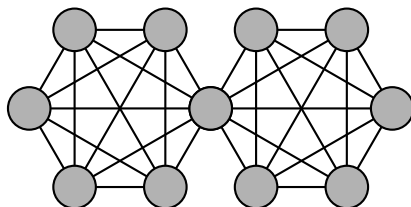
Call v_{i-1} the 'left end' and v_i the 'right end' of the edge e_i . Then there are at least $n/2$ edges on P where the right end is a neighbour of v_0 , and at least $n/2$ edges on P where the left end is a neighbour of v_k . Since there are $k \leq n - 1$ edges in total, there must be at least one edge $e_i = v_{i-1}v_i$ where both is true: v_i is a neighbour of v_0 and v_{i-1} is a neighbour of v_k .

We can use this edge to transform the path P into a cycle C , by walking along the vertices in the order $v_0, v_i, v_{i+1}, \ldots, v_k, v_{i-1}, v_{i-2}, \ldots, v_1, v_0$.

It remains to show that C is indeed a Hamilton cycle. Suppose that it's not, so there are vertices which are not on C . Since G is connected, at least one

of the vertices v on C must have a neighbour w which is not on C . But then we can construct a longer path than P by starting at w , going to v and then going around the cycle. But that's a contradiction since P was chosen to be of maximum length. $\square$

Remark 63. This result is the best possible with a uniform lower bound on the vertex degrees: If we had picked any other integer $k(n) < n/2$, there would be a counterexample. Consider the following construction: Set $k := \lfloor (n-1)/2 \rfloor$ and $\ell = \lceil (n-1)/2 \rceil$, and glue together a copy of K_{k+1} and a copy of $K_{\ell+1}$ by identifying one vertex from each copy. Since $k + \ell = n - 1$ (consider the two cases that n is odd or even), this graph has n vertices and minimum degree k . But it cannot contain a Hamilton cycle (since any Hamilton cycle would have to go through the middle vertex at least twice).



We immediately get the following generalisation of Dirac's theorem:

Theorem 64 (Ore, 1960). *Let G be a graph on $n \geq 3$ vertices, such that for any non-adjacent pair of vertices v, w , we have $\deg(v) + \deg(w) \geq n$. Then G is Hamiltonian.*

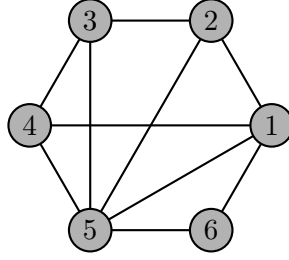
Proof. The proof works in the same way as the proof for Dirac's theorem. $\square$

Even more general is the Bondy-Chvátal theorem, which we will state next (but we require a definition first).

Definition 65. Let $G = (V, E)$ be a finite simple graph. The *closure* $\text{clos}(G)$ of G is defined as the result of the following procedure: For every pair of non-adjacent vertices v, w , draw the edge $\{v, w\}$ if $\deg(v) + \deg(w) \geq n$. Stop when there is no such pair anymore in the graph.

Example 66. If G is a graph on n vertices satisfying the conditions of Ore's theorem, then $\text{clos}(G) = K_n$.

The following is a graph where the closure is also the complete graph K_6 , but vertices 3 and 6 cannot be connected immediately, only after some other edges are inserted.



Lemma 67. *The closure of a graph does not depend on the order in which the edges are inserted (and is thus well-defined).*

Proof. Let $G = (V, E)$ be a finite simple graph. Assume we construct the closure of G in two different ways, where one way adds the edges $e_1, e_2, \dots, e_k$ to G (in this order), and the other way adds the edges $f_1, f_2, \dots, f_{k'}$ to G — note that we do not know yet if $k = k'$.

Then there is a smallest j such that $e_j \neq f_j$ (otherwise, one of the sequences contains the entire other sequence, which is impossible). Let G_{j-1} be the graph obtained by adding the edges $e_1 = f_1, \dots, e_{j-1} = f_{j-1}$, and let $e_j = \{v, w\}$. Then $\deg_{G_{j-1}}(v) + \deg_{G_{j-1}}(w) \geq n$. Since those degrees will not decrease when adding the next edges $f_j, \dots$, eventually $\{v, w\}$ will be inserted as some edge f_ℓ . But then we can modify the sequence $f_1, f_2, \dots, f_{k'}$ by inserting the edge f_ℓ instead of f_j , and inserting the edges $f_j, \dots, f_{\ell-1}$ one step later. Now repeat this modification procedure, until the sequences agree completely. This shows that the two original sequences add all the same edges eventually. $\square$

Theorem 68 (Bondy-Chvátal, 1972). *A graph G is Hamiltonian if and only if $\text{clos}(G)$ is.*

Proof. Suppose we constructed $\text{clos}(G)$ from G by adding edges $e_1, \dots, e_k$, yielding graphs $G = G_0, G_1, \dots, G_{k-1}, G_k = \text{clos}(G)$. If some G_j is Hamiltonian, then all following graphs (which contain G_j as a subgraph) will be Hamiltonian as well. Now suppose that for some j , G_{j+1} is Hamiltonian but G_j is not, and let $e_{j+1} = \{v, w\}$ be the unique edge by which G_j and G_{j+1} differ. In particular, the Hamilton cycle in G_{j+1} must contain $\{v, w\}$. So G_j contains a path P from v to w visiting all vertices in G (often called a *Hamilton path*). But since $\{v, w\}$ is the next edge added, we have $\deg_{G_j}(v) + \deg_{G_j}(w) \geq n$, and all neighbours of v and w must lie on P . But now, we can again use the construction from the proof of Theorem 62 to show that G_j already contained a Hamilton cycle!

It follows that if one of the graphs G_j is Hamiltonian, all of them are. In particular $G = G_0$ is Hamiltonian if and only if $\text{clos}(G) = G_k$ is. $\square$

We turn our attention to two more examples:

The d -dimensional cube. Fix an integer $d \geq 2$. The d -dimensional cube has 2^d vertices, labelled by binary strings of length d – in other words, $V = \{0, 1\}^{\times d}$. Two vertices are neighbours if and only if the corresponding binary strings differ in exactly one position. It follows immediately from this definition that all vertices have degree d .

Proposition 69. *The d -dimensional cube is Hamiltonian for all $d \geq 2$.*

Proof. We use induction over d , but the induction is easier to describe if we prove something slightly stronger: There is a Hamilton cycle which contains the edge between the vertices $(0, 0, \dots, 0)$ and $(1, 0, \dots, 0)$.

Denote the d -dimensional cube by G_d . For $d = 2$, G_2 is isomorphic to the cycle C_4 , which is of course Hamiltonian, and any Hamilton cycle in G_2 will contain all of the edges, so in particular the edge between $(0, 0)$ and $(1, 0)$. Now assume that for some $d \geq 2$, there is a Hamilton cycle in G_d that contains the edge between $(0, 0, \dots, 0)$ and $(1, 0, \dots, 0)$. By deleting this edge from the Hamilton cycle, we get a Hamilton path from $(0, 0, \dots, 0)$ to $(1, 0, \dots, 0)$.

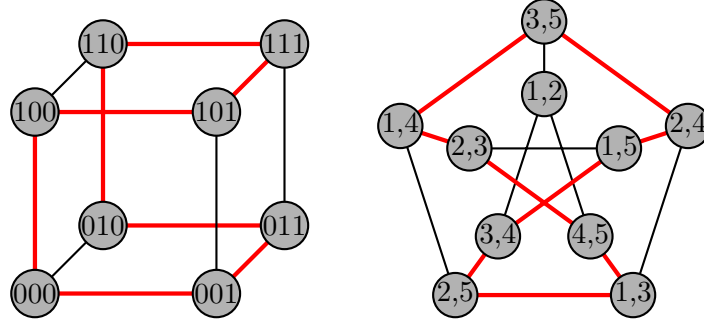
Now consider G_{d+1} . Within G_{d+1} , there is a copy of G_d (namely the subgraph induced by all vertices whose first entry is a 0), and a second disjoint copy of G_d (the subgraph induced by all vertices whose first entry is a 1). Using this, we build a Hamilton cycle including the edge between $(0, 0, \dots, 0)$ and $(1, 0, \dots, 0)$ in G_{d+1} as follows: Start in $(0, 0, 0, \dots, 0)$. Traverse the Hamilton path to $(0, 1, 0, \dots, 0)$ inside the first copy of G_d . Then take the edge to $(1, 1, 0, \dots, 0)$, and traverse the Hamilton path in the second copy of G_d backwards all the way to $(1, 0, 0, \dots, 0)$. Finally, take the edge back to $(0, 0, 0, \dots, 0)$ to close the cycle. $\square$

Note that this Hamilton cycle in particular gives an ordering of the 2^n 0-1-strings of length n (that is, the n bit strings) such that we can go from one string to the next by flipping only one bit. The ordering we constructed in the proof of Proposition 69 is also known as *Gray code*.

The Petersen graph. The Petersen graph has as its vertex set the set of all two-element subsets of $\{1, 2, 3, 4, 5\}$, and two subsets are adjacent in the Petersen graph iff they are disjoint. This gives a graph on $\binom{5}{2} = 10$ vertices, and every vertex has degree $\binom{3}{2} = 3$, and so there are 15 edges in total.

Proposition 70. *The Petersen graph is not Hamiltonian. However, upon removing any one of its vertices, the remaining graph is, and the Petersen graph has a Hamilton path.*

Proof. Consider the five edges connecting the inner five vertices with the outer five vertices (in the figure above). Any Hamilton cycle C needs to use either 2 or 4 of those edges (depending on how often C ‘switches’ between



The Hamiltonian cycle constructed in the proof of Proposition 69 in a 3-cube on the left, and a cycle omitting exactly one vertex in the Petersen graph on the right.

the inner and the outer vertices). If it contains two of them, they need to connect pairs of adjacent vertices on both the inside and the outside, and this is impossible. So C contains four of those edges. Assume the edge between $\{3, 5\}$ and $\{1, 2\}$ is not included. This implies that among the 5 outside edges, the two on top and the bottom one are in C , and the other two are not. However, the vertex $\{1, 2\}$ needs to lie on C , and this forces the edges between $\{1, 2\}$ and $\{3, 4\}$ as well as $\{4, 5\}$ to be on C . But this means that all edges in the cycle on the vertices $\{1, 2\}, \{4, 5\}, \{1, 3\}, \{2, 5\}, \{3, 4\}$ are in C , and this can not be extended to a Hamilton cycle on all vertices.

To see that there exists a cycle through all but one vertex, we first show that the choice of the omitted vertex does not matter by symmetry (because all vertices are ‘the same’). What does this mean? Making this more precise, let $\{i, j\}, \{i', j'\} \subseteq \{1, \dots, 5\}$. Then there exists a permutation π on $\{1, \dots, 5\}$ mapping i to i' and j to j' . This permutation extends to two-element subsets by mapping $\{k, l\}$ to $\{\pi(k), \pi(l)\}$, and this is in fact a bijection on $\mathcal{P}_2(\{1, \dots, 5\})$ that preserves disjointness. In graph-theoretic language, this shows that for any two vertices $\{i, j\}, \{i', j'\}$ of the Petersen graph, there exists an isomorphism of the Petersen graph to itself that maps $\{i, j\}$ to $\{i', j'\}$ (i.e. the graph is vertex-transitive). So the choice of the omitted vertex is irrelevant. Finally, the picture above shows that there is a cycle traversing all vertices but $\{1, 2\}$, and including the edge $\{\{1, 2\}, \{3, 5\}\}$ instead of any of the other two edges incident to $\{3, 5\}$ will result in a Hamilton path. $\square$

Lecture 8 The max-flow-min-cut theorem

Consider a problem of the following form: We have to transport some material from point A to point B, using e.g. the railway network. We can think of this network as a graph, where train stations are vertices, and edges are given by the railway sections in between. However, every railway section only has a certain capacity (limited for example by the number of trains, or by their weights, or ...). Given these constraints, how can we transport as much material as possible?

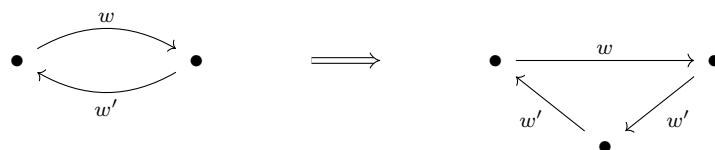
The answer to such a question comes from the study of flows in directed graphs, and will be the topic of this lecture.

Definition 71. A *directed simple graph* is a pair $G = (V, E)$, where $E \subseteq V \times V \setminus \{(v, v) | v \in V\}$. Here, we interpret an edge (v, w) as being an edge from a vertex v to a (different) vertex w . Note that in this definition $(v, w) \neq (w, v)$.

Definition 72. A *flow network* consists of a directed graph $G = (V, E)$ together with a capacity function $c : E \rightarrow [0, \infty)$ and two distinguished vertices: a *source node* $s \in V$ and a *sink node* $t \in V$.

Given a flow network, we extend the *capacity function* c to a function $c : V \times V \rightarrow [0, \infty)$ by setting $c(x, y) = 0$ for $(x, y) \notin E$.

For simplification, we shall assume that if (v, v') is an edge in a flow network, then (v', v) is not an edge. This is not a real restriction, however: If both directed edges are present in a graph, we can split the edge (v, v') by introducing a new vertex z in the middle, and set $c(v, z) = c(z, v') := c(v, v')$.



Intuitively, we can think of the flow network as a network of pipes through which water flows from the source to the sink. The capacity of an edge tells us how much water per unit of time can pass through this pipe. A *flow* will then be given by the information how much water actually passes through every pipe, and it should satisfy two constraints: (1) The flow through each pipe should not exceed the capacity, and (2) at every vertex v other than the source and the sink, the total amount of incoming water should be equal to the total amount of outgoing water. The following definition formalises this:

Definition 73. A *flow* f on the flow network G with capacity function c is a function $f : V \times V \rightarrow [0, \infty)$ such that the following conditions are satisfied:

- (i) $f(v, v') \leq c(v, v')$ for all $v, v' \in V$ (capacity constraint)
- (ii) For $v \in V \setminus \{s, t\}$, we have $\sum_{x \in V} f(x, v) = \sum_{x \in V} f(v, x)$ (conservation constraint).

Note that since we set $c(x, y) = 0$ whenever $(x, y) \notin E$, we also have $f(x, y) = 0$ whenever $(x, y) \notin E$ — so the flow can only go along edges of the network.

Given a flow f (or indeed any function $f : V \times V \rightarrow \mathbb{R}$), and $x \in V$, let

$$I_f(x) = \sum_{y \in V} f(x, y) - \sum_{y \in V} f(y, x).$$

You could think of I_f as the amount of water that needs to be injected at x to maintain balance. Then the conservation constraint above says that for all $x \in V \setminus \{s, t\}$, $I_f(x) = 0$.

The *value* of a flow, denoted by $|f|$, is the total out-flow at the source:

$$|f| = I_f(s) = \sum_{y \in V} f(s, y) - \sum_{x \in V} f(y, s).$$

The idea here is that the source vertex receives some amount $|f|$ of water from outside the network, and so within the flow network more water flows out of s than comes in.

Note that for any flow f (or indeed any function $f : V \times V \rightarrow \mathbb{R}$), we have

$$\sum_{x \in V} I_f(x) = \sum_{x, y \in V} f(x, y) - \sum_{x, y \in V} f(y, x) = 0$$

Since $I_f(x) = 0$ for all $x \in V \setminus \{s, t\}$, it follows that $I_f(s) = -I_f(t)$. That is, $|f| = I_f(s)$ is also equal to the total in-flow $-I_f(t)$ at the sink t . We can therefore always assume that $|f| \geq 0$ (if it is not, simply swap the roles of s and t).

The key question in the theory of flows is: What is the maximum value $|f|$ of a flow f in a given flow network with source s and sink t ?

Definition 74. Let $G = (V, E)$ be a flow network with source s and sink t and capacity function $c : V \times V \rightarrow [0, \infty)$. A *cut* (or *s-t-cut*) is a partition of V into two disjoint sets S, T with $s \in S$ and $t \in T$. The *capacity of the cut* is

$$c(S, T) = \sum_{(v, v') \in S \times T} c(v, v'),$$

that is, the sum of the capacities of the ‘cut edges’ going from S to T .

Intuitively, the capacity $c(S, T)$ is the maximum conceivable flow from S to T . Indeed, for any flow f and any cut S, T we have $|f| \leq c(S, T)$:

$$\begin{aligned}
|f| &= I_f(s) = \sum_{x \in S} I_f(x) = \sum_{x \in S} \sum_{y \in V} (f(x, y) - f(y, x)) \\
&= \underbrace{\sum_{x \in S} \sum_{y \in S} (f(x, y) - f(y, x))}_{=0} + \sum_{x \in S} \sum_{y \in T} (f(x, y) - f(y, x)) \\
&\leq \sum_{x \in S} \sum_{y \in T} f(x, y) \leq \sum_{x \in S} \sum_{y \in T} c(x, y) = c(S, T)
\end{aligned} \tag{*}$$

So the maximum value of a flow is at most the minimum capacity of a cut. The remarkable ‘max-flow min-cut’ theorem tells us that indeed we have equality:

Theorem 75 (Max-Flow Min-Cut). *In any flow network (G, s, t, c) , we have*

$$\sup\{|f| : f \text{ is a flow}\} = \min\{c(S, T) : (S, T) \text{ is a cut}\}.$$

Furthermore, the supremum is attained, that is, there is a flow f where $|f|$ is equal to the supremum above (and so we could replace ‘supremum’ by ‘maximum’).

Before proving this theorem, we need to introduce some more key concepts. The first is the ‘residual network’ of a flow f , which tells us for each pair (u, v) how much ‘slack’ there is to increase the net flow from u to v — either because of leftover capacity in the edge $(u, v) \in E$, or because we can decrease the flow along the edge $(v, u) \in E$ in the opposite direction.

Definition 76. Given a flow network (G, s, t, c) and a flow f , the *residual network* G_f is the flow network with *residual capacity* c_f on the vertex set V constructed in the following way: For vertices u, v , set

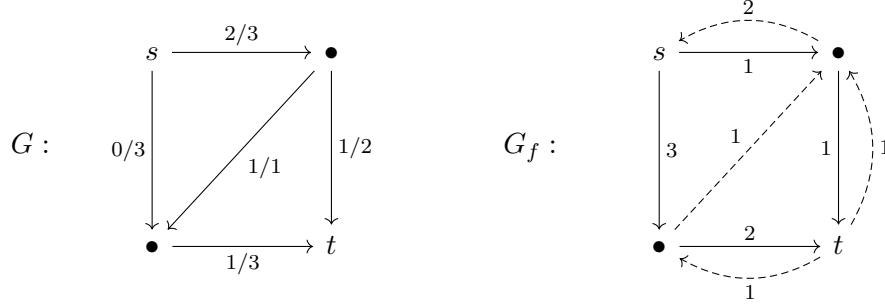
$$c_f(u, v) := \begin{cases} c(u, v) - f(u, v) & \text{if } (u, v) \text{ is an edge in } G \\ f(v, u) & \text{if } (v, u) \text{ is an edge in } G \\ 0 & \text{otherwise} \end{cases}$$

Now let $E(G_f)$ be the set of pairs (u, v) for which $c_f(u, v) > 0$.

Observe how for this definition to make sense, we needed to have no parallel edges in opposite direction in the original graph G .² Note also that in G_f , we can indeed have parallel edges in opposite directions.

²But parallel or even multiple edges in G could still be accommodated by allowing multiple edges in G_f .

Example 77. Consider a flow network G with the flow f as below, where the edge labels are of the form ‘flow’/‘capacity’. Its residual network G_f is shown on the right, where the dashed arrows are the contributions that come from the flow f , whereas the drawn arrows come from the unused capacity of the edge.



Next, we define an ‘augmenting’ path from s to t , which will be a path along which the value of a flow can be improved.

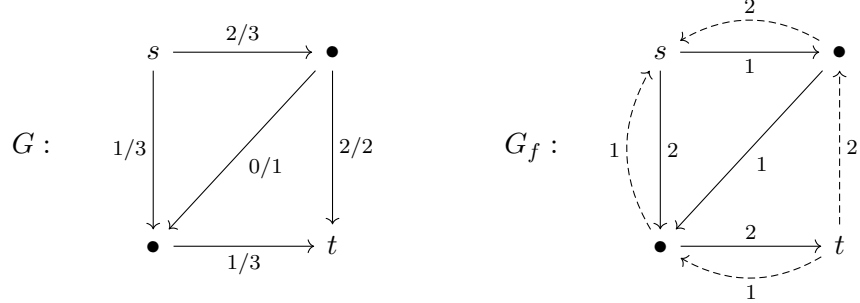
Definition 78. For $\varepsilon > 0$, we call a pair $(u, v) \in V \times V$ ε -slack if $c_f(u, v) \geq \varepsilon$. A (directed) path $v_0 e_1 v_1 e_2 \dots e_k v_k$ (where $e_i = (v_{i-1}, v_i)$) in the residual network G_f is called ε -slack if every pair (v_{i-1}, v_i) is ε -slack, $1 \leq i \leq k$. An s - t path in G_k is called *augmenting* if it is ε -slack for some $\varepsilon > 0$.

Lemma 79. Let f be a flow in a network. If there is an augmenting s - t -path P , then $|f|$ is not maximal. In particular, if P is ε -slack for some $\varepsilon > 0$, then there is a flow f' with $|f'| \geq |f| + \varepsilon$.

Proof. Suppose P is a directed path in G_f along the vertices $s = x_0, x_1, \dots, x_k = t$ that is ε -slack. Then for each pair (x_{i-1}, x_i) , $1 \leq i \leq k$, we can either increase $f(x_{i-1}, x_i)$ by ε (if (x_{i-1}, x_i) is an edge in G) or we can decrease $f(x_i, x_{i-1})$ by ε (if (x_i, x_{i-1}) is an edge in G). Call the resulting function f' , then by the definition of the residual capacities, we still have $0 \leq f'(x, y) \leq c(x, y)$ for all $x, y \in V$. Moreover, for any internal vertices $x_1, \dots, x_{k-1}$ on the path P our changes left $I_f(x_i)$ unchanged, and we still have $I_{f'}(x_i) = 0$. So f' is still a flow. But we have $|f'| = I_{f'}(s) = I_f(s) + \varepsilon = |f| + \varepsilon$. $\square$

Example 80. In Example 77, there is an augmenting path in G_f by going down from s , then taking the diagonal arrow, and then going down to t . This path is 1-slack, and gives the following new flow f' on G (shown left),

and the corresponding residual network G_f :



Now we are ready to prove Theorem 75.

Proof of Theorem 75. Let us first remark quickly on why the supremum is attained. This follows from some very general ‘compactness’ arguments in $\mathbb{R}^k$. Briefly, any flow f can be seen as a vector in $\mathbb{R}^{|E|}$, with possible values for $e = (v, w) \in E$ in the closed interval $[0, c(v, w)]$. Together with the conservation constraint, the set of all flows is a compact subset of $\mathbb{R}^{|E|}$, and $|f|$ is a continuous function on this set, which implies that the supremum is attained.

Now suppose f is a flow so that $|f|$ is maximal. We showed in $(\star)$ that $|f| \leq c(S, T)$ for any cut S, T . So if we can find a cut S, T so that $|f| = c(S, T)$, we are done. Now let

$$S = \{x \in V : \text{there is a } \varepsilon\text{-slack } s\text{-}x\text{-path for some } \varepsilon > 0\},$$

and $T = V \setminus S$. Then $s \in S$ (by considering the path s of length 0). Furthermore, $t \notin S$, because otherwise by Lemma 79, $|f|$ would not be maximal. So $t \in T$, and S, T is a cut.

We claim that for this cut, $|f| = c(S, T)$. For this, again consider the calculation $(\star)$. Equality holds iff $f(x, y) = c(x, y)$ and $f(y, x) = 0$ for all $x \in S, y \in T$, and in that case $|f| = c(S, T)$.

So suppose for a contradiction that this is not the case, i.e. we have some $x \in S, y \in T$ so that either $f(x, y) < c(x, y)$, or $f(y, x) > 0$ (in particular, (x, y) or (y, x) must be an edge in G). But that just means that $c_f(x, y) > 0$ in the residual network. By the definition of S there is an ε -slack path in G_f from s to x , and since $c_f(x, y) > 0$, this can be extended to a ε' -slack path from s to y for some $\varepsilon' > 0$. But $y \in T$, so $y \notin S$, which is a contradiction. So we must indeed have equality in $(\star)$, and $|f| = c(S, T)$. $\square$

The construction of the proof of Theorem 75, Lemma 79 and $(\star)$ imply the following equivalences:

Theorem 81 (Ford-Fulkerson). *For a flow network (G, s, t, c) and a flow f , the following are equivalent:*

- (i) f is a maximal flow.
- (ii) G_f contains no augmenting s - t -path.
- (iii) There is an s - t -cut (S, T) with $|f| = c(S, T)$.

Note that if all capacities $c(x, y)$ are integers, then it even follows from the proofs of Lemma 79 and the Max-Flow Min-Cut theorem that there is an *integer* flow $f : V \times V \rightarrow \mathbb{Z}_0$ so that $|f|$ is maximal. To see this, start with the all 0-flow (which is integral), and improve it with the help of augmenting paths as described before until there is no more such path, hence $|f|$ is maximal. The key observation is that if f and c are integral, then so are all residuals $c_f(x, y)$, and so any augmenting path is 1-slack, which means we can make only integer changes in this procedure. This results in the following theorem.

Theorem 82 (Integer flow theorem). *If G is a flow network with an integer capacity function, i.e. $c : V \times V \rightarrow \mathbb{Z}_{\geq 0}$, then there is a maximal integer flow, i.e. a maximal flow f such that $f(x, y) \in \mathbb{Z}$ for all $x, y \in V$.*

Given a flow network, we can calculate $\max_{S, T} c(S, T)$, but how can we actually find a maximum flow with that value? Our proofs suggest the following method.

Ford-Fulkerson ‘algorithm’ Given a network G , start with the 0-flow f . Repeat the following steps:

1. Construct G_f .
2. Find an augmenting path in G_f , then use it to modify f into a flow with larger value, like in Lemma 79 (with ε as large as possible). If no augmenting path exists, stop.

If this procedure terminates, by Theorem 81 it results in a maximal flow f . If c is integral, then in each step we increase the value $|f|$ by at least 1, so the procedure always terminates. Consequently, this is also true if all capacities $c(u, v)$ are rational numbers. (Why?) But in general the algorithm might not terminate.

Lecture 9 Matchings

Matchings arise when trying to pair up the vertices in a graph: We want to find some disjoint pairs $\{v, w\}$ of vertices so that vw is an edge. This leads to the following definition:

Definition 83. Let $G = (V, E)$ be a finite simple graph. A *matching* on G is a set $M \subseteq E$ so that no two edges in M have a common endpoint (when viewing the edges as subsets of size 2 of V , that means all pairs of edges are disjoint). If $e = \{v, w\}$ is an edge in M , then we also say that the vertices v and w have been matched to each other.

To start with, we only consider *bipartite graphs*, i.e. graphs where we have a vertex partition $V = A \uplus B$ such that there are no edges between two vertices in the same set of the partition, i.e., all edges have one endpoint in A and one endpoint in B .

Given a bipartite graph on a partitioned vertex set $V = A \uplus B$, we can ask if there is a matching that matches all the vertices in A to some vertex in B . For this to work, any subset $Q \subseteq A$ must have at least $|Q|$ neighbours in B (otherwise, you couldn't even match Q into B !). So denote by $N(Q)$ the set of all vertices in B that are a neighbour of some $v \in Q$. The famous next theorem, called Hall's 'marriage theorem', states that this obviously necessary condition is all that we need.

Theorem 84 (Hall, 'Marriage Theorem'). *Let $G = (V, E)$ be a finite simple bipartite graph with $V = A \uplus B$. Then G contains a matching of A into B iff $|N(Q)| \geq |Q|$ for all $Q \subseteq A$.*

There are direct proofs of this result using induction on $|A|$. Perhaps surprisingly, we can also use the integer flow theorem from the last lecture to give a short proof.

Proof. One direction is clear, and it remains to show that the *marriage condition* is sufficient: Assume that $|N(Q)| \geq |Q|$ for all $Q \subseteq A$. Create a flow network as follows: Introduce a new vertex s which is a neighbour to every vertex in A , and another new vertex t that is a neighbour to every vertex in B . Direct the edges in such a way that everything points away from s and towards t , and for edges between A and B orient them so that they go from A to B . Finally, give every edge incident to either s or t capacity 1, and give capacity ∞ to the edges going from A to B (this is for convenience, and can be replaced by a large enough constant integer depending on G).

For $S' = \{s\}$ and $T' = A \cup B \cup \{t\}$ we have $c(S', T') = |A| < \infty$, hence $c(S, T) < \infty$ for any minimum cut (S, T) . In particular, in a minimum cut there cannot be a directed edge (u, v) with $u \in S \cap A$ and $v \in T \cap B$, and

so $N(S \cap A) \subseteq S \cap B$, so $|S \cap B| \geq |N(S \cap A)| \geq |S \cap A|$. With this, let's bound $c(S, T)$:

$$\begin{aligned} c(S, T) &= \sum_{\substack{u \in S \\ v \in T}} c(u, v) = \sum_{v \in T \cap A} c(s, v) + \sum_{u \in S \cap B} c(u, t) \\ &= |T \cap A| + |S \cap B| \geq |T \cap A| + |S \cap A| = |A|. \end{aligned}$$

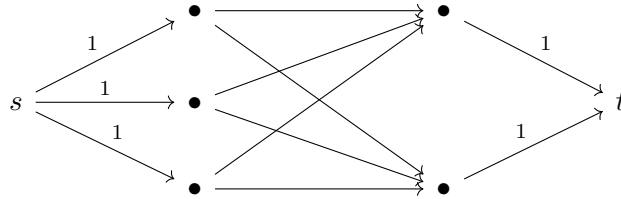
Comparing this with the cut (S', T') shows that a minimum cut has capacity exactly $|A|$. So by Theorem 82 there is a maximal integral flow f with value $|A|$. This flow must have $f(s, x) = 1$ for every $x \in A$, and following along the flow (i.e., which of the edges (x, y) from A to B have $f(x, y) = 1$) then yields a matching of A into B . $\square$

What if we don't quite have $|N(Q)| \geq |Q|$ for every Q , but we're not too far off? Say we have $|N(Q)| \geq |Q| - d$ for all $Q \subseteq A$ and some integer d ? With a small trick we can get a matching of size $|A| - d$ in this case:

Corollary 85. *Let $G = (V, E)$ be a finite simple bipartite graph with $V = A \uplus B$. If $|N(Q)| \geq |Q| - d$ for every subset $Q \subseteq A$ and some $d \in \mathbb{N}$, then G contains a matching with $|A| - d$ edges.*

Proof. Add d new vertices to B , each of them neighbouring all vertices in A . This modified graph satisfies the marriage condition and therefore contains a matching of A into the modified B , and at least $|A| - d$ edges of this matching must be in the original graph. Those form the desired matching. $\square$

The construction of a flow network associated to the bipartite graph $G = (A \uplus B, E)$ was essential in the proof of Hall's Theorem. Here is what the associated flow network for a $K_{3,2}$ looks like. The unlabelled arrows have infinite capacity.



Remark 86. There is a bijection between matchings in $G = (A \uplus B, E)$ and integer flows in the associated flow network. Indeed, given an integer flow f from s to t , taking all edges (a, b) with $a \in A$, $b \in B$ and $f(a, b) = 1$ gives a matching on G because the flow through any vertex in G can be at most 1. The size of this matching equals the value $|f|$ of the flow. Conversely, any matching can be used to construct the corresponding flow by setting $f(a, b) = 1$ if the edge $\{a, b\}$ is in the matching, setting the flow along all

other edges from A to B to 0, and setting the flows along the edges (s, a) and (b, t) to 1 or 0 accordingly to create a valid flow. Again $|f|$ is the size of the matching.

Hall's theorem answers the question whether we can match *every* vertex in A to some vertex in B . But if the answer is no, what is the maximum number of vertices in A that can be matched? By the correspondence between matchings and flows we just described, this is the same thing as the maximum size $|f|$ of a flow on the associated flow network. So of course we look towards the Max-Flow Min-Cut Theorem (Theorem 75) to provide us with the answer! What structures in the underlying bipartite graph G correspond to the capacity of a cut in the associated flow network?

The answer is in the following definition.

Definition 87. Let $G = (V, E)$ be a finite simple graph. A *vertex cover* of G is a subset $C \subseteq V$ such that every edge has an endpoint in C . The *covering number* of G , denoted by $\beta(G)$, is the minimum size $|C|$ of any vertex cover C of G (that is, the size of the smallest vertex cover of G).

There is a bijection between vertex covers C in a bipartite graph $G = (A \uplus B, E)$ and cuts (S, T) with finite capacity in the associated flow network so that $|C| = c(S, T)$: If (S, T) is a finite capacity cut, then $C := (A \cap T) \cup (B \cap S)$ is a vertex cover of G with $|C| = c(S, T)$. (To see this is a vertex cover, suppose that $\{a, b\}$ with $a \in A, b \in B$ is an edge not covered by C . Then $a \in S$ and $b \in T$, but that would imply $c(S, T) = \infty$, a contradiction.) Conversely, any vertex cover C of G gives rise to a cut in the network: Set $S = \{s\} \cup (A \setminus C) \cup (B \cap C)$ and $T = \{t\} \cup (A \cap C) \cup (B \setminus C)$. Again $c(S, T) = |C|$, in particular this is a finite capacity cut.

Theorem 88 (König's theorem). *Let G be a finite simple bipartite graph. Then the maximum size of a matching on G equals the minimum size $\beta(G)$ of a vertex cover of G .*

Proof. We use the bijections between matchings and flows, and between cuts and vertex covers, that we just described. Let M be a matching in G with maximal size $|M|$. Construct the flow network G' to the bipartite graph as in the proof of Hall's theorem. Then, by Remark 86, a maximal flow f on G' has value $|f| = |M|$. By the max-flow min-cut theorem (Theorem 75), a minimum cut (S, T) on G' has capacity $c(S, T) = |M|$. Finally, the bijection constructed after Definition 87 gives $\beta(G) = c(S, T) = |M|$, as required. $\square$

Observe that the inequality $|M| \leq \beta(G)$ is true for any graph, even non-bipartite ones: This is because every edge in a matching M is covered by a vertex in a vertex cover C , and since the edges in M are disjoint, no vertex in C can cover more than one edge of M .

We will now consider matchings in arbitrary graphs. In particular, we will focus on the question of when we can match up *all* vertices in a graph:

Definition 89. A *perfect matching* (sometimes also called *1-factor*) M in a finite simple graph $G = (V, E)$ is a matching in G such that every vertex $v \in V$ is an endpoint of an edge $e \in M$.

Clearly, for a finite simple graph $G = (V, E)$ to have a perfect matching, $|V|$ must be even. But more can be said: Write $G - S := G[V \setminus S]$ for the graph obtained after removing a vertex subset $S \subseteq V$. If G has a perfect matching, then $G - S$ cannot have more than $|S|$ components with an odd number of vertices. To see this, note that in the matching, at least one vertex from each such component has to be matched to a distinct vertex in S . For brevity, we call components with an odd number of vertices *odd* and denote the number of odd components in a graph G by $o(G)$.

Theorem 90 (Tutte's 1-factor theorem). *A finite simple graph $G = (V, E)$ has a perfect matching if and only if $o(G - S) \leq |S|$ for all $S \subseteq V$.*

Proof (omitted in lectures). We already argued that if G has a perfect matching, $o(G - S) \leq |S|$ for all $S \subseteq V$, so it only remains to show the other direction. So suppose that $G = (V, E)$ is a graph that fulfils the condition that $o(G - S) \leq |S|$ for all $S \subseteq V$, but without a perfect matching. Adding an edge to G will not affect the condition: If the new edge connects two even or one even and one odd component in $G - S$, then the $o(G - S)$ doesn't change, and if it connects two odd components, then $o(G - S)$ decreases. So we can assume that G is an edge-maximal graph satisfying $o(G - S) \leq |S|$ for all $S \subseteq V$, but without containing a perfect matching.

Let $n = |V|$, then n must be even (as taking $S = \emptyset$ gives $o(G) = 0$, so G has no odd components, in particular n is even). Now pick

$$S = \{v \in V : \deg(v) = n - 1\}$$

(the set of vertices which are neighbours of all other vertices), and consider $G - S$. We distinguish two cases:

1) All components of $G - S$ are complete. Then all of the even components contain a perfect matching, and in each of the odd components, all but one vertex can be matched up. By choice of S , this remaining vertex must have been a neighbour to all vertices in S . Moreover, since there are at most $|S|$ odd components in $G - S$, we can match up all of these remaining vertices with vertices from S . This might leave some (an even number, since n is even!) vertices in S unmatched, but by our choice of S , $G[S]$ is a complete graph, so they can be matched to each other. This construction gives a perfect matching in G .

2) Now suppose that K is a component of $G - S$ that is not complete (since K is connected, this can only happen if $|K| \geq 3$). In particular, there are vertices $x, a, b \in V(K)$ such that $\{x, a\}$ and $\{a, b\}$ are edges, but $\{x, b\}$ is not (if that were not the case, K would have to be complete³). Moreover, $a \notin S$, so by the definition of S there is some vertex $c \in V(G)$ that is not a neighbour of a . We assumed that G is edge-maximal without a perfect matching. So there are perfect matchings

$$\begin{aligned} M_1 & \quad \text{for } (V, E \cup \{\{x, b\}\}) \\ M_2 & \quad \text{for } (V, E \cup \{\{a, c\}\}). \end{aligned}$$

Of course we must have $\{x, b\} \in M_1$ and $\{a, c\} \in M_2$.

In G , consider a maximal path P starting at c with an edge from M_1 , and then alternating between edges in M_2 and edges in M_1 . This is always possible: Since M_2 matches c with a , but M_1 doesn't have this edge available, M_2 has to match c to some other vertex $d \neq a$. Then M_1 cannot match d to c , as c is already matched, so M_1 matches d to a new vertex $e \neq c$, and so on. This procedure continues until we reach x, a , or b .

Let v denote the last vertex on P . If the last edge of P was from M_1 , then we must have $v = a$ (because x and b are matched to each other in M_1 , and the edge xb is not available in G to extend P further). In that case, there is another vertex d which is M_2 -matched to v and which does not appear on P yet. Define C to be the cycle in $(V, E \cup \{\{a, c\}\})$ consisting of P and the edge $\{a, c\}$.

Similarly, if the last edge of P was from M_2 , then $v \in \{x, b\}$ (since a is matched to c in M_2 , and the edge ac is not available in G to extend P further). In this case, define C to be the cycle in $(V, E \cup \{\{a, c\}\})$ consisting of P and the edges $\{v, a\}, \{a, c\}$.

In both cases, C is a cycle of even length where every second edge belongs to M_2 . Now modify M_2 by replacing its edges on C by the edges not in C – this is again every second edge on C , but avoids the edge $\{a, c\}$. Together with the edges from M_2 that were not on C , we then have a perfect matching in G using only edges from E , a contradiction. $\square$

³Suppose this is not the case. For a vertex v , let $N(v)$ be the neighbourhood of v . Then all vertices in $N(v)$ also have to be neighbours of each other, i.e., $G[v \cup N(v)]$ is complete. Further, no $w \in N(v)$ can have any neighbours outside of $N(v) \cup \{v\}$. So since G is connected, $v \cup N(v)$ is the whole vertex set, so G is complete.

Lecture 10 Connectivity

Back in lecture 1 we defined what it means for a graph G to be connected: For any two vertices v, w , there is a walk (or equivalently, a path) from v to w along the edges of G .

But if a graph G is connected, how *well-connected* is it? For example, if your graph represents an infrastructure network like the electricity grid or the internet, how resilient is it to some links getting cut, or to some of the vertices (e.g. internet servers) being out of service? This could happen accidentally and randomly, such as a tree falling onto a power line, or because an adversary purposefully severs links in your network (like cutting undersea internet cables) or attacks some vertices.

Definition 91. Let $k \in \mathbb{Z}_{\geq 0}$. A finite simple graph $G = (V, E)$ is called *k-connected* if $|V| \geq k + 1$ and for all sets $X \subseteq V$ with $|X| \leq k - 1$, $G - X$ is connected (in other words, you need to remove at least k vertices to disconnect G). The largest k for which G is *k-connected* is the *connectivity* $\kappa(G)$ of G .

With this definition, every non-empty graph is 0-connected, and every connected graph on at least two vertices is 1-connected. So $\kappa(G) = 0$ just means that G is disconnected. Furthermore, note that $\kappa(K_n) = n - 1$.

Definition 92. Let $G = (V, E)$ be a finite simple graph, and let $v, w \in V$, $v \neq w$, and $A, B \subseteq V$. A set $X \subseteq V$ *separates v from w* if $v, w \notin X$ and every path from v to w contains at least one vertex of X . Let $\kappa(v, w)$ be the minimum size $|X|$ of such a set X .

A set $X \subseteq V$ *separates A from B* if every path from a vertex in A to a vertex in B contains at least one vertex from X .

Two paths from v to w are called *(vertex-)independent* if they do not share any common vertices besides the two endpoints.

In other words: If X separates v and w , that means v and w are in different connected components in the graph $G - X$ (where we delete all vertices in X , and all edges with at least one endpoint in X). Furthermore, $\kappa(v, w)$ is the minimum number of vertices we have to delete in order to have v and w in different components in the resulting graph.

Careful: The definition above makes a subtle but important difference between separating vertices v, w (in which case the separating set X may *not* contain either of the two vertices), and separating sets of vertices A, B (in which case X may intersect A and B). In fact, if $A \cap B \neq \emptyset$, then we need to have $A \cap B \subseteq X$ for any separating set X , otherwise the ‘lazy path’ from any vertex $A \cap B$ to itself would prevent separation.

Proposition 93. For a finite simple graph $G = (V, E)$ that is not a complete graph, we have

$$\kappa(G) = \min_{x,y} \kappa(x, y),$$

where the minimum is taken over all non-adjacent pairs of vertices x, y .

Proof. Since G is not complete, $\kappa(G)$ is the minimum size $|X|$ of a set $X \subseteq V$ so that removing the vertices in X disconnects G . In particular, $G - X$ has at least two components, so we can pick x_0, y_0 from different components (in particular, x_0 and y_0 are not adjacent). Then X separates x_0 and y_0 , so $\min_{x,y} \kappa(x, y) \leq |X| = \kappa(G)$.

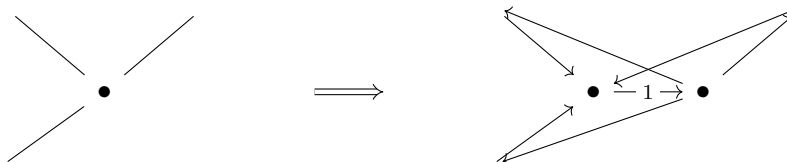
Conversely, let x_0, y_0 be two non-adjacent vertices so that $\kappa(x_0, y_0)$ is minimal. Then there is some set X of vertices with $|X| = \kappa(x_0, y_0) = \min_{x,y} \kappa(x, y)$ that separates x_0 and y_0 . In particular, $G - X$ is disconnected, so $\kappa(G) \leq |X| = \min_{x,y} \kappa(x, y)$. $\square$

Recall that a graph is connected if and only if for any two vertices v, w there is a path from v to w . Is there a similar characterisation for k -connectedness? It turns out, the answer is yes. And we get to apply the Max-Flow Min-Cut theorem again!

Theorem 94 (Menger). Let $G = (V, E)$ be a finite simple graph, and let $v, w \in V$ be non-adjacent. Then the minimum size $\kappa(v, w)$ of a set that separates v from w equals the maximum number of independent paths from v to w .

Proof. We construct a flow network from G . The idea of the construction is that we introduce a ‘bottleneck’ (an edge of capacity 1) inside every vertex which ensures that a flow of size 1 can only pass through that vertex once.

Formally: First, replace every edge $xy \in E$ by two directed edges $x \rightarrow y$ and $y \rightarrow x$ with infinite capacity (this is for convenience and could be replaced by a large integer). Then split every vertex x other than v, w into two vertices x_i, x_o (‘in’ and ‘out’) so that every incoming edge to x is pointed to x_i and every outgoing edge from x starts at x_o . Also add an edge $x_i \rightarrow x_o$ of capacity 1 (this is the aforementioned ‘bottleneck’). Finally, we define v as the source and w as the sink of the flow network.



Similarly as in the proofs of Hall’s and König’s theorem, integer flows f through this flow network correspond to a collection of disjoint paths from v

to w in the graph (and the number of such paths is exactly the value $|f|$ of the flow) — you may think of the 1's in this flow as $|f|$ ‘unit parcels’ that v sends out, and track the path of each parcel to the recipient w , noting that because of the capacity 1 edges, each of the original vertices can only appear on one of these paths.

Now consider cuts. If we let $S' = \{v\} \cup \{x_i : x \in V \setminus \{v, w\}\}$ and $T' = \{w\} \cup \{x_o : x \in V \setminus \{v, w\}\}$, this cut has capacity exactly $|V| - 2$. In particular, the minimum capacity of the network is finite. Now note that each vertex set X that separates v and w in G corresponds to a finite capacity cut (S, T) so that $|X| = c(S, T)$, and vice versa: Starting with X , for $x \in X$ put x_i in S and x_o in T , and for all other vertices $y \notin X$ sort both y_i, y_o into the same set S or T depending on whether they are in the component of v or of w in $G - X$ — and if they are in neither component, put them into S . Then $c(S, T) = |X|$ (check!). Conversely, starting with a finite capacity cut (S, T) , the vertices x so that $x_i \in S, y_o \in T$ form a separating set X of size $c(S, T)$. (Note that these maps are not necessarily bijections, but this does not matter.)

Now the statement follows from the max-flow-min-cut theorem, Theorem 75, and the integer flow theorem, Theorem 82. $\square$

Corollary 95 (Menger, global version). *A finite simple graph $G = (V, E)$ is k -connected if and only if it contains k independent paths between any two vertices.*

Proof. Suppose that G is k -connected. The statement is obviously true for complete graphs on at least $k + 1$ vertices, so we can assume G is not complete. Then by Proposition 93, $\kappa(x, y) \geq k$ for all non-adjacent pairs of vertices x, y . By Menger’s theorem, those vertices v, w are therefore connected by at least k independent paths. But we still need to deal with *adjacent* x, y . So for such a pair, suppose there are at most $k - 1$ independent paths from x to y . After removing the edge $\{x, y\}$ from G , resulting in a graph G' , there are at most $k - 2$ paths left. By Menger’s theorem, we can separate x from y by a set X of size $|X| \leq k - 2$ in G' . Now note that G has at least $k + 1$ vertices (by the definition of k -connectivity), so there is a vertex v not in X , and different from x and y . Then in G' , v is separated by X from either x or y — without loss of generality from x . Then $X \cup \{y\}$ separates v from x in G . But $|X \cup \{y\}| \leq k - 1$, which contradicts that G is k -connected. So x and y must have been connected by k independent paths.

Conversely, if there are k independent paths between any two vertices, then this in particular holds for any pair of non-adjacent vertices. If there is no such pair, then G is a complete graph on at least $k + 1$ vertices, and so $\kappa(G) \geq k$. Otherwise, by Menger’s theorem we have $\kappa(x, y) \geq k$ for any pair of non-adjacent x, y , and so $\kappa(G) \geq k$ by Proposition 93. $\square$

Very often the following corollary of Theorem 94 is useful:

Corollary 96. *Let $v, w_1, \dots, w_k$ be $k+1$ distinct vertices. If G is k -connected, then there are paths $P_1, \dots, P_k$, where the path P_i starts at v and ends in w_i , $1 \leq i \leq k$, which are vertex-disjoint except for the common start vertex v_0 .*

Proof. Add a new vertex w to the graph and connect it to all vertices $w_1, \dots, w_k$. Then we have $\kappa(v, w) = k$ (why?), and so k disjoint v - w paths exist, each of which has to go through one vertex w_i . $\square$

Sometimes also the following version of Menger's theorem is useful. Careful again: This involves the somewhat different definition $\kappa(A, B)$ for sets A, B , and we are now counting *disjoint* paths (i.e., the end points also have to be different).

Theorem 97 (Menger, v2). *Let $G = (V, E)$ be a finite simple graph and $A, B \subseteq V$. Then the minimum size of a set X that separates A from B is equal to the maximum number of disjoint paths with one end in A and the other end in B .*

Proof. Modify the proof of Theorem 94: Add two vertices s, t to G with arrows pointing from s to every vertex in A and from every vertex in B to t . Give these arrows infinite capacity, then proceed to split all of the original vertices as in the proof of Theorem 94. The remaining argument is the same. $\square$

We know what 1-connected graphs 'look like' — they're connected, and we could build them by starting with an edge and adding further edges so that at least one endpoint is already in the graph we've constructed so far. Equivalently, we could start with a path (perhaps of length 1, i.e. an edge), and attach paths so that (at least) one endpoint is in the graph already constructed. Can we describe 2-connected graphs in a similar way?

Proposition 98. *A finite simple graph is 2-connected if and only if it can be constructed from a cycle graph by successively adding paths to it, both of whose endpoints lie in the graph already constructed.*

Proof. Every graph constructed in this way is 2-connected⁴.

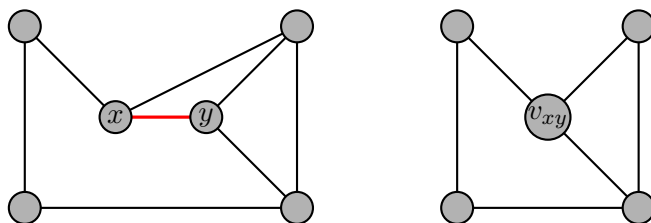
For the other direction, let's call a graph *constructible* if it can be obtained from the described procedure. If $G = (V, E)$ is a finite simple 2-connected graph, then G contains a cycle (as any two vertices in G are joined by two independent paths), and a cycle is constructible.

⁴Exercise, using Menger's theorem, Corollary 96 and induction over the number of paths added.

Now let H be a constructible subgraph of G so that the number of edges in H is maximal. Then H is an induced subgraph: For $x, y \in V(H)$ and an edge xy in G but not in H , this edge would constitute a path with both its endpoints in H — and so H with the edge xy added is constructible, contradicting the maximality of H . So we only need to show that H is spanning, and we are done. So suppose H is not spanning, then as G is connected there is an edge $\{x, y\}$ with $x \in V(H)$ and $y \notin V(H)$. Since G is 2-connected, removing x from G still leaves a path P connecting y to $V(H)$ (by v2 of Menger's theorem). But that means that in G , going from x to y and then back along P to $V(H)$ is a path with both endpoints in $V(H)$, again contradicting the maximality of H . So H is induced and spanning, so $H = G$. $\square$

Let's now think about 3-connected graphs (because we'll need them next time). In the following definition we define the contraction G/e of a graph G along an edge e — the idea is simply to glue together (or *identify*) the two endpoints of the edge e .

Definition 99. Let $G = (V, E)$ be a finite simple graph and $e \in E$. The *contraction* of e is the graph G/e , constructed in the following way: For $e = \{x, y\}$, introduce a new vertex v_{xy} and set $V(G/e) = V \setminus \{x, y\} \cup \{v_{xy}\}$. Draw an edge in G/e between v_{xy} and w whenever w was a neighbour to x or y (or both). Keep all edges of G that were not incident to x or y .



Then we have the following lemma which we state without proof.

Lemma 100. *If a finite simple graph $G = (V, E)$ is 3-connected and has at least 5 vertices, then there exists an edge $e \in E$ such that G/e is again 3-connected.*

Proof (omitted in lectures). Suppose not. Then for every edge $e = xy$, the graph G/e contains a separating set X with $|X| \leq 2$. Since G is 3-connected, this is only possible if X contains v_{xy} and another vertex $z \neq x, y$. Then $T = \{x, y, z\}$ is a separating set for G , and since no proper subset of T separates G , every vertex in T must have a neighbour in every component of $G - T$ (Why?). Now choose the edge xy , the vertex z and a component C of $G - T$ so that $|V(C)|$ is minimal.

Let v be a neighbour of z in C and let $e' = vz$. Then again by assumption, G/e' is not 3-connected, so just as earlier for $T = \{x, y, z\}$, there is a vertex w such that $T' = \{z, v, w\}$ separates G . And once more, z, v and w each have a neighbour in every component of $G - T'$. As x and y are adjacent, at least one component D of $G - T'$ contains neither x nor y . Since $v \in C$, every neighbour of v that is in D must also be in C , so $D \subset C$. Furthermore $D \neq C$, since $v \notin D$, so D is a strict subset of C . But this contradicts the minimality of $|V(C)|$. $\square$

The following theorem by Tutte — which we also quote without proof — says that 3-connected graphs are exactly those that can be transformed into K_4 by a series of edge contractions along edges where both endpoints have degree ≥ 3 .

Theorem 101 (Tutte 1961). *A finite simple graph G is 3-connected if and only if there is a sequence $G_0, \dots, G_n$ of finite simple graphs so that*

- (i) $G_0 = K_4$ and $G_n = G$, and
- (ii) for every $i = 0, \dots, n - 1$, the graph G_{i+1} has an edge $e = xy$ with $\deg_{G_{i+1}}(x), \deg_{G_{i+1}}(y) \geq 3$ and $G_i = G_{i+1}/e$.

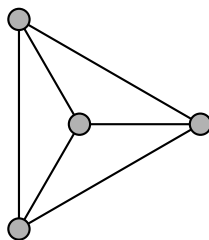
Lecture 11 Planar graphs

For small graphs, we often prefer a picture of the graph over specifying the vertex and edge sets. This raises the question how ‘clean’ of a picture we can draw for a given graph — for example, can we draw the graph without intersecting edges?

More specifically, we draw a graph by placing the vertices at certain points in the plane $\mathbb{R}^2$, and draw an edge by connecting its endpoints with a (polygonal) *arc*, that is, a continuous piecewise linear curve from one endpoint to the other.

Definition 102. Let $G = (V, E)$ be a finite simple graph. A graph is *planar* if it can be drawn (*embedded*) in $\mathbb{R}^2$ such that the vertices are placed at different points, and such that no edges intersect.

For example, K_4 is planar as there is the following embedding to the plane:



The following definition relies on the very intuitive (but difficult to prove) *Jordan curve theorem* which says that a closed continuous simple curve divides the plane into exactly two open regions (inside and outside of the curve), one of which is bounded, and one of which is unbounded. For the purpose of this lecture, we generously ignore the topological details and rely on our intuition.

Definition 103. Let $G = (V, E)$ be a planar graph. Any planar embedding of G divides the plane into several connected components, called *faces*, all but one of which are bounded.

Observe that an embedded tree only yields one face (the unbounded one), and that the boundary of any bounded face contains a cycle in G .

Theorem 104 (Euler formula). *Let $G = (V, E)$ be a connected planar graph. Denote by f the number of faces for some planar embedding of G , let $v = |V|$ and $e = |E|$. Then*

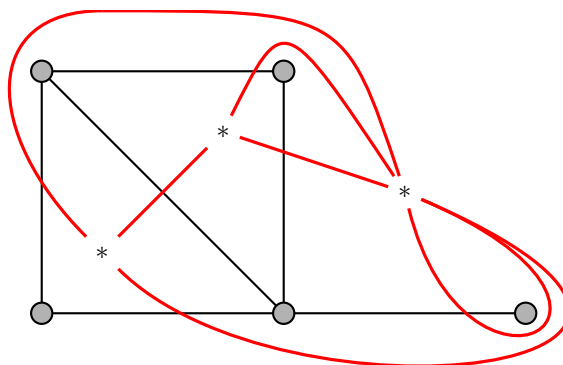
$$v - e + f = 2.$$

In particular, any two planar embeddings of the same graph have the same number of faces.

Our proof uses the following concept, which is important enough to warrant its own definition:

Definition 105. Let $G = (V, E)$ be a planar graph and fix a planar embedding of it. The *planar dual* G^* is the multigraph whose vertices are the faces of the embedding of G and with edge set E . An edge $e \in E$ connects vertices f_1, f_2 in G^* iff e was part of the boundary of the faces f_1, f_2 .

Example 106. Here's an embedded graph together with its planar dual:



Note that the planar dual really depends on the embedding — there are graphs with different planar embeddings, that produce non-isomorphic planar duals!

Convince yourself that, at least for connected graphs, the word ‘dual’ makes sense: If we take the red multigraph G^* in the picture, with the given embedding into the plane, then the dual of G^* is the original graph G . (Formally we only defined the dual for simple graphs, but the definition extends naturally to a planar multigraph.)

Proof of Theorem 104. Let $G = (V, E)$ be a connected planar graph. Fix a certain planar embedding of it, and construct its planar dual G^* . Also fix a spanning tree $T = (V, E_T)$ of G , and consider the spanning subgraph $T^* = (V(G^*), E \setminus E_T)$ of G^* .

If T^* was disconnected, the embedding of T (i.e. the embedding of G , but restricted to the subgraph T) would contain a set of edges that completely encloses a face of G , hence T would contain a cycle. This is impossible due to T being a tree. If T^* contained a cycle, the edges of this cycle would not be in T , and this would mean at least one vertex in T (inside this cycle) would be separated from another (outside the cycle) in T , again contradicting T being a tree.

So both T and T^* are trees. T has v vertices and so it has $v - 1$ edges. T^* has f vertices and $f - 1$ edges. Furthermore, by definition of T^* , the total number of edges in T and T^* is exactly e . So we get $v - 1 + f - 1 = e$, which proves our claim. $\square$

Euler's formula has far-reaching consequences, but most importantly, we can use it to show that certain graphs are not planar:

Corollary 107. *If $G = (V, E)$ is a planar finite simple graph on at least 3 vertices, then $|E| \leq 3|V| - 6$. If $G = (V, E)$ is a planar finite simple graph on at least 3 vertices that does not contain a triangle (as a subgraph, not just in a particular embedding), then $|E| \leq 2|V| - 4$.*

Proof. First note that for both statements, it is enough to consider connected graphs G — if G is not connected, consider the connected components individually.⁵ Furthermore, both statements hold when G is a tree, because then $|E| = |V| - 1$, and we have $|V| \geq 3$.

So from now suppose that G is connected and not a tree. In particular, the boundary edges of the unbounded face f_u contain a cycle — otherwise they would form a tree (because they are connected), but this would imply that the whole graph G is a tree.

The proof is now by a double counting argument. Place a coin on each side of each edge in a planar embedding of G . Then you placed exactly $2|E|$ coins. At the same time, you placed at least $3f$ coins, as every face contains at least 3 coins from the minimum 3 edges necessary to surround it (this includes the unbounded face f_u as its boundary includes a cycle, and the graph is simple, so that cycle has length at least 3). Hence $3f \leq 2|E|$. As G is connected, we can now use Euler's formula and plug in $f = 2 + |E| - |V|$; this yields $|E| \leq 3|V| - 6$.

The second claim follows analogously, only that now every face is bounded by at least 4 edges (and again this includes the outer face f_u , as its boundary edges contain a cycle, which now has to have length at least 4). $\square$

Corollary 108. *The graphs K_5 and $K_{3,3}$ are non-planar.*

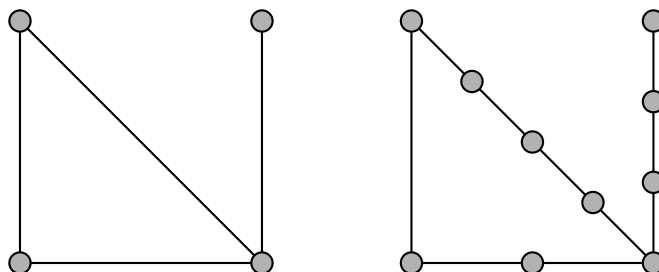
Proof. K_5 contains 5 vertices and 10 edges, so cannot be planar by Corollary 107. $K_{3,3}$ contains 6 vertices and 9 edges, but because it is bipartite it also cannot contain a triangle. Therefore it is non-planar, again by Corollary 107. $\square$

⁵There are some small, annoying details to consider here: A connected component might contain one or two vertices — call such a component ‘tiny’ — and then we can't apply the statement $|E| \leq 2|V| - 4$ or $|E| \leq 3|V| - 6$ to that component individually. But note that in that case we always have $|E| \leq |V| - 1$. Summing up over all tiny and non-tiny components (distinguishing the case where all components are tiny) then gives the result for the whole graph.

As it will turn out, K_5 and $K_{3,3}$ are in a certain sense the only two minimal non-planar graphs: Every non-planar graph ‘contains’ one of these graphs in a certain sense — for this statement to make sense, we have to broaden our definition of containment.

Definition 109. Let G, H be finite simple graphs. A *subdivision* of H is a graph where some of the edges of H are replaced by vertex-independent paths. If a graph G contains a subdivision of H , then H is a *topological minor* of G .

The following pictures show a graph H on four vertices together with a subdivision of H on the right. If a graph G contains such a subdivision, it is said to have H as a topological minor.



There is another definition of a minor (which is the one we usually mean when we talk about graph minors):

Definition 110. Let G, H be finite simple graphs. The graph H is a *minor* of G if there is a subgraph G' of G such that H can be obtained from G' by a sequence of edge contractions (as in Definition 99).

Equivalently, H is a minor of G if it can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

It is clear that if H is a topological minor of G , then it is also a minor of G . Furthermore, if H is a minor of G , and H is non-planar, then so is G . So any graph with K_5 or $K_{3,3}$ as a minor cannot be planar. Perhaps surprisingly, the converse is also true (even with ‘topological minors’):

Theorem 111 (Kuratowski, 1930, Wagner 1937). *For a finite simple graph G , the following are equivalent.*

- (a) G does not have K_5 or $K_{3,3}$ as a topological minor.
- (b) G does not have K_5 or $K_{3,3}$ as a minor.
- (c) G is planar.

The implications (b) $\implies$ (a), (c) $\implies$ (a) and (c) $\implies$ (b) are clear. From the remaining implications, we will only prove the following lemma which is an important step in the proof. Here we assume the graph does not have K_5 or $K_{3,3}$ as a minor and, additionally, that it is 3-connected, and show that such a graph is not planar. For the full proof, see chapter §4.4 of Diestel.

Lemma 112. *Let $G = (V, E)$ be a finite simple 3-connected graph without a K_5 or a $K_{3,3}$ as minor. Then G is planar.*

Proof. Use induction on $|V|$. For $|V| = 4$ the only 3-connected graph is K_4 which is planar, so the claim holds. So, assume that $|V| \geq 5$ and that the claim holds for all graphs on fewer vertices. We now use Lemma 100 from the last lecture which says that if G is 3-connected on at least 5 vertices, there is an edge $e = xy$ in G such that the contraction G/e is again 3-connected. Moreover, G/e is a minor of G and so it can't contain a K_5 or a $K_{3,3}$ minor. So by the induction hypothesis, G/e is planar.

Now consider a planar embedding of G/e . Removing the vertex v_{xy} (the one obtained from contracting $e = xy$) again gives a planar graph. Now in the drawing of $(G/e) - \{v_{xy}\}$, let f be the face that contains the removed vertex v_{xy} , and let C be the boundary of f . Since G/e is 3-connected, $(G/e) - \{v_{xy}\}$ is 2-connected, and so C is the edge set of a cycle.⁶

Let X be the set of neighbours of x in G except for y , then $X \subseteq V(C)$. Symmetrically, let Y be the set of neighbours of y in G except for x , then $Y \subseteq V(C)$. Taking our drawing of G/e and removing all the edges $\{v_{xy}, w\}$ for $w \in Y \setminus X$, we get an embedding of $G - \{y\}$, replacing the vertex v_{xy} by x . Our goal is to show that to this embedding of $G - \{y\}$, we can add y and all its edges back in without destroying planarity, which gives a drawing of G .

So go along the cycle C and enumerate the vertices in X as $x_1, \dots, x_r$ along the cycle C . Observe that these vertices partition the cycle C into paths P_i that go from x_i to x_{i+1} ($i = 1, \dots, r$ with the convention that $x_{r+1} = x_1$). We want to show that all vertices in Y lie on the same path P_i , because then we can draw y inside the face bounded by P_i and the edges $x_i x$ and $x x_{i+1}$, and join it by arcs to all of its neighbours in Y without destroying planarity. So suppose for a contradiction that not all neighbours of y lie on the same path P_i . There are three cases:

- Suppose that $Y \setminus X \neq \emptyset$, which just means that there is a $y_1 \in Y \setminus X$ which lies on the interior of some P_i , and suppose that there is another $y_2 \in Y$ that does not lie on P_i . Then $\{x, y_1, y_2\}$ and $\{y, x_i, x_{i+1}\}$ are vertices of a topological $K_{3,3}$ -minor, a contradiction.

⁶Proposition 4.2.6 in Diestel

- Suppose that $Y \setminus X = \emptyset$, that is, $Y \subset X$, and furthermore that $|Y| \geq 3$. Then let x_i, x_j, x_k be three distinct elements of $Y \subset X$. But now $\{x, y, x_i, x_j, x_k\}$ form the vertex set of a topological K_5 -minor, again a contradiction.
- In the final case, suppose $Y \subset X$ and $|Y| = 2$ (note that Y cannot have fewer than 2 elements, as $\deg(y) \geq 3$ because G is 3-connected, and x is another (the third) neighbour of y). Let x_i and x_k be the elements of $Y \subset X$, and suppose they do not lie on the same path. Then there must be vertices x_j and x_ℓ between them, so that the vertices appear in the order x_i, x_j, x_k, x_ℓ along C . But now $\{x_i, x, x_k\}$ and $\{x_j, y, x_\ell\}$ are the vertices in a topological $K_{3,3}$ -minor, a contradiction.

□

Lecture 12 The chromatic number

Imagine you are hosting a party, and you want to make a seating plan for your guests. But some of your guests really don't get along, and if you place such a pair of people at the same table, things are going to escalate and spoil your party. So any two guests with a conflict must be seated at different tables. The question is, how many different tables do you need in order to do this?

You can model your guests and their conflicts as a graph, where every person is a vertex and you draw an edge between them if they have a conflict. You encode your table assignments as vertex colours (e.g., all the people with red vertices sit together), and the question is now, how many colours (= tables) do you need so that no two endpoints of an edge have the same colour? This is called the *chromatic number* of your network of conflicts.

Definition 113. Let $G = (V, E)$ be a finite simple graph. A (*proper*) k -colouring of G is a map $V \rightarrow \{1, \dots, k\}$, where we think of the numbers $1, \dots, k$ as colours, so that no two adjacent vertices are mapped to the same colour. If G has a proper k -colouring, we call G k -colourable. The *chromatic number* of G , written as $\chi(G)$, is the smallest integer k so that G is k -colourable.

Unless otherwise stated, proper colourings will be the only ones we consider, and so we just call them colourings. (This is also how the term 'colouring' is normally used in graph theory.)

Example 114. We make some simple observations. For any graph G on n vertices, we have

$$1 \leq \chi(G) \leq n.$$

Furthermore, $\chi(G) = 1$ iff G doesn't contain any edges, and $\chi(G) = n$ iff $G = K_n$. Moreover,

$$\chi(C_n) = \begin{cases} 2 & \text{if } n \text{ is even} \\ 3 & \text{if } n \text{ is odd} \end{cases}$$

The chromatic number is closely linked to two other simpler graph parameters: The independence number $\alpha(G)$ and the clique number $\omega(G)$.

Definition 115. Let $G = (V, E)$ be a finite simple graph. A vertex set $S \subseteq V$ is *independent* if there are no edges between any two vertices of S . The *independence number* of G , denoted $\alpha(G)$, is the maximum size of an independent set in G .

Definition 116. Let $G = (V, E)$ be a finite simple graph. A vertex set $C \subseteq V$ is a *clique* if the induced subgraph $V[C]$ is complete. The *clique number* of G , denoted $\omega(G)$ is the largest integer k such that there exists a clique of size k in G .

You may think of a k -colouring as a vertex partition

$$V = S_1 \uplus S_2 \uplus \cdots \uplus S_k$$

where all the sets S_i are independent (just take as sets S_i all the vertices of the same colour). The sets S_i are called the *colour classes* of the colouring. This observation gives us another perspective on the chromatic number of G , which we write down as a lemma.

Definition 117. We call a finite simple graph G *k-partite* if it has a vertex partition

$$S_1 \uplus S_2 \uplus \cdots \uplus S_k$$

so that all the sets S_i are independent (or equivalently, so that no edge of G has both of its endpoints in the same set S_i). Note that some of the sets S_i may be empty.

Lemma 118. *A graph $G = (V, E)$ has a k -colouring if and only if it is k -partite. The chromatic number $\chi(G)$ is the smallest integer k so that G is k -partite.* $\square$

Deciding whether a connected graph G is 2-colourable (i.e., whether G is bipartite) is easy: Start with any vertex v and give it any of the two colours, and then work from there: all neighbours of v have to get the other colour, all of their neighbours are forced to get the first colour again, and so on: either we get a 2-colouring of all vertices by this procedure, or at some point we can't continue and the graph is not 2-colourable.

We also have the following criterion for 2-colourability.

Lemma 119. *A finite simple graph is 2-colourable (i.e., bipartite) if and only if it contains no cycles of odd length.*

Proof. If $G = (V, E)$ is 2-colourable, then going along any cycle in G the colours are alternating, and so the cycle length must be even.

Conversely, suppose G contains no odd length cycles. We use induction on the number $|V|$ of vertices. If $|V| = 1$ clearly G is 2-colourable, so suppose $|V| \geq 2$ and we have proved the claim for all graphs with fewer vertices. We can assume that G is connected (otherwise colour the connected components individually), and then there is a vertex v so that $G - \{v\}$ is still connected (pick a leaf of a spanning tree as v). By induction we can 2-colour $G - \{v\}$.

If all neighbours of v receive the same colour under this colouring, then we can give v the other colour, and this gives a 2-colouring of the whole graph G . So suppose that v has two neighbours x, y that have different colours. Since $G - v$ is connected, there is a path P from x to y within $G - v$, and the colours have to alternate along P . But since x, y have different colours,

that means the length of P is odd, and so the cycle given by P and v (with the edges vx, vy) has odd length, a contradiction. So all neighbours of v did get the same colour, and we have a 2-colouring of G . $\square$

So the question ‘Is $\chi(G) \leq 2$?’ is easy, but in general calculating $\chi(G)$ is hard (even the question ‘is $\chi(G) \leq 3$?’ is very hard). We can give some upper and lower bounds though, and start with two lower bounds:

Lemma 120. *For a finite simple graph G with n vertices, we have*

$$\chi(G) \geq \omega(G) \quad \text{and} \quad \chi(G) \geq \frac{n}{\alpha(G)}.$$

Proof. If G contains a clique on k vertices, then you need at least k different colours to colour the vertices of this clique, and so $\chi(G) \geq \omega(G)$. For the second statement, observe that a k -colouring of G partitions V into k independent sets $S_1, \dots, S_k$, and for each of these sets, $|S_i| \leq \alpha(G)$. So $n = |V| = |S_1| + |S_2| + \dots + |S_k| \leq k\alpha(G)$, which gives $k \geq n/\alpha(G)$. So $\chi(G) \geq n/\alpha(G)$. $\square$

If we want *upper* bounds on the chromatic number, we usually have to construct a colouring. One useful tool is the following algorithm:

Greedy colouring algorithm: Let $G = (V, E)$ be a finite simple graph with an arbitrary order $v_1, v_2, v_3, \dots$ of the vertices in V . Order the possible colours (in our definition of a colouring, the colours are just numbers in $\mathbb{N}$) in the usual way, $1 < 2 < 3 < \dots$. Now go through the vertices one by one in the given order. Give the vertex v_i the smallest colour (with respect to $<$) which has not been used already to colour any neighbours of v_i in $\{v_1, v_2, \dots, v_{i-1}\}$. Then continue with the next vertex.

The number of colours required by the greedy colouring algorithm strongly depends on the vertex order, and can be much larger than the actual chromatic number of the graph. However, we get the following estimate:

Lemma 121. *For a finite simple graph $G = (V, E)$ with maximum degree Δ , we have*

$$\chi(G) \leq \Delta + 1.$$

Proof. Take any vertex order and run the greedy colouring algorithm. When a vertex v is about to be coloured, it has at most Δ already coloured neighbours, which used up at most Δ different colours. So even in the worst case, v can be coloured with colour number $\Delta + 1$, and so $\chi(G) \leq \Delta + 1$. $\square$

Moreover, the greedy algorithm can be optimal if the vertex order is chosen correctly:

Lemma 122. *Let $G = (V, E)$ be a finite simple graph with chromatic number $\chi(G)$. Then there exists a vertex order such that the greedy colouring algorithm with this order requires exactly $\chi(G)$ colours.*

Proof. Fix a $\chi(G)$ -colouring of G , and order the vertices according to their colours: The vertices with colour 1 go first, then the vertices with colour 2, etc. Let $c(v)$ be the colour v had in the original colouring, and $c_g(v)$ the colour it is assigned in the greedy colouring with the given order. Then because every colour class is an independent set, we can show by induction over the vertex order that $c_g(v) \leq c(v)$. So the greedy colouring algorithm will not use more than $\chi(G)$ colours (and of course it will not use less either because it constructs a colouring of G). $\square$

In Example 114, we saw two examples where equality holds, i.e., $\chi(G) = \Delta + 1$: The complete graphs ($\chi(K_n) = n$ and $\Delta = n$) and cycles of odd length ($\Delta = 2$ and $\chi(C_{2k+1}) = 3$). As it turns out, those are the only connected graphs for which this is true:

Theorem 123 (Brooks, 1941). *Let $G = (V, E)$ be a finite simple connected graph with maximum degree Δ . Then $\chi(G) \leq \Delta$, unless G is complete or a cycle of odd length.*

The first part of the proof is the following lemma:

Lemma 124. *Let $G = (V, E)$ is a finite simple connected graph with maximum degree Δ . If there is a vertex v with $\deg(v) < \Delta$, then $\chi(G) \leq \Delta$.*

Proof. Explore the graph using *breadth-first search*, starting from v : List the neighbours of v in some order, and mark them as ‘discovered’. Then go to the first vertex in this list, add all of its previously undiscovered neighbours to the end of the list and mark them as discovered. Delete the current vertex from the list and repeat this procedure with the next vertex on the list, until the list is empty. Since the graph is connected, every vertex will be visited by this process at some point. The order in which the vertices are visited is called the *breadth-first-ordering* of V starting from v .

Now order the vertices in *reverse* of the breadth-first ordering, say $v_n, v_{n-1}, v_{n-2}, \dots, v_2, v_1 = v$. Then every vertex v_i , other than $v_1 = v$, as at least one neighbour coming after it in the list (because from that neighbour is how v_i got added to the breadth-first ordering), and so at most $\Delta - 1$ neighbours which come before v_i in the list. Since $\deg(v_1) \leq \Delta - 1$, the last statement is also true for $v_1 = v$. Now use the greedy colouring algorithm with respect to this ordering, then for all $1 \leq i \leq n$, at most $\Delta - 1$ neighbours of v_i are coloured before v_i . So Δ colours are enough. $\square$

So we only need to prove Brooks’s theorem for regular graphs.

Definition 125. A finite simple graph $G = (V, E)$ is called k -regular if $\deg(v) = k$ for all $v \in V$.

Proof of Brooks's theorem. Let $G = (V, E)$ be a finite simple connected and k -regular graph (so $\Delta = k$) which is neither complete nor a cycle of odd length. For $k = 1$ there is no such graph. For $k = 2$, the only graphs satisfying these conditions are cycles of even length, and for those graphs we have $\chi(G) = 2 = k$. So we may assume $k \geq 3$, and that there are at least 4 vertices.

Case 1: Assume G has connectivity $\kappa(G) = 1$. Then there is some vertex v and two non-empty vertex sets $V_1, V_2 \subseteq V$ with $V = \{v\} \uplus V_1 \uplus V_2$ so that removing v from G separates V_1 from V_2 . Now let $G_1 = G[V_1 \cup \{v\}]$ and $G_2 = G[V_2 \cup \{v\}]$. Observe that $\Delta(G_1) = \Delta(G_2) = k$, but $\deg_{G_1}(v), \deg_{G_2}(v) < k$. So by Lemma 124 there are k -colourings of both G_1 and G_2 . Since the colours in a colouring can be permuted arbitrarily, we can assume that both colourings give colour 1 to the vertex v . But then the colourings can be combined to give a k -colouring of G , and so $\chi(G) \leq k = \Delta$.

Case 2: Assume $\kappa(G) = 2$, and let $\{u, v\}$ be a minimal separating set. Analogously to case 1, let V_1, V_2 be non-empty vertex sets with $V = \{u, v\} \uplus V_1 \uplus V_2$ so that removing u and v separates V_1 and V_2 . Similarly as in case 1, set $G_i = G[V_i \cup \{u, v\}]$ for $i = 1, 2$, then both graphs G_i are k -colourable. If in both graphs G_1, G_2 , at least one of the vertices u, v has degree at most $k - 2$, then we can (if necessary) change the k -colourings of G_1 and G_2 so that they give different colours to u and v . Permuting the colours, we can ensure that those colours match in the two colourings, and then we can once again glue together the colourings to give a k -colouring of G .

Otherwise, for one of the two graphs, say G_2 , both $\deg_{G_2}(u) \geq k - 1$ and $\deg_{G_2}(v) \geq k - 1$, and consequently (since both u, v must have neighbours in both V_1 and V_2 , otherwise $\kappa(G) = 1$) $\deg_{G_1}(u) = \deg_{G_1}(v) = 1$.

Then u, v cannot both have the same neighbour $x \in G_1$: either $|V_1| \geq 2$, but then removing x would disconnect G , contradicting $\kappa(G) = 2$, or $|V_1| = \{x\}$, but then $\deg(x) = 2$, contradicting $k \geq 3$. So v has a unique neighbour v' in G_1 that is not a neighbour of u . But then u and v' form another minimal separating vertex set, where (after redefining G_1, G_2 accordingly) $\deg_{G_1}(u) = \deg_{G_2}(v') = 1$. But now, as $1 \leq k - 2$, we are in the previous case where we can manipulate and permute the k -colourings of G_1 and G_2 so that they can be glued together to a k -colouring in G .

Case 3: For $\kappa(G) \geq 3$, let $u, v, w \in V$ be distinct vertices so that u and v are not neighbours, but w is a neighbour of both u and v (such vertices must exist, otherwise G is complete). Then G with u and v removed is still connected, otherwise case 2 would apply. In this connected graph with u and v removed, construct the breadth-first-ordering starting from $w = v_1$, and append it

with the vertices $v_{n-1} = v, v_n = u$. Then run the greedy colouring algorithm with the reverse vertex order order $u = v_n, v = v_{n-1}, v_{n-2}, \dots, v_2, v_1 = w$. Then u and v will both be assigned colour 1, and for all v_i with $2 \leq i \leq n-2$ there are strictly fewer than k neighbours already coloured when it is time to colour v_i , (by construction of the breadth-first ordering), and for v_1 , two of the k neighbours (u and v) share a colour. So k colours suffice. $\square$

Lecture 13 More on the chromatic number

We say last time in Lemma 120 that $\chi(G) \geq \omega(G)$ — if there is a clique of size k , then in any colouring the vertices of this clique get distinct colours, so the chromatic number is at least k . How far from the truth can this simple lower bound be? That is, suppose that $\omega(G) \leq k$, then how large can $\chi(G)$ possibly be? Sadly, the following holds:

Theorem 126. *For any integer n , there exists a finite simple graph $G_n = (V_n, E_n)$ that does not contain a triangle (and so $\omega(G_n) \leq 2$) such that $\chi(G_n) = n$.*

Proof. For $n \in \{1, 2\}$, take $G_n = K_n$. For $n \geq 3$, we use the following construction:

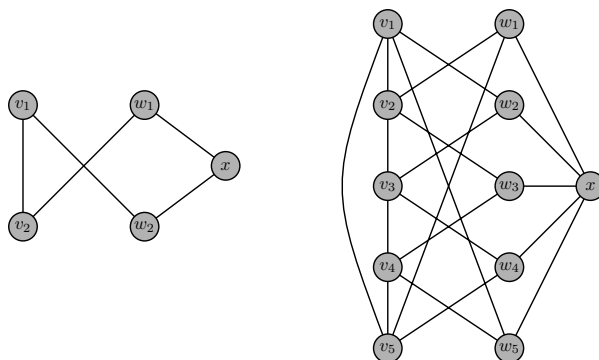
Let $G = (V, E)$ be a finite simple graph. Denote its vertices by $V = \{v_1, \dots, v_m\}$. Introduce further distinct vertices $w_1, \dots, w_m$ and x . The *Mycielskian* $M(G)$ of G is the finite simple graph on the $2m + 1$ vertices $\{v_1, \dots, v_m, w_1, \dots, w_m, x\}$ constructed in the following way: On the vertices $v_1, \dots, v_m$, take a copy of G , that is $M[\{v_1, \dots, v_m\}] = G$. For each vertex $w_i, i = 1, \dots, m$, draw an edge to a vertex v_j iff v_j is a neighbour of v_i . Finally, draw an edge between x and w_i for every $i = 1, \dots, m$.

Lemma 127. *Let G be a finite simple graph.*

- (i) *If G is triangle-free, then so is $M(G)$.*
- (ii) *If $\chi(G) = k$, then $\chi(M(G)) = k + 1$.*

This lemma allows us to conclude the proof: Since K_2 is triangle-free with chromatic number 2, $M(K_2) \cong C_5$ is triangle-free with chromatic number 3, and by iterating Mycielski's construction, we obtain triangle-free graphs of arbitrarily high chromatic number. $\square$

The following picture shows $M(K_2)$ and $M(C_5)$.



Proof of Lemma 127. (i) Let $G = (V, E)$ be a triangle free graph on vertices $v_1, \dots, v_m$. Using the notation from the proof of Theorem 126, we note that $\{w_1, \dots, w_m\}$ is an independent set in $M(G)$. So if there is a triangle in $M(G)$, it cannot contain more than one vertex from $\{w_1, \dots, w_m\}$ (and so it also cannot contain the vertex x). As G is triangle-free, the only remaining possibility for a triangle is to be spanned by vertices v_i, v_j, v_k for some indices i, j, k . Then $\{v_i, v_j\} \in E$, but by construction of $M(G)$, v_i and v_j must also both be neighbours of v_k , but then $G[\{v_i, v_j, v_k\}]$ is a triangle in G , a contradiction. So $M(G)$ is triangle-free.

(ii) Suppose that $G = (V, E)$ has chromatic number k , and fix a k -colouring of G . The construction of $M(G)$ enables us to extend this colouring to $M(G)$ by giving w_i the same colour as v_i . Give x the new colour $k + 1$, then we have constructed a $(k + 1)$ -colouring of $M(G)$. It remains to show that $M(G)$ is not also k -colourable.

So assume for a contradiction that $M(G)$ has a k -colouring. By permuting the colours, we can assume that the vertex x has colour k . Then the vertices $w_1, \dots, w_m$ must be coloured with colours from $\{1, \dots, k - 1\}$. Let $A \subseteq \{v_1, \dots, v_m\}$ be the vertices that are assigned colour k . Now consider the $(k - 1)$ -colouring of the vertices $\{v_1, \dots, v_m\}$ given by changing the colour of each $v_i \in A$ to the colour of w_i (and leaving the remaining colours unchanged). Does this produce a proper colouring of G ? We need to check that no $v_i \in A$ is coloured the same as any of its neighbours in $\{v_1, \dots, v_m\}$. The set A is independent as it was a subset of a colour class in $M(G)$, so any neighbours of a vertex $v_i \in A$ still have their previous colours. Moreover, any neighbour of v_i in G is a neighbour in $M(G)$ of w_i , so any neighbour of v_i must have a colour different from the colour of w_i — so indeed we have a proper $(k - 1)$ -colouring of G , which contradicts $\chi(G) = k$. $\square$

So Theorem 126 states that a graph might have arbitrarily large chromatic number, even if it does not contain any large cliques. Similarly, one could try to outlaw the existence of short cycles: Given $k, \ell \in \mathbb{N}$, is there a graph with $\chi(G) > k$ and no subgraph $C_3, C_4, \dots, C_\ell$? Surprisingly:

Theorem 128 (Erdős, 1959). *For any integers k, ℓ , there exists a finite simple graph G with $\chi(G) > k$ and such that G contains no cycles of length $\leq \ell$.*

The proof of this result is a famous early application of the *probabilistic method*, which we will discuss in a later lecture.

Let us now turn to the problem of colouring *planar* graphs. This problem goes back to the 1850s, when Francis Guthrie tried to colour a map of the counties of England so that neighbouring counties get different colours, and noticed that 4 colours were enough. It's easy to come up with examples of maps that can't be coloured with fewer than 4 colours (for example, a map

of four countries where one country is in the middle and the other three surround it). So Guthrie asked: Is it possible to colour any such map with 4 colours?

In the language of graph theory, this is just asking: If G is a planar graph, is $\chi(G) \leq 4$? Indeed, if you start with a map (with reasonably shaped boundaries, for example arcs), you can put a vertex inside each country and connect two vertices if their countries are neighbours, then the resulting graph G is planar, and $\chi(G) \leq 4$ if and only if the map can be coloured with 4 colours. (Note that G is the planar dual, as in Definition 105, of the graph that is given naturally by the country borders. In particular, starting with some planar G , we could also construct a map that 4-colouring G is equivalent to.)

This problem turned out to be extremely resistant to any attempts of solving it (although there were multiple wrong ‘proofs’), until it was finally resolved more than 200 years later: In 1976 Appel and Haken reduced the theorem to ‘only’ 1936 cases, which they checked with a computer. This was the first time computer assistance was used in the proof of a major theorem, and the proof at the time was not universally accepted (since nobody could manually go through every line of the proof and check its correctness). The proof is now widely accepted (and updated versions of the proof, where some inaccuracies were removed and the number of cases to check was reduced, have since appeared).

Theorem 129 (Four Colour Theorem). *Any planar graph is 4-colourable.*

While proving the Four Colour Theorem is extremely hard, we will be able to prove the following ‘five colour theorem’ within this lecture.

Theorem 130 (Heawood 1891). *Any planar graph is 5-colourable.*

First, an easy lemma:

Lemma 131. *Every planar graph $G = (V, E)$ has a vertex of degree at most 5.*

Proof. We can assume $|V| \geq 3$, and then by Corollary 107, $|E| \leq 3|V| - 6$. By the handshake lemma, $\sum_{v \in V} \deg(v) = 2|E| \leq 6|V| - 12$. So for the average degree,

$$\frac{1}{|V|} \sum_{v \in V} \deg(v) \leq 6 - \frac{12}{|V|} < 6,$$

and so there must be at least one vertex v with $\deg(v) \leq 5$. □

Proof of Theorem 130. We use induction on $|V|$. For $|V| = 1$ we have $\chi(G) = 1$, so assume that for some n , we have proved $\chi(G) \leq 5$ for all planar graphs with $|V| = n$. Let G be a planar graph with $|V| = n + 1$.

Let v be a vertex with minimal degree, then by Lemma 131, $\deg(v) \leq 5$. Let $G' = G - \{v\}$. Then G' is planar and by the induction hypothesis has $\chi(G') \leq 5$.

Case 1: If $\deg(v) \leq 4$, then any 5-colouring of G' can be extended to a 5-colouring of G by giving v a colour that is not used among its neighbours.

Case 2: If $\deg(v) = 5$, and any two neighbours of v have the same colour, then we can repeat the argument from case 1. So assume that all 5 neighbours of v have different colours. Fix a planar embedding of G and call the neighbours $v_1, \dots, v_5$ in a counterclockwise order around v . Suppose vertex v_i has (w.l.o.g.) colour i for $i = 1, \dots, 5$. Let $V_{1,3}$ be the set of all vertices in G' that are coloured either 1 or 3, and let $C_{1,3}^{(1)}$ be the connected component of $G'[V_{1,3}]$ that contains v_1 . Similarly, let $C_{1,3}^{(3)}$ be the connected component of $G'[V_{1,3}]$ that contains v_3 . If $C_{1,3}^{(1)} \neq C_{1,3}^{(3)}$ then the two components have to be disjoint, and we can swap the colours 1 and 3 in one of them, to the effect that v_1 and v_3 end up with the same colour. Then v is only adjacent to vertices with only 4 different colours, and we can again argue like in case 1.

So assume $C_{1,3}^{(1)} = C_{1,3}^{(3)}$, in which case there is a path from v_1 to v_3 using only vertices with colours 1 and 3. This path, together with the edges $\{v_1, v\}, \{v, v_3\}$ and the vertex v form a cycle C , which in the embedding of G divides $\mathbb{R}^2$ into an inside and an outside. However, because G is planar, v_2 and v_4 lie in different components of $\mathbb{R}^2 \setminus C$. Now repeat the same construction as with colours 1 and 3 for colours 2 and 4. Because of planarity, we cannot divide up the plane $\mathbb{R}^2$ by another cycle with vertices of colours 2 and 4, and so the components $C_{2,4}^{(2)}$ and $C_{2,4}^{(4)}$ have to be disjoint. So we can recolour one of them by swapping colours 2 and 4. Then v is once again only adjacent to vertices of 4 different colours. $\square$

Sadly, the existence of planar graphs of minimum degree 5 prevents a modification of this argument into a proof of the Four Colour Theorem.

Lecture 14 Edge colourings and Ramsey theory

How about colouring edges instead of vertices? Then the condition ‘no neighbours have the same colour’ becomes ‘no two edges that share an endpoint have the same colour’.

Definition 132. Let $G = (V, E)$ be a finite simple graph. A (*proper*) k -edge-colouring is a map $E \rightarrow \{1, \dots, k\}$ such that no two edges with a common endpoint are assigned the same colour. The *chromatic index* or *edge-chromatic number*, denoted $\chi'(G)$, is the smallest integer k for which a proper k -edge-colouring exists.

Any proper edge-colouring partitions the edge set into the sets of edges with the same colour (the ‘colour classes’), and by definition, each colour class is a *matching*. So, similarly to how vertex colourings were closely related to independent sets, edge colourings are related to matchings.

If Δ is the maximum degree of G , then

$$\chi'(G) \geq \Delta.$$

To see why, look at a vertex v of degree Δ : all edges with endpoint v must get distinct colours, so we have to use at least Δ different colours. It is a classical result that this lower bound is attained for bipartite graphs:

Theorem 133 (König’s line-colouring theorem, 1916). *For every finite simple bipartite graph $G = (V, E)$ with maximum degree Δ , we have $\chi'(G) = \Delta$.*

Proof. We use induction on the number $m := |E|$ of edges. For $m = 0$ there is nothing to show. So assume that the claim is true for all graphs with $m \geq 0$ edges, and consider a graph G with $m + 1$ edges.

Let Δ be the maximal degree of this G , and fix an arbitrary edge vw from G . Let G' be the graph obtained from G by deleting the edge vw . By the induction hypothesis, there is a proper Δ -edge-colouring of G' . Moreover, each of v, w are incident to at most $\Delta - 1$ edges in G' . So there are colours i and j so that v is not incident to an edge of colour i and w is not incident to an edge of colour j . If $i = j$, then we can colour the edge vw by i , and that’s a Δ -edge-colouring of G .

So suppose $i \neq j$. Let A, B be the vertex bi-partition of G so that $v \in A, w \in B$. Now consider some trail in G' that starts at v and only uses edges of colours i and j . (Recall that a trail cannot repeat edges.) Since v is not incident to an edge of colour i , the first edge on any such trail has to be of colour j . Moreover, the colours i and j alternate along the trail. Furthermore, moving along this trail will alternate between vertices from A and B , where an edge of colour j will be occur exactly when moving from A to B . No such

trail can contain the vertex w , since w is not incident to a vertex of colour j . Now let $C_{i,j}$ be the component that contains v of the subgraph of G' that is given by all edges of colour i or j . Then we just showed that $w \notin C_{i,j}$, so we can swap colours i and j in $C_{i,j}$ without affecting w . After swapping, we have a colouring of G' in which both v, w are not incident to an edge of colour j , so we can extend the colouring as we did in the first case. $\square$

Besides having $\chi'(G) = \Delta$, what else can happen? Not much:

Theorem 134 (Vizing, 1964). *Let $G = (V, E)$ be a finite simple graph with maximum degree Δ . Then either $\chi'(G) = \Delta$ (in which case G is said to be class 1) or $\chi'(G) = \Delta + 1$ (in which case G is said to be class 2).*

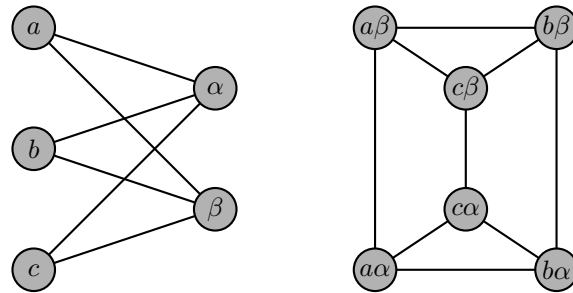
There is no known complete characterisation that distinguishes between class 1 and class 2 graphs. Let's look at an example instead:

Example 135. The complete graphs K_n are of class 1 if n is even, and of class 2 if n is odd. This is because the edge set of a K_{2k} is a disjoint union of $\Delta = 2k - 1$ perfect matchings, each containing k edges⁷, and we can assign one colour to each matching. For K_{2k-1} , after choosing which edges to colour with 1, there will always be at least one vertex remaining with all $\Delta = 2k - 2$ incident edges being uncoloured. So Δ colours cannot be enough.

One final remark about $\chi'(G)$: When considering the *line graph* of a given graph, we can interpret edge colourings as just a special case of vertex colourings.

Definition 136. Let $G = (V, E)$ be a finite simple graph. Its *line graph* $L(G)$ has vertex set E , where two vertices e_1, e_2 are adjacent if and only if the edges e_1 and e_2 shared an endpoint in G .

As an example, here is $K_{2,3}$ and its line graph:



⁷Here's one construction of $2k - 1$ such matchings: Arrange the vertices as the $2k - 1$ endpoints $v_1, \dots, v_{2k-1}$ of a regular $(2k - 1)$ -polygon, with one vertex v^* in the center. Now pick any of the vertices v_i , and take the edge v^*v_i and all $k - 2$ edges between the other vertices which are perpendicular to the edge v^*v_i . This gives a perfect matching, and every edge of K_{2k} appears in exactly one such matching, with $2k - 1$ choices for the vertex v_i .

An edge colouring of G corresponds exactly to a vertex colouring of $L(G)$, and in particular

$$\chi'(G) = \chi(L(G)).$$

A little bit of Ramsey theory. We now look at a completely different type of edge colouring.

Suppose you have a room full of n people. Can you always find a group of k people so that they all know each other, or all are strangers to each other? Or, starting with k , how large does n have to be to guarantee a group of k such people in any room of n people?

In graph theoretic language, you have a complete graph K_n , and you give each edge one of two colours, say red and blue. (Edges with a common endpoint don't have to get different colours anymore.) Can we find a clique of size k so that all edges between them have the same colour (all red, or all blue)?

Definition 137. Let $k \in \mathbb{N}$. The k -th Ramsey number $R(k)$ is defined to be the smallest integer n such that every red-/blue-colouring of K_n contains a monochromatic K_k .

Are we sure such numbers $R(k)$ even exists? Yes, by Ramsey's theorem (the upper bound was given by Erdős and Székereš):

Theorem 138 (Ramsey 1930, Erdős-Székereš 1935). *For any $k \geq 2$, we have $R(k) \leq 2^{2^{k-1}}$. In particular, Ramsey numbers exist.*

The motto of Ramsey theory is often phrased as:

Complete disorder is impossible.

For example, in any red-blue colouring of K_n , no matter how chaotic and disordered it looks, we can always find a small highly structured part: a clique of size k which is either all-red or all-blue.

Proof of Theorem 138. Set $n = 2^{2^{k-1}}$, and consider any red-blue colouring of the vertices of K_n . Now consider the following construction of vertices p_i (called pivots), vertex sets V_i , and of a string S consisting of letters 'R' and 'B':

1. Set $V_0 = V$, and pick any vertex from V_0 as the pivot p_1 . Start with the empty string S .
2. For $i = 1, 2, \dots$, suppose you already have V_{i-1} and p_i . Now consider the sets

$$\begin{aligned} V_i^R &= \{v \in V_{i-1} \mid v \neq p_i, \text{ the edge } p_i v \text{ is coloured red}\}, \\ V_i^B &= \{v \in V_{i-1} \mid v \neq p_i, \text{ the edge } p_i v \text{ is coloured blue}\}. \end{aligned}$$

(That is, the ‘red neighbourhood’ and the ‘blue neighbourhood’ of p_i within V_{i-1} .) Pick the *larger* one of these two sets (arbitrarily if both have the same size), and set it as V_i . Let p_{i+1} be any vertex from S_i . Append R to the string S if you chose $V_i = V_i^R$, and B otherwise.

3. Repeat this procedure $2k - 3$ times, creating sets $V_0 \supset V_1 \supset V_2 \supset \dots \supset V_{2k-3}$, pivots $p_1, \dots, p_{2k-2}$ and a string S of length $2k - 3$.

Observe that, because in each step we picked the larger neighbourhood, we have $|V_i| \geq 2^{2k-3-i}$ for all i . In particular, $|V_{2k-3}| \geq 2^0 = 1$, so $V_{2k-3} \neq \emptyset$ (so we could choose all the pivots, including the final one p_{2k-2}).

By construction, all edges from a pivot p_i to all the following pivots p_j with $j > i$ have the same colour, and that colour is the i^{th} letter in the word S . As $|S| = 2k - 3$, by the pigeonhole principle, one of the letters R or B, let’s say R, appears at least $k - 1$ times. Let K be the set of pivots p_i so that the letter number i in S is R, as well as the last pivot p_{2k-2} . Then $|K| \geq k$, and all the edges between vertices in K are red. So we have found a monochromatic clique of size at least k . $\square$

So up to subexponential factors, we have $R(k) \leq 4^{k+o(k)}$. The base 4 of the exponent remained the best upper bound for over 80 years (any improvements were only by subexponential factors), until a breakthrough result in 2023 by Campos, Griffiths, Morris and Sahasrabudhe, who showed $R(k) \leq 3.999^k$.

We’ll discuss lower bounds on $R(k)$ later when we talk about the probabilistic method. For now, let’s look at some small examples.

Example 139. We have $R(1) = 1$ and $R(2) = 2$. We’ll show next that $R(3) = 6$.

Proposition 140. *We have $R(3) = 6$, that is, every red/blue-colouring of K_6 contains a monochromatic triangle, but there is a red/blue-colouring of K_5 without a monochromatic triangle.*

Proof. The part about K_5 is left as an exercise. For K_6 , pick any vertex v , then amongst its five neighbours, there must be at least three, call them x, y, z , so that the edges vx, vy, vz all have the same colour, let’s say red. Now consider two cases:

Case 1: If at least one of the edges xy, yz, zx is also red, then together with the vertex v that gives us a red triangle.

Case 2: If all the edges xy, yz, zx are blue, then that gives us the blue triangle x, y, z . $\square$

For higher k , it is known that $R(4) = 18$, but above that no exact numbers are known: We only know $43 \leq R(5) \leq 46$ and $102 \leq R(6) \leq 160$, and

determining exact values for $R(k)$ seems completely out of reach of our current methods.

We conclude this discussion with a quote by Paul Erdős.

“Aliens invade the earth and threaten to obliterate it in a year’s time unless human beings can find the Ramsey number for red five and blue five [= $R(5)$]. We could marshal the world’s best minds and fastest computers, and within a year we could probably calculate the value. If the aliens demanded the Ramsey number for red six and blue six, however, we would have no choice but to launch a preemptive attack.”

Lecture 16 The probabilistic method and random graphs

The next two lectures will be about probabilistic combinatorics / probabilistic graph theory. This can mean two things: (1) Using randomness as a tool to prove deterministic statements. (2) To study the properties of randomly generated graphs for their own sake. We start with an example for (1), and then define random graphs (which we will use for both (1) and (2)).

The probabilistic method and Ramsey numbers

A simple inequality from probability is the *union bound*: If we have two events A, B , then $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$. More generally, for n events $A_1, \dots, A_n$,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

To see the union bound in action, we study Ramsey numbers again. Recall that in the last lecture, we showed that $R(k) \leq 2^{2k-3}$, which is exponential in k . We will now prove an exponential lower bound, which is an early (and perhaps the most well-known) application of the probabilistic method.

Theorem 141 (Erdős, 1947). *Suppose that n, k are integers such that $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$, then $R(k) > n$. In particular, if $k \geq 3$,*

$$R(k) \geq 2^{\frac{k}{2}}.$$

Proof. To prove that n is a lower bound for $R(k)$, we need to show there is a red/blue colouring of the edges of K_n without any monochromatic K_k . We generate a colouring *randomly*: we colour every edge red with probability $1/2$ and blue with probability $1/2$, independently for different edges.

A key observation is this: If the probability that there is no monochromatic K_k is greater than 0, then there must be *at least one* colouring where this is true. So let's try to calculate this probability. Order the $\binom{n}{k}$ copies of K_k in K_n in some arbitrary way, and let A_i be the event that the copy number i is monochromatic, $i = 1, \dots, \binom{n}{k}$. Then

$$\mathbb{P}(A_i) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{k}{2}} = 2^{1-\binom{k}{2}},$$

because there are 2 choices for the colour red/blue, and the probability that all the edges of the copy of K_k come out as this colour is $(\frac{1}{2})^{\binom{k}{2}}$.

The probability that there is at least one monochromatic K_k can be bounded with the union bound:

$$\mathbb{P}(\exists \text{ monochromatic } K_k) = \mathbb{P}\left(\bigcup_{i=1}^{\binom{n}{k}} A_i\right) \leq \sum_{i=1}^{\binom{n}{k}} \mathbb{P}(A_i) \leq \binom{n}{k} 2^{1-\binom{k}{2}} < 1.$$

So in our random colouring, the probability that there is no monochromatic K_k is greater than 0. As argued above, this means that there is at least one such ‘good’ colouring, and it follows that $R(k) > n$.

For the second statement, let $n = \lfloor 2^{k/2} \rfloor$. Then

$$\binom{n}{k} 2^{1-\binom{k}{2}} \leq \frac{n^k}{k!} 2^{1-\binom{k}{2}} \leq \frac{2^{k^2/2}}{k!} 2^{1-k^2/2+k/2} = 2 \frac{2^{k/2}}{k!},$$

which is < 1 for $k \geq 3$ (for example by induction). So $R(k) \geq 2^{k/2}$ by the first part of the theorem. $\square$

The probabilistic argument leads to a strange situation: Because the probability that a random colouring is good is greater than 0, we know that at least one good colouring exists. But we have no idea what it looks like, and the proof gives no clue on how to generate one algorithmically (beyond the brute-force approach of trying out every possible colouring, which quickly becomes unfeasible). The best *explicit* construction of a colouring only gives the lower bound $R(k) \geq 2^{(\log n)^C}$ for some constant C (this bound grows barely faster than polynomial in k).

The lower bound $2^{\frac{k}{2}}$ can be improved a little bit by a smarter probabilistic argument but not by much, in particular for the base of the exponent we only have $R(k) \geq \sqrt{2}^{k+o(k)}$. Improving this to any constant greater than $\sqrt{2}$ would be a major breakthrough, similar to the 2023 breakthrough we mentioned last time which improved the upper bound from $4^{k+o(k)}$ to 3.999^k (which has since been optimised to 3.8^k).

The Erdős-Rényi random graph

Let’s now study random graphs. Take n vertices, and let $N = \binom{n}{2}$ be the number of potential edges between them. For each of the edges, flip a coin, and include the edge in your graph if the coin comes up heads, and don’t include it if it comes up tails. This could be a fair coin (edge probability $1/2$), but more generally we think of a biased coin (some edge probability $p \in [0, 1]$).

This produces the classic *Erdős-Rényi random graph* $G(n, p)$: More formally, we have n vertices, and include each of the $N = \binom{n}{2}$ potential edges between

them with probability $p \in [0, 1]$, independently from all other edges.⁸

What the resulting graph is likely to look like depends on the parameters. Of course $G(n, p)$ is not one graph, but rather a probability distribution on the set of all possible graphs on n labelled vertices. If $G = (V, E)$ is a fixed labelled graph on n vertices, then the probability that the random graph $G(n, p)$ comes out as exactly G is

$$p^{|E|}(1 - p)^{N - |E|},$$

because the edges are chosen independently of each other. In particular, for $p = \frac{1}{2}$, every graph on n labelled vertices is chosen with the same probability $(\frac{1}{2})^N$, and so $G(n, 1/2)$ is just a graph chosen uniformly from all possible graphs on n labelled vertices (that's why, especially in older papers, it is sometimes called 'the random graph on n vertices').

We will use the notation $G \sim G(n, p)$ to say that we consider a graph G which is the result of including every potential edge independently with probability p . (Formally, G is a random variable with the distribution $G(n, p)$.)

Some more simple observations: The number of edges of a graph $G \sim G(n, p)$ is a binomial random variable with parameters $N = \binom{n}{2}$ and p . The degree of a fixed vertex in G has the distribution $\text{Bin}(n - 1, p)$.

The first moment method

The first moment method is a generalisation of the union bound.

Definition 142. The *first moment* of a random variable X is simply its expectation $\mathbb{E}[X]$.

Recall some properties of the expectation: $\mathbb{E}$ is linear, meaning that if a is a constant,

$$\mathbb{E}[aX] = a\mathbb{E}[X],$$

and if X, Y are two random variables,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Importantly, this last equation *always* holds — even if X and Y are not independent. We will often be interested in random variables X that *count*

⁸Technically, the original paper by Erdős and Rényi from 1959 introduced the slightly different model $G(n, m)$, where among n labelled vertices, a set of m edges is chosen uniformly at random from all $\binom{N}{m}$ possible edge sets of that size. The random graph $G(n, p)$ was originally introduced by Gilbert, also in 1959. The random graphs $G(n, m)$ and $G(n, p)$ behave very similarly to each other if $p = m/N$.

something, like the number of triangles in a random graph, and in that case $\mathbb{E}[X]$ takes a particular form: Suppose that

$$X = \sum_{i=1}^k \mathbb{1}_{A_i},$$

where the random variables $\mathbb{1}_{A_i}$ are the *indicators* for some events A_i , that is,

$$\mathbb{1}_{A_i} = \begin{cases} 1 & \text{if } A_i \text{ holds,} \\ 0 & \text{otherwise.} \end{cases}$$

Then X counts how many of the events A_i holds — for example, each A_i could be that a particular copy of a triangle is present in a random graph.

Lemma 143. *If $X = \sum_{i=1}^k \mathbb{1}_{A_i}$ for some events $A_1, \dots, A_k$, then*

$$\mathbb{E}[X] = \mathbb{P}(A_1) + \dots + \mathbb{P}(A_k).$$

Proof. Note that for any event A , $\mathbb{E}(\mathbb{1}_A) = \mathbb{P}(A)$. By linearity of the expectation,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^k \mathbb{1}_{A_i}\right] = \sum_{i=1}^k \mathbb{E}[\mathbb{1}_{A_i}] = \sum_{i=1}^k \mathbb{P}(A_i).$$

□

As an example, consider triangles in $G(n, p)$.

Lemma 144. *Let X be the number of triangles in a random graph $G \sim G(n, p)$, then*

$$\mathbb{E}[X] = \binom{n}{3} p^3.$$

Proof. List all $\binom{n}{3}$ potential triangles in some arbitrary order, and let A_i be the event that the i^{th} triangle is present in G , then $\mathbb{P}(A_i) = p^3$. The claim now follows by Lemma 143. □

Recall Markov's inequality:

Theorem 145 (Markov's inequality). *For a non-negative random variable X and $c > 0$, we have*

$$\mathbb{P}(X \geq c) \leq \mathbb{E}[X]/c.$$

There's an important special case which applies when $X \geq 0$ only takes integer values (for example when X counts something). For this we just take $c = 1$ in Markov's inequality:

Theorem 146 (First moment method). *Let X be a random variable which only takes non-negative integer values. Then*

$$\mathbb{P}(X > 0) = \mathbb{P}(X \geq 1) \leq \mathbb{E}[X].$$

Using this theorem, or more generally Markov's inequality, is called the first moment method. Here is an example, which follows directly from Lemma 144 and Theorem 146.

Corollary 147. *Let $G \sim G(n, p)$. Then the probability that G contains a triangle is at most $\binom{n}{3}p^3$.*

We will revisit this result in the next lecture.

High girth and chromatic number

We'll now use random graphs and the first moment method to obtain a deterministic statement that we mentioned earlier (Theorem 128).

Definition 148. The *girth* $g(G)$ of a finite simple graph G is the shortest length of a cycle in G (where we set $g(G) = \infty$ if there are no cycles).

If G has a large girth, then locally G looks like a tree: If we start at any vertex and explore the graph (by looking at its neighbours, its neighbours' neighbours, and so on), then the structure will look tree-like for a long time because there are no short cycles. Trees have chromatic number ≤ 2 . Surprisingly, however, there are graphs with arbitrarily large girth *and* chromatic number:

Theorem 149 (Erdős, 1959). *For any k and ℓ there exists a graph G with $\chi(G) \geq k$ and $g(G) \geq \ell$.*

The idea of the proof is to consider $G(n, p)$ for suitable n and p , and then show separately that (a) it is very likely that there are only few short cycles, and (b) it is very likely that the chromatic number is large. Then it is likely that properties (a) and (b) *both* hold, and after removing some vertices (to kill the few cycles that exist), we obtain a graph with the required properties.

Proof. We can assume $k, \ell \geq 3$. There are

$$\frac{n(n-1)(n-2)\dots(n-r+1)}{2r}$$

cycles of length r in K_r (we select the r vertices in the order they appear along the cycle, and divide by r because there are r possible starting vertices, and 2 possible directions of the cycle, so that this choice counts in the same

cycle). Each such cycle is present in $G \sim G(n, p)$ with probability p^r , so by Lemma 143, if we let X_r be the number of cycles of length r in G ,

$$\mathbb{E}[X_r] = \frac{n(n-1)(n-2)\dots(n-r+1)}{2r} p^r \leq \frac{n^r p^r}{2r}.$$

Now let X be the number of cycles of length less than ℓ , and set $p = p(n) = n^{-1+1/\ell}$, then⁹

$$\mathbb{E}[X] = \sum_{r=3}^{\ell-1} \mathbb{E}[X_r] \leq \sum_{r=3}^{\ell-1} \frac{n^r p^r}{2r} \leq \sum_{r=3}^{\ell-1} \frac{n^{r/\ell}}{2r} \leq O(n^{(\ell-1)/\ell}) = O(n^{1-1/\ell}) = o(n).$$

So by Markov's inequality (Theorem 145), it is unlikely that there are many 'short' cycles of length less than ℓ :

$$\mathbb{P}(X \geq n/2) \leq \mathbb{E}[X]/(n/2) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Now we consider the chromatic number. Recall (Lemma 120) that for any graph G , $\chi(G) \geq n/\alpha(G)$, where $\alpha(G)$ is the size of the largest independent set in G . So to show that the chromatic number is large, we will show that the independence number is likely relatively small. Let $t = t(n) = \lfloor n^{1-1/(2\ell)} \rfloor$, and let Y count the number of independent vertex sets of size exactly t in $G(n, p)$ where again $p = p(n) = n^{-1+1/\ell}$. Then by Lemma 143, it can be shown (we skip the calculation here) that if n is large enough,

$$\mathbb{E}[Y] = \binom{n}{t} (1-p)^{\binom{t}{2}} \leq \left(\frac{en}{t} n^{-2} \right)^t \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

So by Theorem 146,

$$\begin{aligned} \mathbb{P}(\alpha(G) \geq t) &= \mathbb{P}(G \text{ contains an independent set of size } t) \\ &= \mathbb{P}(Y \geq 1) \leq \mathbb{E}[Y] \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

By the union bound,

$$\mathbb{P}(X \geq n/2 \text{ or } \alpha(G) \geq t) \leq \mathbb{P}(X \geq n/2) + \mathbb{P}(\alpha(G) \geq t) \rightarrow 0.$$

So

$$\mathbb{P}(X < n/2 \text{ and } \alpha(G) \leq t) \rightarrow 1,$$

and so in particular, if n is large enough, this probability is positive and so there exists *some* graph G_0 on n vertices with fewer than $n/2$ short cycles and $\alpha(G_0) < t$.

⁹We use the asymptotic O - and o -notation here: $f(n) = O(g(n))$ means that there is a constant $C > 0$ and an integer n_0 so that $f(n) \leq Cg(n)$ for all $n \geq n_0$ (so, " $f(n)$ is of order at most $g(n)$ "). Furthermore, $f(n) = o(g(n))$ means that $f(n)/g(n) \rightarrow 0$ as $n \rightarrow \infty$.

Now in the final step, we construct a graph G^* from G_0 by deleting one vertex from every short cycle in G_0 . Then G^* has more than $n/2$ vertices, no cycles of length $< \ell$ and so $g(G^*) \geq \ell$, and furthermore $\alpha(G^*) \leq \alpha(G_0) < t$. It follows that

$$\chi(G^*) \geq \frac{n/2}{\alpha(G^*)} \geq \frac{n/2}{t} \geq \frac{n}{n^{1-1/(2\ell)}} = n^{1/(2\ell)},$$

which is larger than k if n is large enough. So we have found a graph G^* with the required properties. $\square$

Lecture 17 More on random graphs

We now shift our perspective on the random graph $G(n, p)$: We let the edge probability p depend on n , that is, we look at a *sequence* of random graphs $G(n, p)$, where $p = p(n)$ is a *function* of n . Usually $p(n) \rightarrow 0$ as $n \rightarrow \infty$. The question now is what the resulting graph looks like, depending on how quickly $p(n) \rightarrow 0$. In particular, we are interested in properties that either hold with probability tending to 0 or with probability tending to 1 as $n \rightarrow \infty$. Usually we suppress the dependence on n and only write p instead of $p(n)$.

Definition 150. Let $(E_n)_{n \geq 1}$ be a sequence of events (for example, E_n could be the event that $G \sim G(n, p)$ contains a triangle). Then we say that E_n holds *with high probability (whp)* if

$$\mathbb{P}(E_n) \rightarrow 1 \text{ as } n \rightarrow \infty.$$

(Alternatively, *asymptotically almost surely (aas)* is used, which means the same thing.)

Let us get back to triangle counts. Let X be the number of triangles in $G(n, p)$. In Lemma 144 we saw that the probability that $G(n, p)$ contains a triangle, i.e., that $X \geq 1$, is at most $\binom{n}{3}p^3$. This directly implies the following statement.

Lemma 151. Suppose¹⁰ that $p = p(n) \ll n^{-1}$ and let $G \sim G(n, p)$. Then

$$\mathbb{P}(X \geq 1) \rightarrow 0 \text{ as } n \rightarrow \infty,$$

that is, with high probability G does not contain any triangles.

Proof. $\mathbb{P}(X \geq 1) \leq \binom{n}{3}p^3 \leq n^3p^3 \rightarrow 0$ as $n \rightarrow \infty$. □

What if we want to prove the opposite statement, i.e., that for some function $p = p(n)$, $G \sim G(n, p)$ contains at least one triangle whp? Then we can use the *second moment method*. The following theorem is a consequence of Chebyshev's inequality.

Theorem 152 (Second moment method). *Let X be a non-negative random variable so that $\mathbb{E}[X]$ and $\mathbf{Var}(X)$ are finite.*¹¹ *Then*

$$\mathbb{P}(X = 0) \leq \frac{\mathbf{Var}(X)}{\mathbb{E}[X]^2}.$$

¹⁰We write $f \ll g$ if $f(x)/g(x) \rightarrow 0$ as $x \rightarrow \infty$, and analogously for $f \gg g$. We also use $f \sim g$ to express $f(x)/g(x) \rightarrow 1$.

¹¹When we are talking about $G(n, p)$, this is almost always trivially true, because the underlying probability space is finite.

We usually calculate the variance by

$$\mathbf{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

and $\mathbb{E}[X^2]$ is called the *second moment* of X , hence the name of the method.

Similarly as for $\mathbb{E}[X]$, when X is the random variable that counts how many of some events $A_1, \dots, A_k$ hold, we can express $\mathbb{E}[X^2]$ as a sum of probabilities:

Lemma 153. *If $X = \sum_{i=1}^k \mathbb{1}_{A_i}$ for some events $A_1, \dots, A_k$, then*

$$\mathbb{E}[X^2] = \sum_{i,j} \mathbb{P}(A_i \cap A_j) = \mathbb{E}[X] + \sum_{\substack{i,j \\ i \neq j}} \mathbb{P}(A_i \cap A_j)$$

Proof. Using the fact that $\mathbb{1}_{A_i} \mathbb{1}_{A_j} = \mathbb{1}_{A_i \cap A_j}$ and linearity of expectation:

$$\begin{aligned} \mathbb{E}[X^2] &= \mathbb{E}\left[\left(\sum_i \mathbb{1}_{A_i}\right)^2\right] = \sum_{i,j} \mathbb{E}[\mathbb{1}_{A_i} \mathbb{1}_{A_j}] = \sum_{i,j} \mathbb{P}(A_i \cap A_j) \\ &= \sum_i \mathbb{P}(A_i) + \sum_{\substack{i,j \\ i \neq j}} \mathbb{P}(A_i \cap A_j) = \mathbb{E}[X] + \sum_{\substack{i,j \\ i \neq j}} \mathbb{P}(A_i \cap A_j) \end{aligned}$$

□

Let's apply the second moment method to the number X of triangles in $G(n, p)$.

Proposition 154. *As $n \rightarrow \infty$, we have*

$$\mathbf{P}[G(n, p) \text{ contains a } K_3] \rightarrow \begin{cases} 0 & \text{if } p(n) \ll \frac{1}{n} \\ 1 & \text{if } p(n) \gg \frac{1}{n} \end{cases}$$

Proof. The first claim is Lemma 151. For the second claim, we again think of the number X of triangles in $G(n, p)$ as a sum of indicator random variables, like we did last time: We number the $\binom{n}{3}$ potential triangles in $G(n, p)$ from 1 to $\binom{n}{3}$ in some arbitrary way, let A_i be the event that the i^{th} triangle is present in $G(n, p)$, then $X = \sum_{i=1}^{\binom{n}{3}} \mathbb{1}_{A_i}$. Like last time, by Lemma 143,

$$\mathbb{E}[X] = \binom{n}{3} p^3 \sim \frac{n^3 p^3}{6}.$$

To prove the second claim, let's also calculate the variance of X , using Lemma 153 to work out $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \mathbb{E}[X] + \sum_{\substack{i,j \\ i \neq j}} \mathbb{P}[\text{Both triangles } i, j \text{ are present}].$$

The triangles i and j in the last line could share 0, 1, or 2 vertices, depending on their position relative to each other:

- If they have no vertices in common, then $\mathbb{P}(A_i \cap A_j) = p^6$, and this happens in $\binom{n}{3}\binom{n-3}{3}$ many configurations.
- If they have one vertex in common, then $\mathbb{P}(A_i \cap A_j) = p^6$, and this happens in $3\binom{n}{3}\binom{n-3}{2}$ many configurations.
- If they have two vertices in common, then they share an edge as well, so $\mathbb{P}(A_i \cap A_j) = p^5$, and this happens in $3\binom{n}{3}(n-3)$ many configurations.

Therefore, continuing the calculation above:

$$\mathbb{E}[X^2] = \mathbb{E}[X] + \binom{n}{3} \left(\binom{n-3}{3} p^6 + 3 \binom{n-3}{2} p^6 + 3(n-3) p^5 \right)$$

For the variance, we obtain

$$\begin{aligned} \mathbf{Var}[X] &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= \binom{n}{3} p^3 + \binom{n}{3} \binom{n-3}{3} p^6 + 3 \binom{n}{3} \binom{n-3}{2} p^6 \\ &\quad + 3(n-3) \binom{n}{3} p^5 - \binom{n}{3}^2 p^6 \\ &\leq \binom{n}{3} p^3 + 3 \binom{n}{3} \binom{n-3}{2} p^6 + 3(n-3) \binom{n}{3} p^5 \\ &\sim \frac{(np)^3}{6} + \frac{(np)^6}{4n} + \frac{(np)^5}{2n}. \end{aligned}$$

Hence, the expression $\mathbf{Var}[X]/\mathbb{E}[X]^2$ from the right-hand side of Theorem 152 is asymptotically less or equal to

$$\frac{\frac{(np)^3}{6} + \frac{(np)^6}{4n} + \frac{(np)^5}{2n}}{\frac{(np)^6}{36}} = \frac{6}{(np)^3} + \frac{9}{n} + \frac{18}{n^2 p}.$$

Now if $p \gg 1/n$, then $np(n) \rightarrow \infty$, so

$$\mathbb{P}(X = 0) \leq \frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} \rightarrow 0 \quad \text{for } n \rightarrow \infty$$

using the second moment method (Theorem 152), and so $\mathbb{P}(X > 0) \rightarrow 1$ and we have proved the second claim. $\square$

Definition 155. Let (P) be some property. A function $f(n)$ is called a *threshold* or *threshold function* for (P) if

$$\mathbb{P}(G(n, p) \text{ satisfies } (P)) \rightarrow \begin{cases} 0 & \text{if } p(n) \ll f(n) \\ 1 & \text{if } p(n) \gg f(n) \end{cases}$$

Thus, Proposition 154 can be rephrased as ‘ $1/n$ is a threshold for the property of the graph having at least one triangle’. We call (P) is a *monotone increasing property*, if, whenever a graph has property (P) and we add more edges to it, then the resulting graph still has property (P) . It can be shown that every monotone property has a threshold function. So properties such as containing a fixed subgraph H , being connected, being non-planar, or being Hamiltonian all have threshold functions.

Remark 156. In the setting of Proposition 154, what happens if $p = \frac{c}{n}$ for some $c > 0$? In this case, the argument can be refined a bit using Janson’s inequality, to show that

$$\mathbb{P}(G(n, c/n) \text{ contains a triangle}) \rightarrow 1 - e^{-c^3/6}$$

as $n \rightarrow \infty$. In fact, one can show that in $G(n, c/n)$, the number of triangles X is asymptotically distributed as a Poisson distribution with parameter $c^3/6$.

By the same methods as in Proposition 154 (though the computations are a little bit more tedious) one can obtain the critical function for the containment of a K_4 :

$$\mathbf{P}[G(n, p) \supseteq K_4] \rightarrow \begin{cases} 0 & \text{if } p(n) \ll n^{-2/3} \\ 1 & \text{if } p(n) \gg n^{-2/3} \end{cases}$$

What about containing general a subgraph H ? Suppose H has v_H vertices and e_H edges. Let $\text{Aut}(H)$ be the set of isomorphisms $H \rightarrow H$ (such isomorphisms are also called automorphisms, hence the notation).

The following lemma can be proved by the same methods we used when working out the expected number of triangles.

Lemma 157. *With the notation just introduced, the following holds:*

(i) *There are*

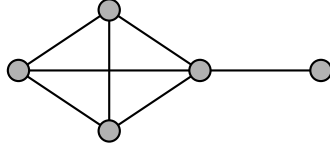
$$\binom{n}{v_H} \frac{v_H!}{|\text{Aut}(H)|} \sim \frac{n^{v_H}}{|\text{Aut}(H)|}$$

copies of H in a labelled K_n .

(ii) *If X_H is the number of copies of H in $G(n, p)$, then*

$$\mathbb{E}[X_H] \sim \frac{n^{v_H} p^{e_H}}{|\text{Aut}(H)|} \rightarrow \begin{cases} 0 & \text{if } p \ll n^{-v_H/e_H} \\ \infty & \text{if } p \gg n^{-v_H/e_H} \end{cases}$$

This would suggest that n^{-v_H/e_H} is a threshold function for containment of H , and for K_3 and K_4 this is indeed the case. But it’s not true in general! Consider the following graph H :



We have $v_H = 5$, $e_H = 7$, so the heuristic above would suggest $n^{-5/7}$ is a threshold for containment of H . But this is less than $n^{-2/3}$, the threshold for containment of K_4 . If we select some function $n^{-5/7} \ll p(n) \ll n^{-2/3}$, then for this p , $G(n, p)$ whp does not contain any copy of K_4 , so it can't contain any copy of H , because H contains a K_4 !

What went wrong here? H contained a subgraph, K_4 , which is 'denser', in the sense that its e/v ratio is larger than that of H (K_4 has more edges per vertex than H). So the threshold $n^{-v_{K_4}/e_{K_4}} = n^{-2/3}$ is the 'bottleneck' for containment of H .

In general, let's define a function which captures this 'bottleneck' for every graph H : Set

$$m(H) := \max_{H' \subseteq H, v_{H'} > 0} \frac{e_{H'}}{v_{H'}},$$

that is, $m(H)$ is the edge density of the densest subgraph of H . Then $n^{-1/m(H)}$ is a threshold function for containment of H , which can be proved via the same approach that we used for triangles.

Theorem 158 (Bollobás, 1981). *Let H be a finite simple graph with at least one edge. Then*

$$\mathbb{P}[G(n, p) \supseteq H] \rightarrow \begin{cases} 0 & \text{if } p \ll n^{-1/m(H)} \\ 1 & \text{if } p \gg n^{-1/m(H)} \end{cases}.$$

The evolution of $G(n, p)$.

Imagine now that we fix the number n of vertices, but increase the edge probability p . This should look as if we were adding more edges into the graph randomly. This idea can be made rigorous by using a *coupling*: Consider a labelled K_n where every edge e is equipped with an independent random variable X_e that is uniformly distributed on $[0, 1]$. Then for any fixed p , the Erdős-Rényi random graph $G(n, p)$ has the same distribution as the random graph on the n vertices where we include exactly those edges e with $X_e \leq p$. Using this coupling, we can thus ensure a version of $G(n, p)$ that is monotonous in p . What happens as we increase p from 0 to 1? We will let $n \rightarrow \infty$ here, so p is really a function of n .

- For $p(n) \ll n^{-2}$, the graph starts out very boring: Whp, it does not even have a single edge. The function n^{-2} is the threshold for the existence of an edge.

- For fixed k , the function $n^{-k/(k-1)}$ is the threshold for the existence of a tree on k vertices.
- For $p(n) \ll 1/n$ there are whp no cycles in $G(n, p)$, whereas for $p(n) \gg 1/n$ they are whp guaranteed.
- For $p(n) = \frac{c}{n}$ with $0 < c < 1$, there are many components that are either trees or contain a single cycle, and all of them are small (of logarithmic size): If $C_{\max}$ denotes the number of vertices in the largest component, then we have

$$\frac{C_{\max}}{\log n} \rightarrow \frac{1}{c - 1 - \log c}$$

in probability.¹²

- For $p(n) = \frac{1}{n}$, the largest component has grown to contain about $n^{2/3}$ vertices.
- For $p(n) = \frac{c}{n}$ with $c > 1$, the largest component contains a positive proportion $\zeta_c n$ of vertices. Moreover, there is only one such component (as the probability is large enough to guarantee an edge between any two vertex sets of positive proportion), and we call it the *giant component*. The giant component contains enough edges to be non-planar. All other components are trees on $O(\log(n))$ vertices.
- For $p(n) = \frac{\log(n)+c}{n}$ where $c > 0$, all vertices are either isolated or in the giant component. The number of isolated vertices is Poisson distributed with parameter e^{-c} .
- A lot goes down when p is around $\frac{\log(n)}{n}$: This is a threshold function for the graph having minimum degree 1 (no isolated vertices), minimum degree k for k fixed, connectivity, k -connectivity, containment of a perfect matching and Hamiltonicity.
- The functions $n^{-2/(k-1)}$ are the threshold functions for the appearance of K_k .
- The functions $\frac{(\log n)^{1/r}}{n^{1-1/r}}$ are threshold functions for the graph to have diameter r .
- By the time we reach constant edge density p , let's say $p = 1/2$, the graph has diameter 2. Its largest clique and its largest independent set are both of size $\sim 2 \log_2 n$ whp, and its chromatic number is $\sim n/(2 \log_2 n)$.

¹²A sequence X_n converges to $a \in \mathbb{R}$ in probability if $\mathbf{P}[|X_n - a| > \varepsilon] \rightarrow 0$ as $n \rightarrow \infty$ pointwise for all $\varepsilon > 0$.

- If we let p approach 1, then whp the graph contains almost all edges except around $N(1 - p)$ of them. If $1 - p \ll n^{-2}$, whp the graph is complete.

Lecture 18 Extremal graph theory and Szemerédi's regularity lemma

In this lecture we give a brief glimpse into *extremal graph theory*. Here we ask questions such as: How many edges can a graph on n vertices still have if we impose some restriction on it, for example that it is triangle-free? Or how many edges would we need to delete, at most, to make the graph triangle-free? And what do the extremal graphs look like, i.e., those that have the maximum number of edges while being triangle free?

Turán's theorem

For a fixed finite simple graph H , we say that G is H -free if it does not contain a copy of H as a subgraph, and we let

$$\text{ex}(n, H)$$

be the maximum number of edges amongst all H -free graphs on n vertices. As a warm up, let's look at triangles:

Theorem 159 (Mantel, 1907). $\text{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$.

Proof. By induction. For $n = 1, 2$ the result is trivial, so suppose $n \geq 3$, and that we have proved the result for all $k < n$. Suppose that G is triangle-free on n vertices, and let x and y be any two neighbouring vertices. Then x, y cannot have any common neighbour, because otherwise G would contain a triangle. In particular, $(d(x) - 1) + (d(y) - 1) \leq n - 2$, that is, $d(x) + d(y) \leq n$. Now remove the vertices x and y from G and call the resulting graph on $n - 2$ vertices G' . Then as G' is triangle-free, by induction we have

$$e(G) = d(x) + d(y) - 1 + e(G') \leq n - 1 + \frac{(n - 2)^2}{4} = \frac{n^2}{4}$$

(the -1 is because the edge xy is counted twice in $d(x) + d(y)$). Since $e(G)$ is an integer, it follows that $e(G) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$.

How about the other direction? We need to find a triangle-free graph with $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges. Take a *complete bipartite graph* $K_{a,b}$ with parts of size a and b where $a + b = n$, which has ab edges. Clearly such a graph is triangle-free. It's not hard to show that ab is maximal if a and b are as equal as possible, that is, if $a = b = \frac{n}{2}$ for even n , and $a = \left\lfloor \frac{n}{2} \right\rfloor$ and $b = \left\lceil \frac{n}{2} \right\rceil$ for odd n , and that in that case, $ab = \left\lfloor \frac{n^2}{4} \right\rfloor$. $\square$

We saw one example of a triangle-free graph that maximises the number of edges: The complete bipartite graph $K_{a,b}$, $a + b = n$, with a and b as

equal as possible. Are there any others? And what about K_4 -free, K_5 -free, ..., graphs?

Definition 160. The *Turán graph* $T_r(n)$ is the complete r -partite graph on n vertices where the part sizes are as equal as possible. Let $t_r(n)$ be the number of edges of the Turán graph $T_r(n)$. The numbers $t_r(n)$ are called the *Turán numbers*.

So all parts of $T_r(n)$ contain either $\lfloor \frac{n}{r} \rfloor$ or $\lceil \frac{n}{r} \rceil$ vertices. Note that for $n \leq r$, $T_r(n) = K_n$.

Theorem 161 (Turán, 1941). *For any $r, n \geq 1$,*

$$\text{ex}(n, K_{r+1}) = t_r(n),$$

and up to isomorphism, the Turán graph $T_r(n)$ is the only K_{r+1} -free graph on n vertices with $t_r(n)$ edges.

Proof. We first show that if x and y are non-adjacent, and y and z are non-adjacent, then x and z are non-adjacent (in other words, being non-adjacent is an equivalence relation on the vertex set V). It follows from this that G must be a complete multipartite graph. So suppose for a contradiction that $xy \notin E$, $yz \notin E$, but $xz \in E$.

If $d(y) < d(x)$, then create a new graph G' from G by deleting y and creating a new copy of x , say x' , which is a neighbour of all neighbours of x (but not of x itself). Then G' is still K_{r+1} -free, but $e(G') > e(G)$, a contradiction. So we must have $d(y) \geq d(x)$, and by the same argument (with x and z switched out), $d(y) \geq d(z)$. So now create a new graph G'' by deleting both x and z and instead introducing two new copies y' , y'' of y . Again G'' is K_{r+1} -free, and, since xz is an edge counted in both $d(x)$ and $d(z)$,

$$e(G'') = e(G) - (d(x) - 1) - (d(z) - 1) - 1 + 2d(y) \geq e(G) + 1 > e(G),$$

again a contradiction.

So we know that G is complete multipartite. Clearly G cannot have more than r non-empty parts, otherwise G would contain a K_{r+1} . As a final step, we need to verify that the Turán graph $T_r(n)$ has the most edges amongst all complete r -partite graphs on n vertices. To see this, consider a complete r -partite graph where the sizes of two parts differ by at least 2. Then you can increase the number of edges by moving one vertex from the smaller to the larger part, making the part sizes more equal. You can continue like this until you get $T_r(n)$.

So we must have $G \cong T_r(n)$, and $e(G) = t_r(n)$. □

The regularity lemma and a counting lemma

Szemerédi’s regularity lemma is perhaps the most commonly used tool in extremal graph theory. Roughly speaking, it says that if a graph G is large enough, then we can ‘tame’ it: We can find a vertex partition so that the edges between the parts behave ‘random-like’, up to a small number of exceptions. We can then use this partition, for example, to count the number of subgraphs such as triangles in G .

To state the regularity lemma, we need a few definitions.

Definition 162. Let $G = (V, E)$ be a finite simple graph and $X, Y \subset V$ vertex sets. Let $e(X, Y)$ be the number of edges with the first endpoint in X and the other endpoint in Y ,

$$e(X, Y) = |\{(x, y) | xy \in E, x \in X, y \in Y\}|,$$

then we call

$$d(X, Y) = \frac{e(X, Y)}{|X||Y|}$$

the *density* of the pair (X, Y) .

Since $e(X, Y) \leq |X||Y|$, we always have $0 \leq d(X, Y) \leq 1$.

While we mostly think of the sets X, Y as disjoint, and of the edges and density ‘between’ them, the definition also allows X and Y to overlap, or even $X = Y$ (note we are counting edges twice in $e(X, Y)$ in that case).

Definition 163. Let G be a finite simple graph. Then we say that a pair (A, B) of vertex sets is ε -regular if, for all $A' \subset A$ and $B' \subset B$ with $|A'| \geq \varepsilon|A|$ and $|B'| \geq \varepsilon|B|$,

$$|d(A', B') - d(A, B)| \leq \varepsilon.$$

A vertex partition $V = V_1 \cup \dots \cup V_k$ is called ε -regular if

$$\sum_{(V_i, V_j) \text{ not } \varepsilon\text{-regular}} \frac{|V_i||V_j|}{n^2} \leq \varepsilon.$$

So if (A, B) is ε -regular, any large enough subsets $A' \subset A, B' \subset B$ have roughly the same edge density between them as A, B themselves. This is also what would happen in a random graph if we included every edge between A and B independently with probability $p \approx d(A, B)$, so we may think of ε -regular pairs as behaving ‘random-like’.¹³ A partition is ε -regular if most pairs of parts are ε -regular — but we allow for some non- ε -regular pairs, as long as there are not too many or too large ones.

We are ready to state the regularity lemma (but won’t prove it).

¹³This type of property in a (non-random) graph is often called *pseudorandomness*.

Theorem 164 (Szemerédi’s regularity lemma, 1978). *For every $\varepsilon > 0$ there is an M such that every graph has an ε -regular partition with at most M parts.*

There are stronger (and more complicated) versions of the regularity lemma: In particular, we could make the parts of the partition, except for one small part, have the same size. But this simpler version will be enough here.

Note that the regularity lemma tells us nothing about small graphs (with few vertices), because if $n \leq M$, a vertex partition into M parts of sizes 0 and 1 is always ε -regular. It’s also not particularly useful for sparse graphs (with few edges) where the edge density between parts is less than ε anyway.

How large is $M = M(\varepsilon)$ in relation to ε ? Sadly, the M from the proof is of the form of a tower $2^{2^{\dots^2}}$, where the height of the tower is of order $O(\varepsilon^{-C})$ for some constant $C > 0$. Gowers showed in 1997 that you can’t do better than this type of bound¹⁴. So the regularity lemma only makes sense for very large graphs.

The regularity lemma becomes particularly useful in combination with counting lemmas. Here is a triangle counting lemma. It says that between ε -regular pairs of parts there are at least as many triangles as, roughly, you expect to have in a random graph.

Lemma 165. *Let G be a graph with three disjoint subsets X, Y, Z . Suppose that all three pairs (X, Y) , (Y, Z) and (Z, X) are ε -regular with densities $d(X, Y) = \alpha$, $d(Y, Z) = \beta$, $d(Z, X) = \gamma$. Then, if $\alpha, \beta, \gamma > 2\varepsilon$, the number of triangles xyz with $x \in X$, $y \in Y$, $z \in Z$ is at least*

$$(1 - 2\varepsilon)(\alpha - \varepsilon)(\beta - \varepsilon)(\gamma - \varepsilon)|X||Y||Z|.$$

Note that if we included the edges between X, Y, Z independently at random with probabilities α, β, γ , then the expected number of triangles would be $\alpha\beta\gamma|X||Y||Z|$.

Proof. For $x \in X$, let $d_Y(x)$ and $d_Z(x)$ be the number of neighbours of x in Y and Z , respectively. Then there are fewer than $\varepsilon|X|$ vertices x with $d_Y(x) < (\alpha - \varepsilon)|Y|$ — otherwise, take as X' the set of such x and $Y' = Y$, then $d(X', Y') < \alpha - \varepsilon$, which contradicts ε -regularity of the pair (X, Y) . In the same way, we can show that there are fewer than $\varepsilon|X|$ vertices x with $d_Z(x) < (\gamma - \varepsilon)|Z|$.

Now for any x with $d_Y(x) \geq (\alpha - \varepsilon)|Y| \geq \varepsilon|Y|$ and $d_Z(x) \geq (\gamma - \varepsilon)|Z| \geq \varepsilon|Z|$, since the pair (Y, Z) is regular, there are at least

$$(\beta - \varepsilon)(\alpha - \varepsilon)|Y|(\gamma - \varepsilon)|Z|$$

¹⁴He gave examples of graphs where you need M to be a tower of height about $\varepsilon^{-1/16}$.

edges between a neighbour y of x in Y and a neighbour z of x in Z . So there are at least that many triangles xyz containing this particular x . Since there are at least $(1 - 2\varepsilon)|X|$ vertices x with that property, taking the sum over all such x , we have at least

$$(1 - 2\varepsilon)(\alpha - \varepsilon)(\beta - \varepsilon)(\gamma - \varepsilon)|X||Y||Z|.$$

triangles in total. $\square$

The triangle removal lemma

We now use the regularity lemma together with the counting lemma to prove the triangle removal lemma. It says that if a graph only contains few triangles, we can make it triangle-free by only deleting a small number of edges.

Theorem 166 (Triangle removal lemma). *For every $\varepsilon > 0$ there is a $\delta > 0$ so that any graph G on n vertices with at most δn^3 triangles can be made triangle-free by removing at most εn^2 edges.*

Proof. By the regularity lemma, G has an $\frac{\varepsilon}{4}$ -regular partition $V_1 \cup \dots \cup V_M$ for some $M = M(\varepsilon/4)$. Now remove an edge xy from G if:

1. $x \in V_i$ and $y \in V_j$, where (V_i, V_j) is not a $\frac{\varepsilon}{4}$ -regular pair (exceptional pairs)
2. $x \in V_i$ and $y \in V_j$, where $d(V_i, V_j) < \frac{\varepsilon}{2}$ (low density)
3. $x \in V_i$ with $|V_i| \leq \frac{\varepsilon}{4M}n$ (small parts)

As $\sum \frac{|V_i||V_j|}{n^2} \leq \frac{\varepsilon}{4}$, where the sum is taken over all non- $\frac{\varepsilon}{4}$ -regular pairs, the first condition removes at most $\frac{\varepsilon}{4}n^2$ edges. Condition 2 clearly removes at most $\frac{\varepsilon}{2}n^2$ edges. And at most $M\frac{\varepsilon}{4M}n = \frac{\varepsilon}{4}n$ vertices x fulfil condition 3, so this removes at most $\frac{\varepsilon}{4}n^2$ edges. In total, we have removed at most εn^2 edges.

Are there any triangles left? Suppose there is such a triangle, xyz . After our removals this is only possible if $x \in V_i$, $y \in V_j$, $z \in V_k$ for some parts V_i, V_j, V_k of size at least $\frac{\varepsilon}{4M}n$ and which are pairwise $\frac{\varepsilon}{4}$ -regular with density at least $\frac{\varepsilon}{2}$. But then, by the triangle counting lemma, there are at least

$$\left(1 - 2 \cdot \frac{\varepsilon}{4}\right) \left(\frac{\varepsilon}{2} - \frac{\varepsilon}{4}\right)^3 \left(\frac{\varepsilon}{4M}n\right)^3 = \left(1 - \frac{\varepsilon}{2}\right) \frac{\varepsilon^6}{4^6 M^3} n^3$$

triangles. If we pick γ smaller than this, say $\gamma = \frac{\varepsilon^6}{4^{10} M^3}$, this gives a contradiction because there were at most γn^3 triangles in the graph to begin with. So G must be triangle-free after we deleted the edges. $\square$

Lecture 19 The Rado graph

In previous lectures, we considered the random graph $G(n, p)$ on n vertices where every edge is included independently with probability p . What happens if we do the same thing on (countably) infinitely many vertices? That is, we construct a random graph $G(p)$ on the vertex set $\mathbb{N}_0$ by including every possible edge ij (where $i, j \in \mathbb{N}_0$ with $i \neq j$) independently with probability p . It turns out that this infinite random graph behaves very differently than the finite version. In particular, we will see that with probability 1, the resulting graph is always exactly the same one (up to isomorphism), for every $0 < p < 1$: the Rado graph.

The Rado graph has many interesting properties, and we will define and study it from a deterministic point of view first. Then we will show that $G(p)$ is isomorphic to the Rado graph with probability 1, regardless of p .

Construction of the Rado graph

We first give an explicit construction of the Rado graph, but why we choose this particular construction will make much more sense in a little while when we learn about the characterisation of the Rado graph as the unique graph on $\mathbb{N}_0$ with the extension property.

Let $V = \mathbb{N}_0$, and also write down the representation of $i \in \mathbb{N}_0$ in the binary system. Then for $i < j$, draw an edge between the vertices i and j if the digit representing 2^i in the binary expression for j is a 1, otherwise don't. Here's an example: $13 = 2^3 + 2^2 + 2^0 = (1101)_2$, so vertex 13 has neighbours 0, 2 and 3. It is also a neighbour of all i which have a 1 in the 2^{13} place in their binary representation, for example all $2^{13}, 2^{13} + 1, 2^{13} + 2, \dots$ and infinitely many others. Call this graph the *Rado graph* R .

Another way to understand this construction is to consider the vertices neighbouring 0, 1, 2, ...: Neighbours of 0 must have a 1 in their last binary position, so 0 has an edge to all odd numbers. Similarly, neighbours of 1 have a 1 in their second-to-last binary position, and those are exactly the numbers 2, 3, 6, 7, 10, 11, ..., i.e. the numbers congruent to 2 or 3 mod 4.

The extension property

The key property of the Rado graph is the extension property. This is an innocent looking property which turns out to be extremely powerful.

Definition 167. A simple graph $G = (V, E)$ has the *extension property* (P) if, for all finite disjoint subsets $U, W \subseteq V$, there exists a vertex $x \notin U \cup W$ such that x is a neighbour to every vertex in U , but not a neighbour to any vertex in W .

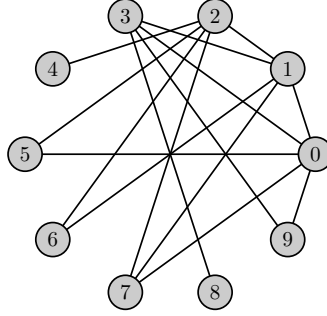


Figure 1: The subgraph of (our construction of) the Rado graph induced by the vertices $1, \dots, 9$.

It follows immediately that any graph $G = (V, E)$ with property (P) must be infinite, since for a finite graph, we could for example set $U = V, W = \emptyset$, and property (P) cannot hold because there is no further vertex x outside V .

Lemma 168. *The Rado graph R satisfies the extension property.*

Proof. Consider two finite, disjoint subsets $U, W \subseteq \mathbb{N}_0$, say $U = \{u_1, \dots, u_m\}$ and $W = \{w_1, \dots, w_n\}$ for $m, n \geq 0$. Our goal is to construct an integer r such that r is a neighbour to $u_1, \dots, u_m$ according to the construction of Rado's graph, but not a neighbour to $w_1, \dots, w_n$. So, set $\ell := 1 + \max(U \cup W)$, and define

$$r := 2^\ell + \sum_{i=1}^m 2^{u_i}.$$

With this definition, we ensure that r is larger than any number in either U or W . Moreover, all numbers in U are neighbours to r , as their binary digits occur in the binary expansion of r , by construction. Similarly, since W is disjoint to U , none of the digits of r belonging to $w_1, \dots, w_n$ can be 1; hence r is not a neighbour to any of those vertices. $\square$

To illustrate how powerful the extension property is, we use it to prove the following theorem.

Theorem 169. *Let R be the Rado graph, and let G be any finite simple graph. Then there is an induced subgraph H of R such that $H \cong G$.*

Proof of Theorem 169. Let $G = (V, E)$ be any finite simple graph, and order its vertices $v_1, \dots, v_n$ in some arbitrary way. We will inductively construct a sequence of induced subgraphs $R_1, \dots, R_n$ of R such that for each k , $R_k \cong V[\{v_1, \dots, v_k\}] =: G_k$.

Start by mapping v_1 to any vertex w_1 of R , and set $R_1 = R[\{w_1\}]$. Now, assume that for $k = 1, \dots, n-1$ we have already constructed $R_k = R[\{w_1, \dots, w_k\}]$

in such a way that the isomorphism between G_k and R_k maps v_j to w_j for all $j = 1, \dots, k$. Now consider v_{k+1} and set

$$\begin{aligned} U &:= \{w_j : 1 \leq j \leq k \text{ and } v_j \text{ is adjacent to } v_{k+1}\} \\ W &:= \{w_j : 1 \leq j \leq k \text{ and } v_j \text{ is not adjacent to } v_{k+1}\} \end{aligned}$$

Since R satisfies the extension property, there exists a vertex w_{k+1} that is a neighbour to all $w_j \in U$ and not a neighbour to any $w_j \in W$. So we can map v_{k+1} to w_{k+1} and extend the isomorphism between G_k and R_k to an isomorphism of the induced subgraphs G_{k+1} and $R_{k+1} := R[w_1, \dots, w_{k+1}]$.

We continue this construction until we map $G = G_n$ to R_n . $\square$

Observe how we did not require anything about the Rado graph in the proof of Theorem 169 other than the fact that it satisfies property (P). So we could have phrased the theorem as:

Corollary 170. *Any graph with the extension property contains all finite simple graphs as induced subgraphs.*

Characterisation of the Rado graph

Here we give a characterisation of the Rado graph as the unique graph on countably many vertices that satisfies the extension property. We could have also started with this characterisation to *define* the Rado graph, but then would have needed to prove (for example with our earlier explicit construction) that such a graph exists at all.

Theorem 171. *Let G and H be two simple graphs on countable vertex sets V and W , respectively. Assume that both G and H satisfy (P). Then $G \cong H$. So up to isomorphism, R is the only graph on a countable vertex set that satisfies (P).*

Proof. Similarly to the proof of Theorem 169, we can use property (P) to extend isomorphisms between smaller subgraphs. But we have to be slightly careful: If we simply try to mimick the proof from Theorem 169 exactly, then we could for example map G to an induced subgraph of H — but we want to use *all* vertices in H in order to show that H is isomorphic to G , not just that some induced subgraph of H is isomorphic to G .

The trick is to go ‘back and forth’ between G and H in the construction of the isomorphism, which is possible because they both have the extension property, to make sure we use all vertices in both graphs. Here is the argument in detail:

Start by ordering the vertex set V of G by $v_0, v_1, v_2, \dots$ in some arbitrary way, and the vertex set W of H in some arbitrary way by $w_0, w_1, w_2, \dots$. Let $G_0 := G[\{v_0\}]$ and $H_0 := H[\{w_0\}]$, then clearly $G_0 \cong H_0$.

Now assume that we have extended this isomorphism to two induced subgraphs G_n and H_n , with vertices $a_0, a_1, \dots, a_n$ and $b_0, b_1, \dots, b_n$, respectively, so that the isomorphism maps a_j to b_j for all $0 \leq j \leq n$. Now proceed along the following two steps:

1. Let a_{n+1} be the first vertex in G (in the ordering $v_0, v_1, v_2, \dots$ of V) that is not yet included in G_n . Set $G_{n+1} := G[\{a_0, a_1, \dots, a_{n+1}\}]$. Because H has the extension property, by the same argument as in the proof of Theorem 169, there is a vertex b_{n+1} in the vertex set W of H that extends the isomorphism to $G_{n+1} \cong H_{n+1} := H[\{b_0, b_1, \dots, b_{n+1}\}]$.

2. Let b_{n+2} be the earliest vertex in H (in the ordering $w_0, w_1, w_2, \dots$ of W) not yet included in H_{n+1} . Analogously to step 1, there exists a vertex a_{n+2} in G that extends the isomorphism to $H_{n+2} := H[\{b_0, b_1, \dots, b_{n+2}\}]$ and $G_{n+2} := G[\{a_0, a_1, \dots, a_{n+2}\}]$.

Now, starting from $G_0 \cong H_0$, we can extend the isomorphism by repeating steps 1 and 2 over and over again. Eventually, this procedure maps every vertex in V to some vertex in W (and vice versa): every vertex v_n appears as one of the vertices $\{a_0, a_1, \dots, a_{2n}\}$, and every w_n appears as one of the vertices $\{b_0, b_1, \dots, b_{2n}\}$. So this procedure extends to an isomorphism $G \cong H$. $\square$

Remark 172. The proof also implies that the Rado graph R is *vertex transitive*: This means that for all vertices v, w , there is an automorphism (an isomorphism from R to itself) that maps v to w .

So, in a strong sense, all vertices in the Rado graph are the same and have the same properties. This is not at all obvious from our explicit construction of the Rado graph at the beginning of the lecture!

Remark 173. Now that we have characterised the Rado graph in this way, let's revisit the construction. If we did not know already that a graph with the extension property exists, how would we find one? We want to make sure that for all finite disjoint subsets U, W , there's a vertex that's adjacent to all vertices in U , but no vertex in W .

We could construct such a graph iteratively: Start with a single vertex. When you've already constructed a graph on k vertices $v_1, \dots, v_k$, list all possible disjoint subsets U and W of $V_k = \{v_1, \dots, v_k\}$. Or in fact, we might as well just look at all bipartitions $U, V_k \setminus U$. For every such bipartition $(U, V_k \setminus U)$, add a new vertex to the graph and connect it by edges exactly to the vertices in U . Then in your new graph (which has lots of new vertices), you at least have the extension property for all pairs of subsets of $\{v_1, \dots, v_k\}$. Now repeat this step infinitely often: the union of all your graphs is now an

infinite graph with the extension property.

The construction we saw at the beginning of the lecture is a slick way of formalising this idea: If you look at the first k vertices, you can encode all bipartitions (U, W) of that vertex set by a binary string $1001\dots 01$ of length k , with a 1 for all the elements in U and a 0 for all the elements in W . The vertex with that string as its binary representation (and maybe an additional 1 in the beginning to make sure it is not one of the first k vertices) is then exactly joined by edges to all vertices in U , but none in W , and there's a vertex for each possible bipartition.

More properties of the Rado graph

We already saw that the Rado graph contains every finite simple graph as an induced subgraph, and that it is vertex transitive.

The extension property is very strong and also implies all of the following:

- $\text{diam}(R) = 2$

Proof. Just take any two vertices v, w , and set $U = \{v, w\}$ and $W = \emptyset$. Then by the extension property there's an x that is a common neighbour of v and w , so the graph distance between any v, w is at most 2. $\square$

- For all finite disjoint vertex sets U, W , there are infinitely many vertices that are neighbours to every $u \in U$, and non-adjacent to every $w \in W$.

Proof. Let x_1 be one such vertex, which is guaranteed to exist by the extension property, and add x_1 to either U or W (it does not matter which). Now apply the extension property again to your new sets, then you find a new vertex $x_2 \neq x_1$ which is adjacent to all of U and none of W . Repeating this process gives an infinite sequence $x_1, x_2, \dots$ of such vertices. $\square$

- If a finite number of edges is either added to or deleted from R , then the resulting graph is still isomorphic to R .

Proof. This follows from the previous statement: Since for any U, W , there are infinitely many vertices to 'choose from' for the extension property, changing finitely many edges can only rule out finitely many of them, so the resulting graph still has a vertex x which is a neighbour to every vertex in U and no vertex in W , so the extension property still holds. $\square$

- For $\mathbb{N}_0 = V_1 \uplus V_2$, at least one of $R[V_1]$ and $R[V_2]$ is isomorphic to R .

Proof. We need to show at least one of $R[V_1]$ and $R[V_2]$ has the extension property. So suppose that $R[V_1]$ does not have the extension property. That means that there are disjoint sets $U, W \subset V_1$ so that there is no vertex $x \in V_1$ which is adjacent to all vertices in U , but none in W . So any such vertex x must lie in V_2 .

Now let U', W' be any disjoint subsets of V_2 , and set $U'' = U \cup U'$, $W'' = W \cup W'$. By the extension property of R , there is a vertex $x \in V$ which is a neighbour to every vertex in U'' , but no vertex in W'' . In particular it is a neighbour to every vertex in U and no vertex in W , so by our assumption x must be in V_2 . But that means that $R[V_2]$ has the extension property. $\square$

The infinite random graph

As promised at the start, we will now show that the *infinite random graph* $G(p)$ is isomorphic to the Rado graph with probability 1.

Let's repeat the definition. We fix $0 \leq p \leq 1$, let $V = \mathbb{N}_0$, and for any pair of different integers i, j , we include the edge ij in $G(p)$ with probability p , independently of all other edges.

The cases $p = 0$ and $p = 1$ are not particularly interesting: For $p = 0$, $G(p)$ will contain no edges at all (with probability 1), whereas for $p = 1$, $G(p)$ is the complete graph on $\mathbb{N}_0$ (again, with probability 1). For all values in between, one would expect $G(p)$ depend both on the parameter p as well as on the randomness involved in the construction. However, this is not the case:

Theorem 174. *For $0 < p < 1$, the graph $G(p)$ satisfies the extension property with probability 1.*

By Theorem 171, this immediately implies that $G(p) \cong R$ almost surely.

We will need the following two basic properties of probability measures in the proof.¹⁵

If $A_1, A_2, A_3, \dots$ is a sequence of events, and we set $A = \bigcap_{i=1}^{\infty} A_i$, then

$$\mathbb{P}(A) = \lim_{i \rightarrow \infty} \mathbb{P}\left(\bigcap_{j=1}^i A_j\right).$$

Call this **Fact 1**.

And a useful special case is that if $A_1, A_2, \dots$ is a countable collection of events, each of which holds almost surely, that is, $\mathbb{P}(A_n) = 1$ for all n , then $A = \bigcap_{n=1}^{\infty} A_n$ also holds almost surely, that is, $\mathbb{P}(A) = 1$. Call this **Fact 2**.

¹⁵They follow directly from continuity of probability measures, which in turn follows from countable additivity, one of the defining properties of a probability measure.

Proof of Theorem 174. Let $U, W \subseteq \mathbb{N}_0$ be finite, disjoint vertex subsets of $G(p)$. Let $r := \max U \cup W$, and define events A_n to be

$$A_n := \{(r+n) \sim u, (r+n) \not\sim w \text{ for all } u \in U, w \in W\}.$$

In other words, A_n describes the event that the vertex $r+n$ is a vertex as required for property (P). For A_n to occur, we need $|U|$ edges to be in $G(p)$ (each of them is present with probability p), and $|W|$ edges to not be in $G(p)$ (which happens with probability $1-p$ each), so

$$\mathbb{P}(A_n) = p^{|U|}(1-p)^{|W|} =: q \in (0, 1).$$

All the events A_n are independent from each other (because they depend on disjoint edge sets, and the edges appear independently). Now let

$$B_k = \bigcap_{n=1}^k A_n^c$$

be the event that there is *no* such vertex amongst the vertices $r+1, \dots, r+k$, then by independence of the events A_n ,

$$\mathbb{P}(B_k) = (1-q)^k \rightarrow 0 \text{ as } k \rightarrow \infty.$$

Let $B = \bigcap_{n=1}^{\infty} A_n^c$ the the event that there is no such vertex at all amongst $r+1, r+2, r+3, \dots$. By Fact 1, we therefore have

$$\mathbb{P}(B) = 0.$$

So with probability 1, at least one such vertex exists for the given sets U, W . Given finite disjoint $U, W \subset \mathbb{N}_0$, let $C_{U,W}$ be the event that there is a vertex which is adjacent to all vertices in U and none in W , then we just showed $\mathbb{P}(C_{U,W}) = 1$. Since there are only countably many such pairs of sets U, W ,¹⁶ by Fact 2,

$$\mathbb{P}(G(p) \text{ satisfies } (P)) = \mathbb{P}\left(\bigcap_{U,W} C_{U,W}\right) = 1.$$

□

¹⁶The number of finite subsets of a countable set is countable.

Some references

These lecture notes are adapted from Fabian Burghart's lecture notes, who lectured this course between 2020 and 2022. His references are given below. Additionally I used the following material (especially in lectures 12-17 on colourings, Ramsey theory and random graphs, lecture 18 on extremal graph theory, and in some of lecture 8 on the max-flow min-cut theorem):

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- Oliver Riordan: *C8.4 Probabilistic Combinatorics*, Lecture notes, Oxford University 2019.
- David Conlon: *Extremal graph theory*, lecture notes for a course.

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